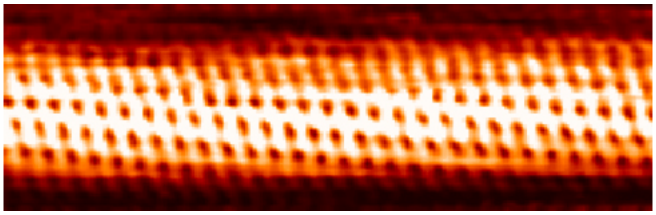
Introduction to Atomic Physics



Individual carbon atoms are visible in this image of a carbon nanotube made by a scanning tunneling electron microscope. (credit: Taner Yildirim, National Institute of Standards and Technology, via Wikimedia Commons)

From childhood on, we learn that atoms are a substructure of all things around us, from the air we breathe to the autumn leaves that blanket a forest trail. Invisible to the eye, the existence and properties of atoms are used to explain many phenomena—a theme found throughout this text. In this chapter, we discuss the discovery of atoms and their own substructures; we then apply quantum mechanics to the description of atoms, and their properties and interactions. Along the way, we will find, much like the scientists who made the original discoveries, that new concepts emerge with applications far beyond the boundaries of atomic physics.

Discovery of the Atom

- Describe the basic structure of the atom, the substructure of all matter.

How do we know that atoms are really there if we cannot see them with our eyes? A brief account of the progression from the proposal of atoms by the Greeks to the first direct evidence of their existence follows.

People have long speculated about the structure of matter and the existence of atoms. The earliest significant ideas to survive are due to the ancient Greeks in the fifth century BCE, especially those of the philosophers Leucippus and Democritus. (There is some evidence that philosophers in both India and China made similar speculations, at about the same time.) They considered the question of whether a substance can be divided without limit into ever smaller pieces. There are only a few possible answers to this question. One is that infinitesimally small subdivision is possible. Another is what Democritus in particular believed—that there is a smallest unit that cannot be further subdivided. Democritus called this the **atom**. We now know that atoms themselves can be subdivided, but their identity is destroyed in the process, so the Greeks were correct in a respect. The Greeks also felt that atoms were in constant motion, another correct notion.

The Greeks and others speculated about the properties of atoms, proposing that only a few types existed and that all matter was formed as various combinations of these types. The famous proposal that the basic elements were earth, air, fire, and water was brilliant, but incorrect. The Greeks had identified the most common examples of the four states of matter (solid, gas, plasma, and liquid), rather than the basic elements. More than 2000 years passed before observations could be made with equipment capable of revealing the true nature of atoms.

Over the centuries, discoveries were made regarding the properties of substances and their chemical reactions. Certain systematic features were recognized, but similarities between common and rare elements resulted in efforts to transmute them (lead into gold, in particular) for financial gain. Secrecy was endemic. Alchemists discovered and rediscovered many facts but did not make them broadly available. As the Middle Ages ended, alchemy gradually faded, and the science of chemistry arose. It was no longer possible, nor considered desirable, to keep discoveries secret. Collective knowledge grew, and by the beginning of the 19th century, an important fact was well established—the masses of reactants in specific chemical reactions always have a particular mass ratio. This is very strong indirect evidence that there are basic units (atoms and molecules) that have these same mass ratios. The English chemist John Dalton (1766–1844) did much of this work, with significant contributions by the Italian physicist Amedeo Avogadro (1776–1856). It was Avogadro who developed the idea of a fixed number of atoms and molecules in a mole, and this special number is called Avogadro's number in his honor. The Austrian physicist Johann Josef Loschmidt was the first to measure the value of the constant in 1865 using the kinetic theory of gases.

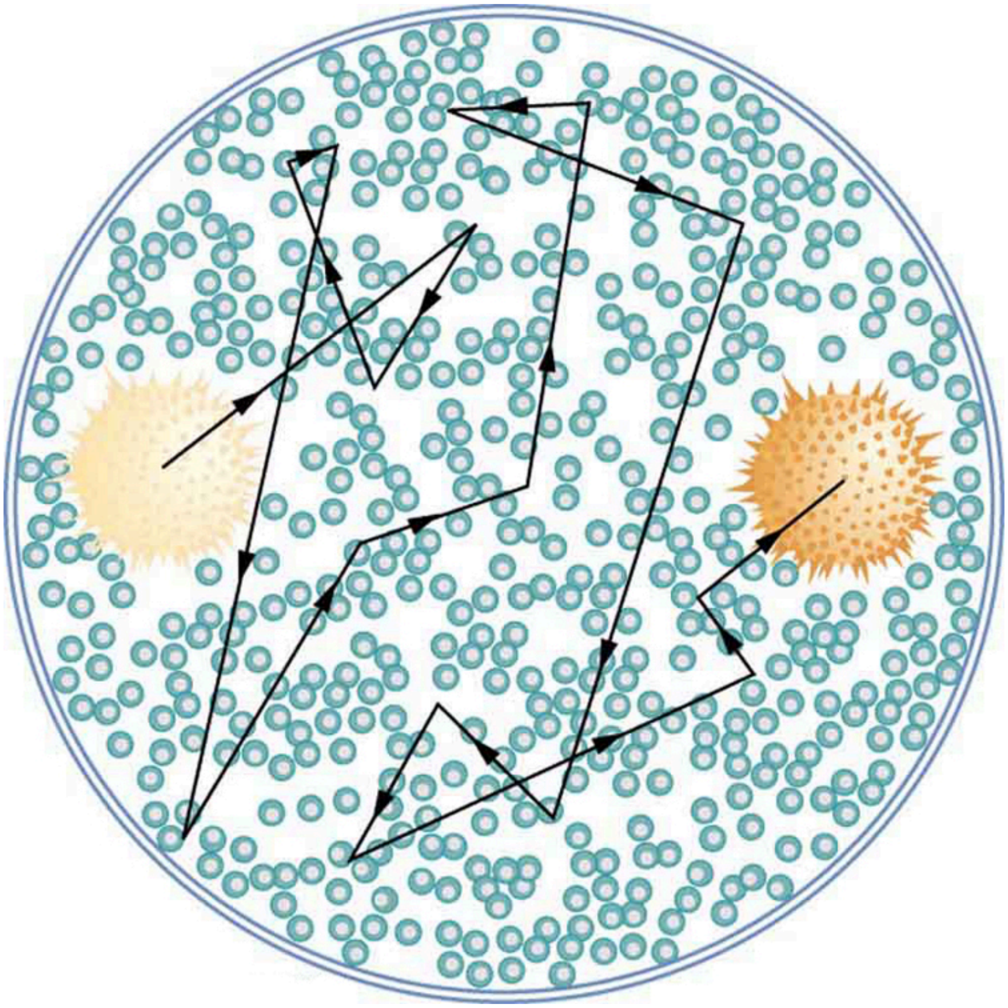
Patterns and Systematics

The recognition and appreciation of patterns has enabled us to make many discoveries. The periodic table of elements was proposed as an organized summary of the known elements long before all elements had been discovered, and it led to many other discoveries. We shall see in later chapters that patterns in the properties of subatomic particles led to the proposal of quarks as their underlying structure, an idea that is still bearing fruit.

Knowledge of the properties of elements and compounds grew, culminating in the mid-19th-century development of the periodic table of the elements by Dmitri Mendeleev (1834–1907), the great Russian chemist. Mendeleev proposed an ingenious array that highlighted the periodic nature of the properties of elements. Believing in the systematics of the periodic table, he also predicted the existence of then-unknown elements to complete it. Once these elements were discovered and determined to have properties predicted by Mendeleev, his periodic table became universally accepted.

Also during the 19th century, the kinetic theory of gases was developed. Kinetic theory is based on the existence of atoms and molecules in random thermal motion and provides a microscopic explanation of the gas laws, heat transfer, and thermodynamics (see Introduction to Temperature, Kinetic Theory, and the Gas Laws and Introduction to Laws of Thermodynamics). Kinetic theory works so well that it is another strong indication of the existence of atoms. But it is still indirect evidence—individual atoms and molecules had not been observed. There were heated debates about the validity of kinetic theory until direct evidence of atoms was obtained.

The first truly direct evidence of atoms is credited to Robert Brown, a Scottish botanist. In 1827, he noticed that tiny pollen grains suspended in still water moved about in complex paths. This can be observed with a microscope for any small particles in a fluid. The motion is caused by the random thermal motions of fluid molecules colliding with particles in the fluid, and it is now called **Brownian motion**. (See [Figure 1].) Statistical fluctuations in the numbers of molecules striking the sides of a visible particle cause it to move first this way, then that. Although the molecules cannot be directly observed, their effects on the particle can be. By examining Brownian motion, the size of molecules can be calculated. The smaller and more numerous they are, the smaller the fluctuations in the numbers striking different sides.

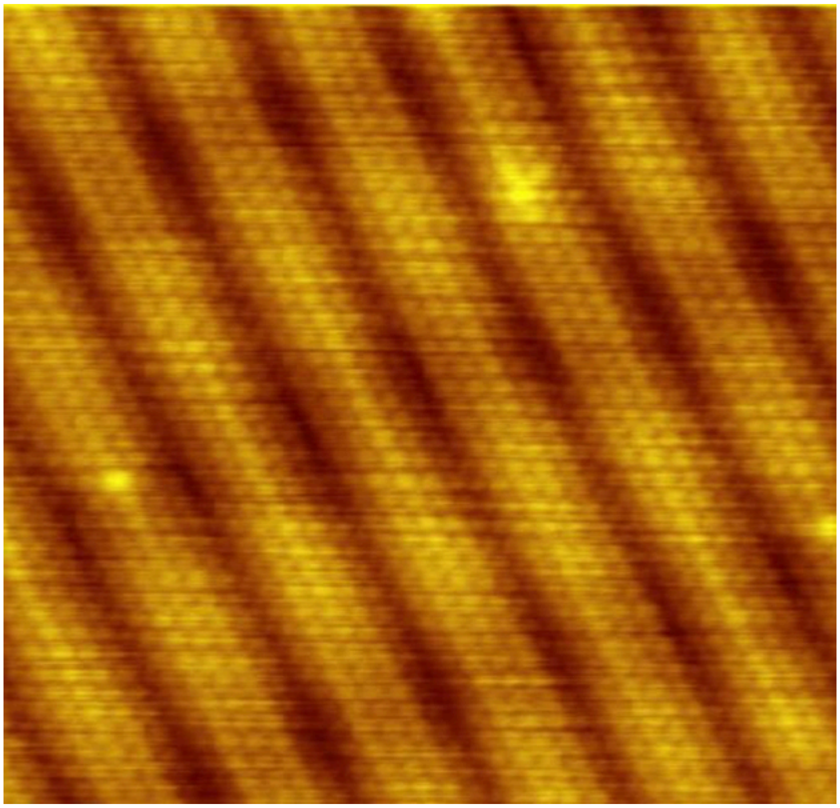


The position of a pollen grain in water, measured every few seconds under a microscope, exhibits Brownian motion. Brownian motion is due to fluctuations in the number of atoms and molecules colliding with a small mass, causing it to move about in complex paths. This is nearly direct evidence for the existence of atoms, providing a satisfactory alternative explanation cannot be found.

It was Albert Einstein who, starting in his epochal year of 1905, published several papers that explained precisely how Brownian motion could be used to measure the size of atoms and molecules. (In 1905 Einstein created special relativity, proposed photons as quanta of EM radiation, and produced a theory of Brownian motion that allowed the size of atoms to be determined. All of this was done in his spare time, since he worked days as a patent examiner. Any one of these very basic works could have been the crowning achievement of an entire career—yet Einstein did even more in later years.) Their sizes were only approximately known to be 10^{-10}m , based on a comparison of latent heat of vaporization and surface tension made in about 1805 by Thomas Young of double-slit fame and the famous astronomer and mathematician Simon Laplace.

Using Einstein's ideas, the French physicist Jean-Baptiste Perrin (1870–1942) carefully observed Brownian motion; not only did he confirm Einstein's theory, he also produced accurate sizes for atoms and molecules. Since molecular weights and densities of materials were well established, knowing atomic and molecular sizes allowed a precise value for Avogadro's number to be obtained. (If we know how big an atom is, we know how many fit into a certain volume.) Perrin also used these ideas to explain atomic and molecular agitation effects in sedimentation, and he received the 1926 Nobel Prize for his achievements. Most scientists were already convinced of the existence of atoms, but the accurate observation and analysis of Brownian motion was conclusive—it was the first truly direct evidence.

A huge array of direct and indirect evidence for the existence of atoms now exists. For example, it has become possible to accelerate ions (much as electrons are accelerated in cathode-ray tubes) and to detect them individually as well as measure their masses (see More Applications of Magnetism for a discussion of mass spectrometers). Other devices that observe individual atoms, such as the scanning tunneling electron microscope, will be discussed elsewhere. (See [Figure 2].) All of our understanding of the properties of matter is based on and consistent with the atom. The atom's substructures, such as electron shells and the nucleus, are both interesting and important. The nucleus in turn has a substructure, as do the particles of which it is composed. These topics, and the question of whether there is a smallest basic structure to matter, will be explored in later parts of the text.



Individual atoms can be detected with devices such as the scanning tunneling electron microscope that produced this image of individual gold atoms on a graphite substrate. (credit: Erwin Rossen, Eindhoven University of Technology, via Wikimedia Commons)

Section Summary

- Atoms are the smallest unit of elements; atoms combine to form molecules, the smallest unit of compounds.
- The first direct observation of atoms was in Brownian motion.
- Analysis of Brownian motion gave accurate sizes for atoms (10^{-10}m on average) and a precise value for Avogadro's number.

Conceptual Questions

Name three different types of evidence for the existence of atoms.

Explain why patterns observed in the periodic table of the elements are evidence for the existence of atoms, and why Brownian motion is a more direct type of evidence for their existence.

If atoms exist, why can't we see them with visible light?

Problems & Exercises

Using the given charge-to-mass ratios for electrons and protons, and knowing the magnitudes of their charges are equal, what is the ratio of the proton's mass to the electron's? (Note that since the charge-to-mass ratios are given to only three-digit accuracy, your answer may differ from the accepted ratio in the fourth digit.)

Show Solution

Strategy

We know from the chapter that the charge-to-mass ratio for the electron is $\frac{q_e}{m_e} = -1.76 \times 10^{11}$ C/kg and for the proton is $\frac{q_p}{m_p} = 9.58 \times 10^{7}$ C/kg . Since the magnitudes of the charges are equal ($$ q_e = q_p = e), we can use these ratios to find the mass ratio. We'll set up the ratio \frac{m_p}{m_e} $$ and express it in terms of the known charge-to-mass ratios.

Solution

Since the magnitudes of the charges are equal, we have $q_e = q_p = e$, where $e = 1.60 \times 10^{-19} \text{ C}$.

From the charge-to-mass ratio s, we can write:

$m_e = \frac{|q_e|}{|q_e/m_e|}$ and $m_p = \frac{|q_p|}{|q_p/m_p|}$

The ratio of the proton mass to the electron mass is:

$$\frac{m_p}{m_e} = \frac{|q_p|/(|q_p/m_p|)}{|q_e|/(|q_e/m_e|)} = \frac{|q_p|}{|q_e|} \cdot \frac{|q_e/m_e|}{|q_p/m_p|}$$

Since $|q_p| = |q_e|$, this simplifies to:

$$\frac{m_p}{m_e} = \frac{|q_e/m_e|}{|q_p/m_p|} = \frac{1.76 \times 10^{11} \text{ C/kg}}{9.58 \times 10^{7} \text{ C/kg}}$$

$$\frac{m_p}{m_e} = 1.84 \times 10^{3}$$

Therefore, the proton is approximately **1840 times** more massive than the electron.

Discussion

This result shows that the proton is nearly 2000 times heavier than the electron, despite both particles having the same magnitude of electric charge. This enormous mass difference has profound implications for atomic structure. It explains why the nucleus, which contains protons (and neutrons, which have similar mass), accounts for essentially all the mass of an atom even though it occupies only a tiny fraction of the atom's volume. The electron, being so much lighter, can be easily accelerated and deflected by electric and magnetic fields, which is why cathode rays (electron beams) were relatively easy to manipulate in Thomson's experiments. The modern accepted value of the proton-to-electron mass ratio is 1836.15, which differs slightly from our result in the fourth digit, as expected given the three-digit accuracy of the charge-to-mass ratios provided. This mass ratio is one of the fundamental constants of nature and plays a crucial role in determining the properties of atoms, molecules, and ultimately all matter in the universe.

(a) Calculate the mass of a proton using the charge-to-mass ratio given for it in this chapter and its known charge. (b) How does your result compare with the proton mass given in this chapter?

Show Solution

Strategy

From the chapter, we know the charge-to-mass ratio for the proton is $\frac{q_p}{m_p} = 9.58 \times 10^{7} \text{ C/kg}$ and the magnitude of the proton's charge is $q_p = 1.60 \times 10^{-19} \text{ C}$ (the same magnitude as the electron's charge). We can rearrange the charge-to-mass ratio to solve for the proton mass: $m_p = \frac{q_p}{(q_p/m_p)}$.

Solution

(a) Substituting the known values:

$$m_p = \frac{q_p}{(q_p/m_p)} = \frac{1.60 \times 10^{-19} \text{ C}}{9.58 \times 10^{7} \text{ C/kg}}$$

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

(b) This result is identical to the proton mass given in the chapter ($m_p = 1.67 \times 10^{-27} \text{ kg}$) to three significant figures, showing excellent agreement.

Discussion

The fact that we can calculate the proton mass from its charge-to-mass ratio and its charge demonstrates the consistency of the experimental measurements. This mass is approximately 1836 times the mass of an electron, which makes sense given that the charge-to-mass ratio of the electron is about 1836 times larger than that of the proton (while both have the same magnitude of charge). The proton's much larger mass compared to the electron is fundamental to understanding atomic structure—it explains why the nucleus, despite being tiny, contains essentially all the mass of an atom. This large mass ratio also justifies Bohr's assumption that the nucleus remains essentially stationary while the electron orbits around it, which simplifies the analysis of atomic structure significantly.

If someone wanted to build a scale model of the atom with a nucleus 1.00 m in diameter, how far away would the nearest electron need to be?

Show Solution

Strategy

From the chapter and Rutherford's experiments, we know that the nucleus has a diameter of approximately 10^{-15} m, while the atom itself has a diameter of approximately 10^{-10} m (the typical distance to the nearest electron). To find the scale model distance, we need to find the ratio of atomic diameter to nuclear diameter, then multiply this ratio by the model nuclear diameter of 1.00 m.

Solution

The ratio of atomic diameter to nuclear diameter is:

$$\text{Scale factor} = \frac{\text{atomic diameter}}{\text{nuclear diameter}} = \frac{10^{-10}\ \text{m}}{10^{-15}\ \text{m}} = 10^{5}$$

In the scale model, if the nucleus has a diameter of 1.00 m, the nearest electron would be at a distance:

$$\text{Model distance} = \text{Scale factor} \times \text{model nuclear diameter}$$
$$\text{Model distance} = 10^{5} \times 1.00\ \text{m} = 100{,}000\ \text{m} = 100\ \text{km}$$

However, this calculation assumes the electron is at the edge of the atom. For a more accurate calculation using the Bohr radius (the most probable distance for the electron in hydrogen's ground state), we have:

- Nuclear diameter: $\sim 10^{-15}$ m
- Bohr radius (atomic size): $a_0 \sim 0.5 \times 10^{-10}$ m

The ratio is:

$$\frac{0.5 \times 10^{-10}\ \text{m}}{10^{-15}\ \text{m}} = 0.5 \times 10^{5} = 5 \times 10^{4}$$
$$\text{Model distance} = 5 \times 10^{4} \times 1.00\ \text{m} = 50{,}000\ \text{m} = 50\ \text{km}$$

The nearest electron would be approximately **50 km** away.

Discussion

This calculation dramatically illustrates how empty atoms actually are. If the nucleus were scaled up to the size of a basketball (1 m diameter), the nearest electron would be 50 kilometers away! This means that atoms are mostly empty space—a fact that was completely unexpected before Rutherford's gold foil experiment. This emptiness explains why most alpha particles in Rutherford's experiment passed straight through the gold foil without deflection: they simply didn't encounter anything substantial. Only the rare alpha particles that happened to pass very close to a nucleus experienced significant deflection. This model also helps us understand why matter can't be compressed much under normal conditions—when you push on an object, you're really dealing with the electromagnetic forces between electron clouds, not actual contact between solid matter. The vast emptiness of atoms also explains why neutrinos (which interact very weakly with matter) can pass through the entire Earth without hitting anything. If atoms were solid throughout, the world would be a very different place: we couldn't see through anything, and all matter would be incredibly dense.

Glossary

atom
: basic unit of matter, which consists of a central, positively charged nucleus surrounded by negatively charged electrons

Brownian motion
: the continuous random movement of particles of matter suspended in a liquid or gas

Discovery of the Parts of the Atom: Electrons and Nuclei

- Describe how electrons were discovered.
- Explain the Millikan oil drop experiment.
- Describe Rutherford's gold foil experiment.
- Describe Rutherford's planetary model of the atom.

Just as atoms are a substructure of matter, electrons and nuclei are substructures of the atom. The experiments that were used to discover electrons and nuclei reveal some of the basic properties of atoms and can be readily understood using ideas such as electrostatic and magnetic force, already covered in previous chapters.

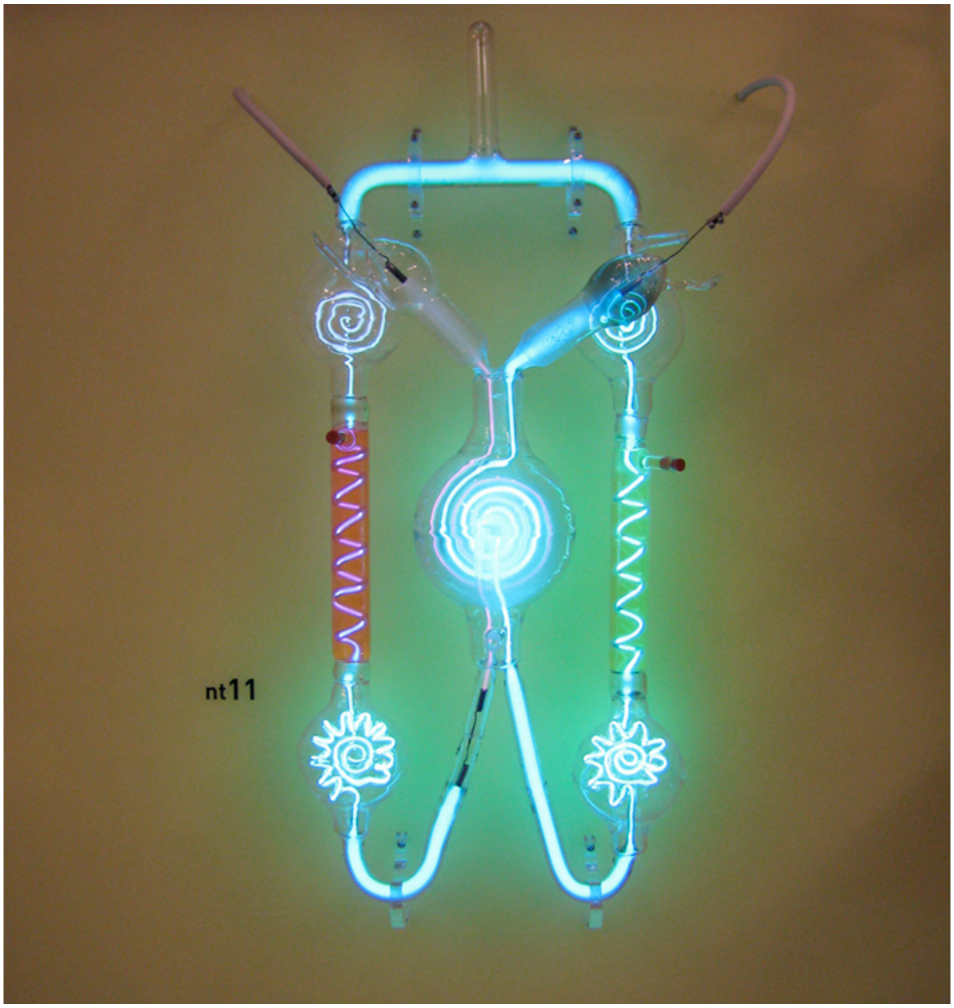
Charges and Electromagnetic Forces

In previous discussions, we have noted that positive charge is associated with nuclei and negative charge with electrons. We have also covered many aspects of the electric and magnetic forces that affect charges. We will now explore the discovery of the electron and nucleus as substructures of the atom and examine their contributions to the properties of atoms.

The Electron

Gas discharge tubes, such as that shown in [Figure 1], consist of an evacuated glass tube containing two metal electrodes and a rarefied gas. When a high voltage is applied to the electrodes, the gas glows. These tubes were the precursors to today's neon lights. They were first studied seriously by Heinrich Geissler, a German inventor and glassblower, starting in the 1860s. The English scientist William Crookes, among others, continued to study what for some time were called Crookes tubes, wherein electrons are freed from atoms and molecules in the rarefied gas inside the tube and are accelerated from the cathode (negative) to the anode (positive) by the high potential. These "*cathode rays*" collide with the gas atoms and molecules and excite them, resulting in the emission of electromagnetic (EM) radiation that makes the electrons' path visible as a ray that spreads and fades as it moves away from the cathode.

Gas discharge tubes today are most commonly called **cathode-ray tubes**, because the rays originate at the cathode. Crookes showed that the electrons carry momentum (they can make a small paddle wheel rotate). He also found that their normally straight path is bent by a magnet in the direction expected for a negative charge moving away from the cathode. These were the first direct indications of electrons and their charge.



A gas discharge tube glows when a high voltage is applied to it. Electrons emitted from the cathode are accelerated toward the anode; they excite atoms and molecules in the gas, which glow in response. Once called Geissler tubes and later Crookes tubes, they are now known as cathode-ray tubes (CRTs) and are found in older TVs, computer screens, and x-ray machines. When a magnetic field is applied, the beam bends in the direction expected for negative charge. (credit: Paul Downey, Flickr)

The English physicist J. J. Thomson (1856–1940) improved and expanded the scope of experiments with gas discharge tubes. (See [Figure 2] and [Figure 3].) He verified the negative charge of the cathode rays with both magnetic and electric fields. Additionally, he collected the rays in a metal cup and found an excess of negative charge. Thomson was also able to measure the ratio of the charge of the electron to its mass, q_e/m_e —an important step to finding the actual values of both q_e and m_e. [Figure 4] shows a cathode-ray tube, which produces a narrow beam of electrons that passes through charging plates connected to a high-voltage power supply. An electric field $\vec{\mathbf{E}}$ is produced between the charging plates, and the cathode-ray tube is placed between the poles of a magnet so that the electric field $\vec{\mathbf{E}}$ is perpendicular to the magnetic field $\vec{\mathbf{B}}$ of the magnet. These fields, being perpendicular to each other, produce opposing forces on the electrons. As discussed for mass spectrometers in More Applications of Magnetism, if the net force due to the fields vanishes, then the velocity of the charged particle is $v = E/B$. In this manner, Thomson determined the velocity of the electrons and then moved the beam up and down by adjusting the electric field.

J. J. Thomson (credit: www.firstworldwar.com, via Wikimedia Commons)

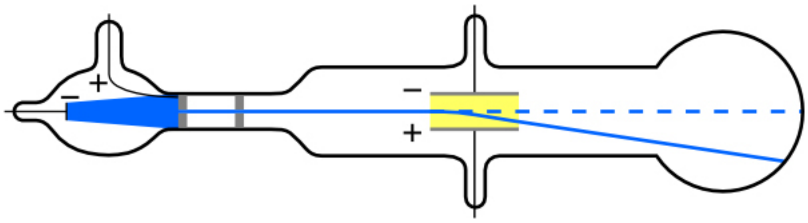
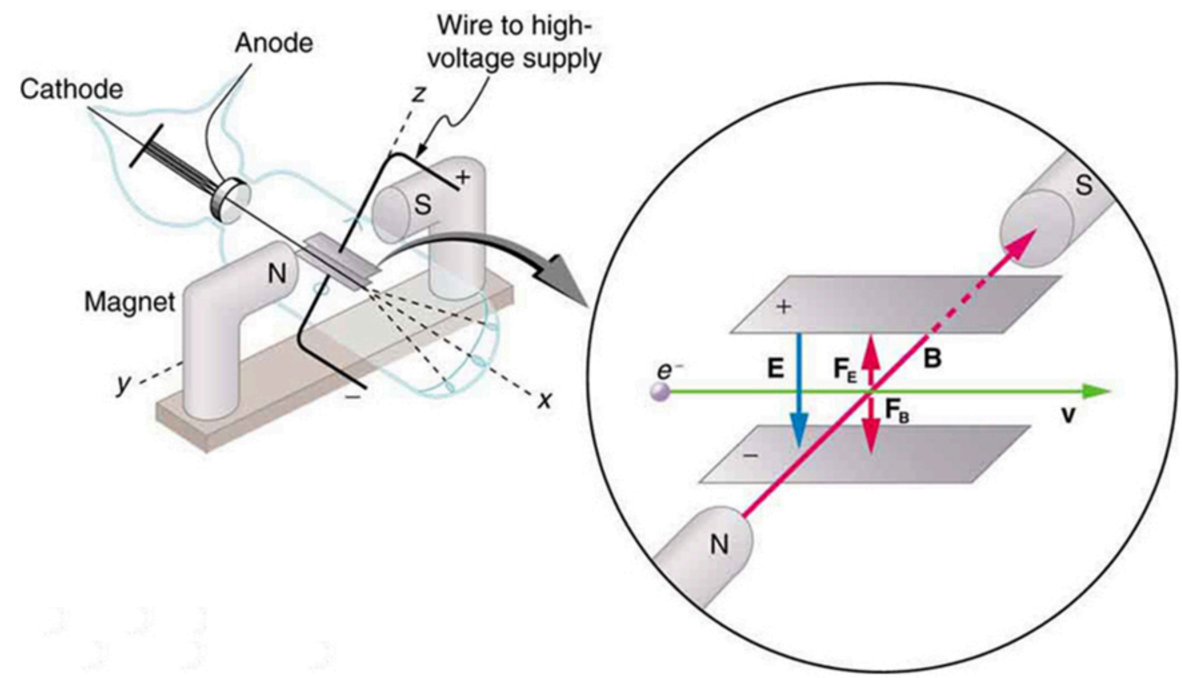


Diagram of Thomson’s CRT. (credit: Kurzon, Wikimedia Commons)



This schematic shows the electron beam in a CRT passing through crossed electric and magnetic fields and causing phosphor to glow when striking the end of the tube.

To see how the amount of deflection is used to calculate q_e/m_e, note that the deflection is proportional to the electric force on the electron:

$$F = q_e E.$$

But the vertical deflection is also related to the electron's mass, since the electron's acceleration is

$$a = \frac{F}{m_e}.$$

The value of F is not known, since q_e was not yet known. Substituting the expression for electric force into the expression for acceleration yields

$$a = \frac{F}{m_e} = \frac{q_e E}{m_e}.$$

Gathering terms, we have

$$\frac{q_e}{m_e} = \frac{a}{E}.$$

The deflection is analyzed to get a, and E is determined from the applied voltage and distance between the plates; thus, $\frac{q_e}{m_e}$ can be determined. With the velocity known, another measurement of $\frac{q_e}{m_e}$ can be obtained by bending the beam of electrons with the magnetic field. Since $F_{\text{mag}} = q_e v B = m_e a$, we have $q_e/m_e = a/vB$. Consistent results are obtained using magnetic deflection.

What is so important about q_e/m_e, the ratio of the electron's charge to its mass? The value obtained is

$$\frac{q_e}{m_e} = -1.76 \times 10^{11}\,\text{C/kg (electron)}.$$

This is a huge number, as Thomson realized, and it implies that the electron has a very small mass. It was known from electroplating that about $10^8\,\text{C/kg}$ is needed to plate a material, a factor of about 1000 less than the charge per kilogram of electrons. Thomson went on to do the same experiment for positively charged hydrogen ions (now known to be bare protons) and found a charge per kilogram about 1000 times smaller than that for the electron, implying that the proton is about 1000 times more massive than the electron. Today, we know more precisely that

$$\frac{q_p}{m_p} = 9.58 \times 10^7\,\text{C/kg}(proton),$$

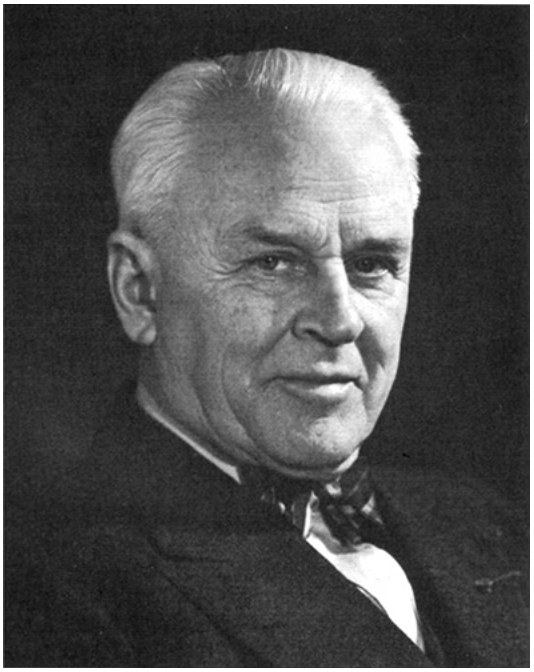
where q_p is the charge of the proton and m_p is its mass. This ratio (to four significant figures) is 1836 times less charge per kilogram than for the electron. Since the charges of electrons and protons are equal in magnitude, this implies $m_p = 1836 m_e$.

Thomson performed a variety of experiments using differing gases in discharge tubes and employing other methods, such as the photoelectric effect, for freeing electrons from atoms. He always found the same properties for the electron, proving it to be an independent particle. For his work, the important pieces of which he began to publish in 1897, Thomson was awarded the 1906 Nobel Prize in Physics. In retrospect, it is difficult to appreciate how astonishing it was to find that the atom has a substructure. Thomson himself said, “It was only when I was convinced that the experiment left no escape from it that I published my belief in the existence of bodies smaller than atoms.”

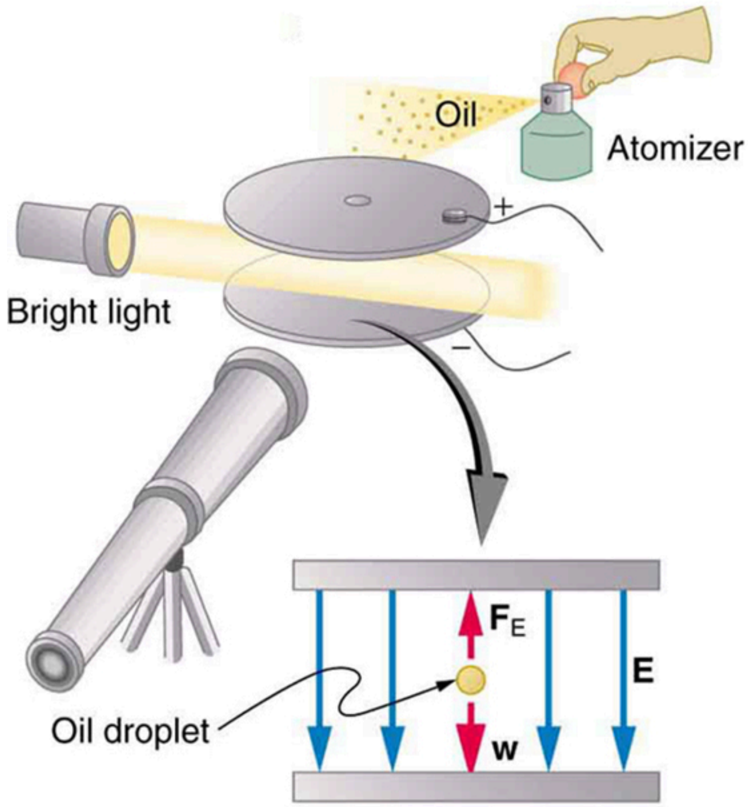
Thomson attempted to measure the charge of individual electrons, but his method could determine its charge only to the order of magnitude expected.

Since Faraday’s experiments with electroplating in the 1830s, it had been known that about 100 000 C per mole was needed to plate singly ionized ions. Dividing this by the number of ions per mole (that is, by Avogadro’s number), which was approximately known, the charge per ion was calculated to be about $1.6 \times 10^{-19} \mathrm{C}$, close to the actual value.

An American physicist, Robert Millikan (1868–1953) (see [Figure 5]), decided to improve upon Thomson’s experiment for measuring q_e and was eventually forced to try another approach, which is now a classic experiment performed by students. The Millikan oil drop experiment is shown in [Figure 6].



Robert Millikan (credit: Unknown Author, via Wikimedia Commons)



The Millikan oil drop experiment produced the first accurate direct measurement of the charge on electrons, one of the most fundamental constants in nature. Fine drops of oil become charged when sprayed. Their movement is observed between metal plates with a potential applied to oppose the gravitational force. The balance of gravitational and electric forces allows the calculation of the charge on a drop. The charge is found to be quantized in units of $1.6\times 10^{-19} \text{C}$, thus determining directly the charge of the excess and missing electrons on the oil drops.

In the Millikan oil drop experiment, fine drops of oil are sprayed from an atomizer. Some of these are charged by the process and can then be suspended between metal plates by a voltage between the plates. In this situation, the weight of the drop is balanced by the electric force:

$$m_{\text{drop}} g = q_e E$$

The electric field is produced by the applied voltage, hence, $E = V/d$, and V is adjusted to just balance the drop's weight. The drops can be seen as points of reflected light using a microscope, but they are too small to directly measure their size and mass. The mass of the drop is determined by observing how fast it falls when the voltage is turned off. Since air resistance is very significant for these submicroscopic drops, the more massive drops fall faster than the less massive, and sophisticated sedimentation calculations can reveal their mass. Oil is used rather than water, because it does not readily evaporate, and so mass is nearly constant. Once the mass of the drop is known, the charge of the electron is given by rearranging the previous equation:

$$q = m_{\text{drop}} g E = m_{\text{drop}} \text{gd} V,$$

where d is the separation of the plates and V is the voltage that holds the drop motionless. (The same drop can be observed for several hours to see that it really is motionless.) By 1913 Millikan had measured the charge of the electron q_e to an accuracy of 1%, and he improved this by a factor of 10 within a few years to a value of $-1.60 \times 10^{-19} \text{C}$. He also observed that all charges were multiples of the basic electron charge and that sudden changes could occur in which electrons were added or removed from the drops. For this very fundamental direct measurement of q_e and for his studies of the photoelectric effect, Millikan was awarded the 1923 Nobel Prize in Physics.

With the charge of the electron known and the charge-to-mass ratio known, the electron's mass can be calculated. It is

$$m = q_e (q_e m_e).$$

Substituting known values yields

$$m_e = -1.60\times 10^{-19} \text{C} - 1.76\times 10^{11} \text{C/kg}$$

or

$$m_e = 9.11\times 10^{-31} \text{kg} (electron's mass),$$

where the round-off errors have been corrected. The mass of the electron has been verified in many subsequent experiments and is now known to an accuracy of better than one part in one million. It is an incredibly small mass and remains the smallest known mass of any particle that has mass. (Some particles, such as photons, are massless and cannot be brought to rest, but travel at the speed of light.) A similar calculation gives the masses of other particles, including the proton. To three digits, the mass of the proton is now known to be

$$m_p = 1.67 \times 10^{-27}\,\text{kg}\,(proton's\,mass),$$

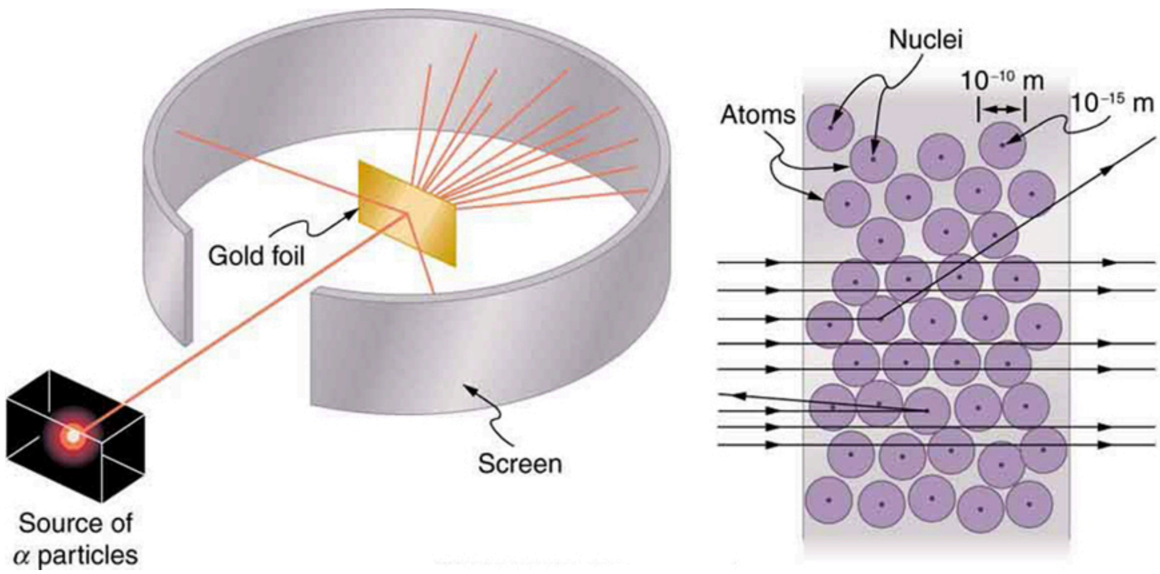
which is nearly identical to the mass of a hydrogen atom. What Thomson and Millikan had done was to prove the existence of one substructure of atoms, the electron, and further to show that it had only a tiny fraction of the mass of an atom. The nucleus of an atom contains most of its mass, and the nature of the nucleus was completely unanticipated.

Another important characteristic of quantum mechanics was also beginning to emerge. All electrons are identical to one another. The charge and mass of electrons are not average values; rather, they are unique values that all electrons have. This is true of other fundamental entities at the submicroscopic level. All protons are identical to one another, and so on.

The Nucleus

Here, we examine the first direct evidence of the size and mass of the nucleus. In later chapters, we will examine many other aspects of nuclear physics, but the basic information on nuclear size and mass is so important to understanding the atom that we consider it here.

Nuclear radioactivity was discovered in 1896, and it was soon the subject of intense study by a number of the best scientists in the world. Among them was New Zealander Lord Ernest Rutherford, who made numerous fundamental discoveries and earned the title of "father of nuclear physics." Born in Nelson, Rutherford did his postgraduate studies at the Cavendish Laboratories in England before taking up a position at McGill University in Canada where he did the work that earned him a Nobel Prize in Chemistry in 1908. In the area of atomic and nuclear physics, there is much overlap between chemistry and physics, with physics providing the fundamental enabling theories. He returned to England in later years and had six future Nobel Prize winners as students. Rutherford used nuclear radiation to directly examine the size and mass of the atomic nucleus. The experiment he devised is shown in [Figure 7]. A radioactive source that emits alpha radiation was placed in a lead container with a hole in one side to produce a beam of alpha particles, which are a type of ionizing radiation ejected by the nuclei of a radioactive source. A thin gold foil was placed in the beam, and the scattering of the alpha particles was observed by the glow they caused when they struck a phosphor screen.

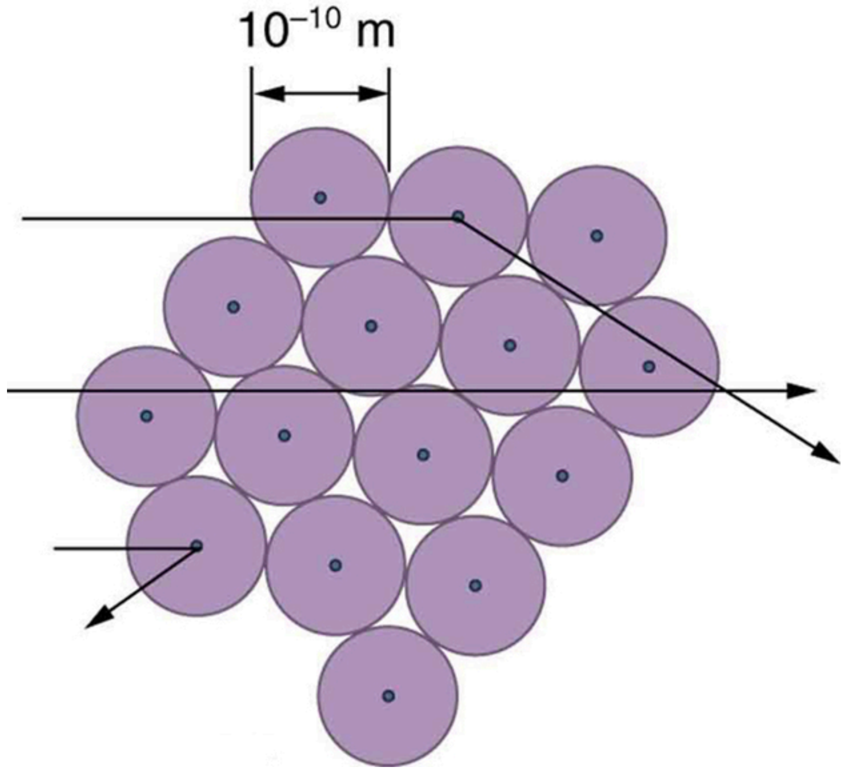


Rutherford's experiment gave direct evidence for the size and mass of the nucleus by scattering alpha particles from a thin gold foil. Alpha particles with energies of about 5 MeV are emitted from a radioactive source (which is a small metal container in which a specific amount of a radioactive material is sealed), are collimated into a beam, and fall upon the foil. The number of particles that penetrate the foil or scatter to various angles indicates that gold nuclei are very small and contain nearly all of the gold atom's mass. This is particularly indicated by the alpha particles that scatter to very large angles, much like a soccer ball bouncing off a goalie's head.

Alpha particles were known to be the doubly charged positive nuclei of helium atoms that had kinetic energies on the order of $5\,\text{MeV}$ when emitted in nuclear decay, which is the disintegration of the nucleus of an unstable nuclide by the spontaneous emission of charged particles. These particles interact with matter mostly via the Coulomb force, and the manner in which they scatter from nuclei can reveal nuclear size and mass. This is analogous to observing how a bowling ball is scattered by an object you cannot see directly. Because the alpha particle's energy is so large compared with the typical energies associated with atoms (MeV versus eV), you would expect the alpha particles to simply crash through a thin foil much like a supersonic bowling ball would crash through a few dozen rows of bowling pins. Thomson had envisioned the atom to be a small sphere in which equal amounts of positive and negative charge were distributed evenly. The incident massive alpha particles would suffer only small deflections in such a model. Instead, Rutherford and his collaborators found that alpha particles occasionally were scattered to large angles, some even back in the direction from which they came! Detailed analysis using conservation of momentum and energy—particularly of the small number that came straight back—implied that gold nuclei are very small compared with the size of a gold atom, contain almost all of the atom's mass, and are tightly bound. Since the gold nucleus is several times more massive than the alpha particle, a head-on collision would scatter the alpha particle straight back toward the source. In addition, the smaller the nucleus, the fewer alpha particles that would hit one head on.

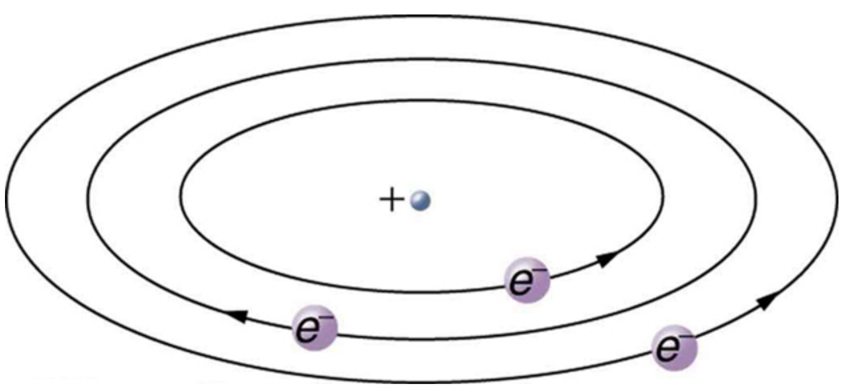
Although the results of the experiment were published by his colleagues in 1909, it took Rutherford two years to convince himself of their meaning. Like Thomson before him, Rutherford was reluctant to accept such radical results. Nature on a small scale is so unlike our classical world that even those at the forefront of discovery are sometimes surprised. Rutherford later wrote: “It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you. On consideration, I realized that this scattering backwards … [meant] … the greatest part of the mass of the atom was concentrated in a tiny nucleus.” In 1911, Rutherford published his analysis together with a proposed model of the atom. The size of the nucleus was determined to be about 10^{-15}m , or 100 000 times smaller than the atom. This implies a huge density, on the order of 10^{15}g/cm^3 , vastly unlike any macroscopic matter. Also implied is the existence of previously unknown nuclear forces to counteract the huge repulsive Coulomb forces among the positive charges in the nucleus. Huge forces would also be consistent with the large energies emitted in nuclear radiation.

The small size of the nucleus also implies that the atom is mostly empty inside. In fact, in Rutherford’s experiment, most alphas went straight through the gold foil with very little scattering, since electrons have such small masses and since the atom was mostly empty with nothing for the alpha to hit. There were already hints of this at the time Rutherford performed his experiments, since energetic electrons had been observed to penetrate thin foils more easily than expected. [Figure 8] shows a schematic of the atoms in a thin foil with circles representing the size of the atoms (about 10^{-10}m) and dots representing the nuclei. (The dots are not to scale—if they were, you would need a microscope to see them.) Most alpha particles miss the small nuclei and are only slightly scattered by electrons. Occasionally, (about once in 8000 times in Rutherford’s experiment), an alpha hits a nucleus head-on and is scattered straight backward.



An expanded view of the atoms in the gold foil in Rutherford’s experiment. Circles represent the atoms (about 10^{-10}m in diameter), while the dots represent the nuclei (about 10^{-15}m in diameter). To be visible, the dots are much larger than scale. Most alpha particles crash through but are relatively unaffected because of their high energy and the electron’s small mass. Some, however, head straight toward a nucleus and are scattered straight back. A detailed analysis gives the size and mass of the nucleus.

Based on the size and mass of the nucleus revealed by his experiment, as well as the mass of electrons, Rutherford proposed the **planetary model of the atom**. The planetary model of the atom pictures low-mass electrons orbiting a large-mass nucleus. The sizes of the electron orbits are large compared with the size of the nucleus, with mostly vacuum inside the atom. This picture is analogous to how low-mass planets in our solar system orbit the large-mass Sun at distances large compared with the size of the sun. In the atom, the attractive Coulomb force is analogous to gravitation in the planetary system. (See [Figure 9].) Note that a model or mental picture is needed to explain experimental results, since the atom is too small to be directly observed with visible light.

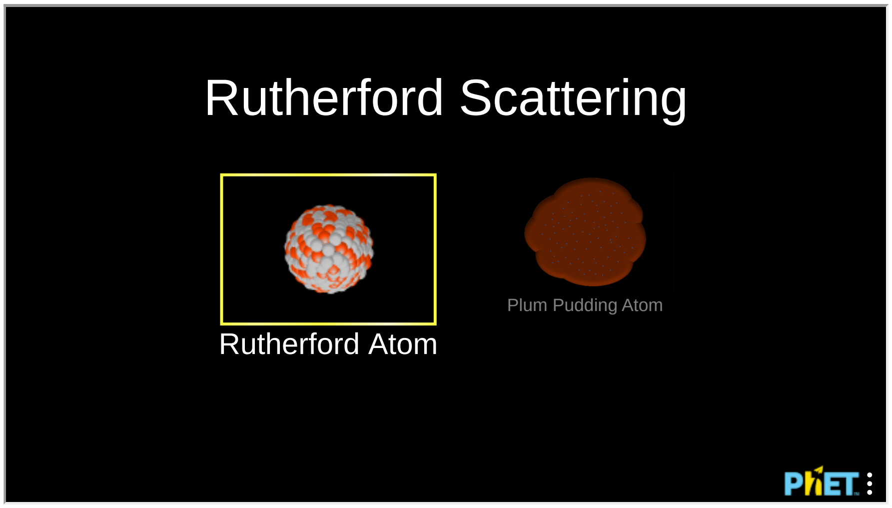


> Rutherford’s planetary model of the atom incorporates the characteristics of the nucleus, electrons, and the size of the atom. This model was the first to recognize the structure of atoms, in which low-mass electrons orbit a very small, massive nucleus in orbits much larger than the nucleus. The atom is mostly empty and is analogous to our planetary system.

Rutherford’s planetary model of the atom was crucial to understanding the characteristics of atoms, and their interactions and energies, as we shall see in the next few sections. Also, it was an indication of how different nature is from the familiar classical world on the small, quantum mechanical scale. The discovery of a substructure to all matter in the form of atoms and molecules was now being taken a step further to reveal a substructure of atoms that was simpler than the 92 elements then known. We have continued to search for deeper substructures, such as those inside the nucleus, with some success. In later chapters, we will follow this quest in the discussion of quarks and other elementary particles, and we will look at the direction the search seems now to be heading.

PhET Explorations: Rutherford Scattering

How did Rutherford figure out the structure of the atom without being able to see it? Simulate the famous experiment in which he disproved the Plum Pudding model of the atom by observing alpha particles bouncing off atoms and determining that they must have a small core.



Section Summary

- Atoms are composed of negatively charged electrons, first proved to exist in cathode-ray-tube experiments, and a positively charged nucleus.
- All electrons are identical and have a charge-to-mass ratio of
  $\frac{q_e}{m_e} = -1.76 \times 10^{11}\ \text{C/kg}.$
- The positive charge in the nuclei is carried by particles called protons, which have a charge-to-mass ratio of
  $\frac{q_p}{m_p} = 9.57 \times 10^{7}\ \text{C/kg}.$
- Mass of electron,
  $m_e = 9.11 \times 10^{-31}\ \text{kg}.$
- Mass of proton,
  $m_p = 1.67 \times 10^{-27}\ \text{kg}.$
- The planetary model of the atom pictures electrons orbiting the nucleus in the same way that planets orbit the sun.

Conceptual Questions

What two pieces of evidence allowed the first calculation of m_e , the mass of the electron?

(a) The ratios q_e/m_e and q_p/m_p.

(b) The values of q_e and E_B.

(c) The ratio q_e/m_e and q_e.

Justify your response.

How do the allowed orbits for electrons in atoms differ from the allowed orbits for planets around the sun? Explain how the correspondence principle applies here.

Problem Exercises

Rutherford found the size of the nucleus to be about $10^{-15}\ \text{m}$. This implied a huge density. What would this density be for gold?

Show Solution

Strategy

To find the density of the gold nucleus, we need to calculate the mass of the nucleus and divide by its volume. For gold (Au), the atomic mass is approximately 197 u (unified atomic mass units), where $1\text{ u} = 1.66 \times 10^{-27}\text{ kg}$. Nearly all the mass of an atom is in the nucleus, so we can use the atomic mass as the nuclear mass. We'll treat the nucleus as a sphere with diameter $d = 10^{-15}\text{ m}$ (radius $r = 0.5 \times 10^{-15}\text{ m}$), and calculate its volume using $V = 43\pi r^3$.

Solution

The mass of a gold nucleus (atomic mass number A = 197) is:

$$m = 197\text{ u} = 197 \times 1.66 \times 10^{-27}\text{ kg} = 3.27 \times 10^{-25}\text{ kg}$$

The volume of the nucleus, assuming a spherical shape with radius $r = 0.5 \times 10^{-15}\text{ m}$:

$$V = 43\pi r^3 = 43\pi (0.5 \times 10^{-15}\text{ m})^3$$
$$V = 43\pi (1.25 \times 10^{-46}\text{ m}^3) = 5.24 \times 10^{-46}\text{ m}^3$$

The density is:

$$\rho = mV = 3.27 \times 10^{-25}\text{ kg}5.24 \times 10^{-46}\text{ m}^3 = 6.2 \times 10^{20}\text{ kg/m}^3$$

Rounding to one significant figure: $\rho \approx 6 \times 10^{20}\text{ kg/m}^3$

Discussion

This density is absolutely enormous—about 6×10^{17} times greater than the density of ordinary matter! For comparison, gold metal has a density of about $1.9 \times 10^4\text{ kg/m}^3$, water has a density of 10^3 kg/m^3 , and even the densest stable element (osmium) is only about $2.3 \times 10^4\text{ kg/m}^3$. A single cubic centimeter of nuclear matter would have a mass of about $6 \times 10^{14}\text{ kg}$ or 600 million metric tons! This incredible density was one of the most surprising discoveries from Rutherford's experiment. It explained why alpha particles could be scattered backwards—they were encountering something incredibly massive and concentrated in an extremely small volume. This nuclear density is essentially the same for all nuclei (it's approximately constant at about $2 \times 10^{17}\text{ kg/m}^3$), which means that nuclear volume is proportional to the number of nucleons (protons plus neutrons). Only in exotic astrophysical objects like neutron stars do we find densities comparable to nuclear density. The existence of such high density confined to such a tiny volume also implied the existence of previously unknown nuclear forces—the gravitational and electromagnetic forces alone could not explain how the nucleus stays together, given the enormous electrostatic repulsion between protons packed into such a small space.

In Millikan's oil-drop experiment, one looks at a small oil drop held motionless between two plates. Take the voltage between the plates to be 2033 V, and the plate separation to be 2.00 cm. The oil drop (of density 0.81g/cm^3) has a diameter of $4.0 \times 10^{-6}\text{m}$. Find the charge on the drop, in terms of electron units.

Show Solution

Strategy

In Millikan's oil-drop experiment, a charged oil drop is suspended motionless when the upward electric force exactly balances the downward gravitational force. The condition for equilibrium is $qE = m_{\text{drop}}g$, where $E = V/d$ is the electric field. The mass of the spherical drop can be found from its volume and density: $m_{\text{drop}} = \rho V = \rho 43\pi r^3$. We'll solve for the charge q and then express it in terms of the elementary charge $e = 1.60 \times 10^{-19}\text{ C}$.

Solution

First, calculate the mass of the oil drop. The radius is $r = d2 = 4.0 \times 10^{-6}\text{ m}2 = 2.0 \times 10^{-6}\text{ m}$.

The volume of the spherical drop is:

$$V = 43\pi r^3 = 43\pi (2.0 \times 10^{-6}\text{ m})^3 = 3.35 \times 10^{-17}\text{ m}^3$$

Converting the density to SI units: $\rho = 0.81\text{ g/cm}^3 = 810\text{ kg/m}^3$

The mass is:

$$m_{\text{drop}} = \rho V = (810\ \text{kg/m}^3)(3.35\times10^{-17}\ \text{m}^3) = 2.71\times10^{-14}\ \text{kg}$$

The electric field between the plates is:

$$E = \frac{V}{d} = \frac{2033\ \text{V}}{0.0200\ \text{m}} = 1.017\times10^5\ \text{V/m}$$

For equilibrium, $qE = m_{\text{drop}} g$, so:

$$q = \frac{m_{\text{drop}} g}{E} = \frac{(2.71\times10^{-14}\ \text{kg})(9.80\ \text{m/s}^2)}{1.017\times10^5\ \text{V/m}} = 2.61\times10^{-18}\ \text{C}$$

To express this in terms of electron units:

$$n = \frac{q}{e} = \frac{2.61\times10^{-18}\ \text{C}}{1.60\times10^{-19}\ \text{C}} = 16.3 \approx 16$$

The charge on the drop is approximately **16 electron units** (or 16e).

Discussion

The result that the charge is approximately 16 times the elementary charge demonstrates Millikan's crucial discovery: electric charge is quantized in units of the electron charge. In his experiments, Millikan observed that the charge on any oil drop was always an integer multiple of the elementary charge e , never a fractional value. This provided direct evidence that charge comes in discrete units. Our calculation gives 16.3, which rounds to 16, showing that this particular drop carries 16 excess electrons (or is missing 16 electrons, depending on the sign). The slight deviation from a perfect integer (16.3 vs 16) can be attributed to experimental uncertainties in measuring the drop's diameter, the voltage, the plate separation, or the density. Millikan's painstaking measurements of thousands of oil drops, combined with careful statistical analysis, allowed him to determine the elementary charge to within 1% accuracy, a remarkable achievement that earned him the 1923 Nobel Prize in Physics.

(a) An aspiring physicist wants to build a scale model of a hydrogen atom for her science fair project. If the atom is 1.00 m in diameter, how big should she try to make the nucleus?

(b) How easy will this be to do?

Show Solution

Strategy

For a hydrogen atom, the atomic diameter is approximately 10^{-10} m (the Bohr radius is about 0.5×10^{-10} m , making the diameter about 10^{-10} m), and the nuclear diameter is approximately 10^{-15} m . To find the scale model size, we need to determine the ratio of these dimensions and apply it to the 1.00 m model atom diameter.

Solution

(a) The scale factor for the model is:

$$\text{Scale factor} = \frac{\text{Model atom diameter}}{\text{Actual atom diameter}} = \frac{1.00\ \text{m}}{10^{-10}\ \text{m}} = 10^{10}$$

The nucleus size in the model should be:

$$\text{Model nucleus diameter} = \text{Scale factor} \times \text{Actual nucleus diameter}$$
$$\text{Model nucleus diameter} = 10^{10} \times 10^{-15}\ \text{m} = 10^{-5}\ \text{m} = 10\ \mu\text{m}$$

The nucleus in the model should be approximately **10.0 μm** in diameter (or 0.010 mm).

(b) Making a nucleus of approximately this size is relatively easy. A diameter of 10 μm is about 1/10 the thickness of a human hair (which is typically 50-100 μm), or about the size of a small dust particle. Such an object could be created using:

- A very fine grain of sand or dust
- A small pinhead of paint
- A tiny bead or sphere from precision manufacturing
- Modern 3D printing techniques

However, making it **exactly** 10.0 μm would be quite difficult and would require precision instruments like micrometers or microscopy to measure and verify the size. For a science fair project, an approximate size would be acceptable and educational.

Discussion

This problem beautifully illustrates the vast emptiness of the atom. In a model where the atom is 1 meter across (about the width of a doorway), the nucleus would be only 10 micrometers—barely visible to the naked eye! This is like having a grain of dust in the center of a sphere the size of a large room. The rest of the atom is essentially empty space where the electron probability cloud exists. This visualization helps explain several phenomena: (1) Why most alpha particles in Rutherford's experiment passed straight through the gold foil—they were passing through mostly empty space; (2) Why X-rays and other high-energy radiation can penetrate matter—there's very little solid material to stop them; (3) Why matter seems solid despite being mostly empty—the electromagnetic forces from electron clouds extend throughout the atomic volume, creating effective "boundaries" even though there's no solid material there. This scale model also demonstrates that if we could somehow remove all the empty space from atoms, all the matter in a human body could be compressed into a volume smaller than a grain of sand (though its mass would remain the same). Such “collapsed” matter actually exists in neutron stars, where gravitational forces are strong enough to overcome the electromagnetic forces that normally keep atomic matter spread out.

Glossary

cathode-ray tube
: a vacuum tube containing a source of electrons and a screen to view images

planetary model of the atom
: the most familiar model or illustration of the structure of the atom

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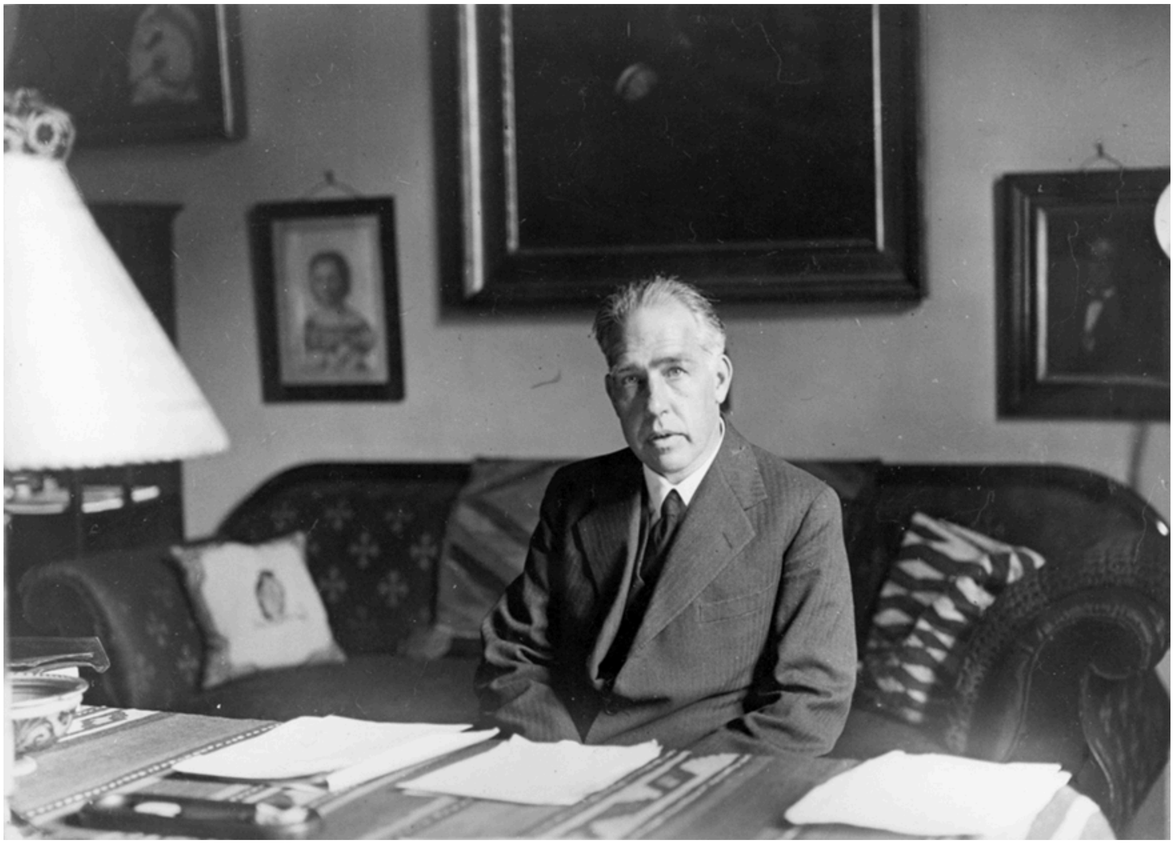
Bohr's Theory of the Hydrogen Atom

- Describe the mysteries of atomic spectra.
- Explain Bohr's theory of the hydrogen atom.
- Explain Bohr's planetary model of the atom.
- Illustrate energy state using the energy-level diagram.
- Describe the triumphs and limits of Bohr's theory.

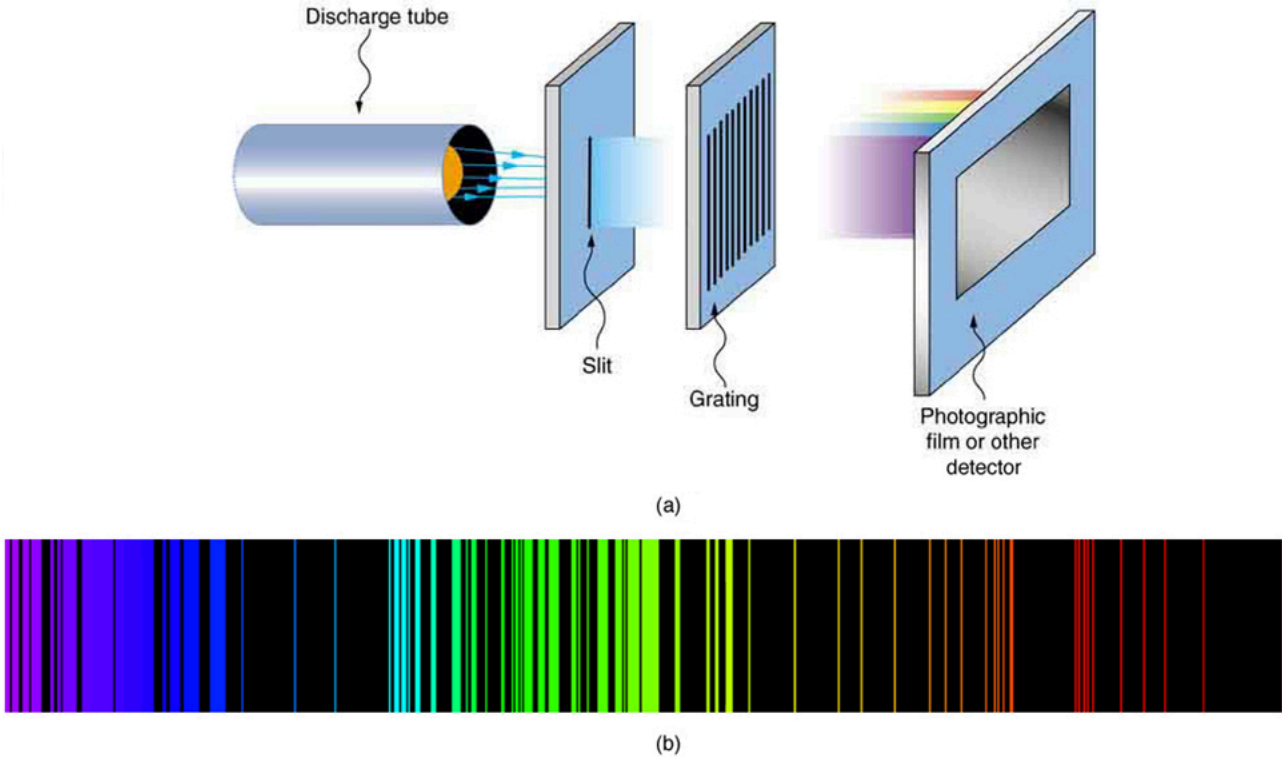
The great Danish physicist Niels Bohr (1885–1962) made immediate use of Rutherford's planetary model of the atom. ([Figure 1]). Bohr became convinced of its validity and spent part of 1912 at Rutherford's laboratory. In 1913, after returning to Copenhagen, he began publishing his theory of the simplest atom, hydrogen, based on the planetary model of the atom. For decades, many questions had been asked about atomic characteristics. From their sizes to their spectra, much was known about atoms, but little had been explained in terms of the laws of physics. Bohr's theory explained the atomic spectrum of hydrogen and established new and broadly applicable principles in quantum mechanics.



Niels Bohr, Danish physicist, used the planetary model of the atom to explain the atomic spectrum and size of the hydrogen atom. His many contributions to the development of atomic physics and quantum mechanics, his personal influence on many students and colleagues, and his personal integrity, especially in the face of Nazi oppression, earned him a prominent place in history. (credit: Unknown Author, via Wikimedia Commons)

Mysteries of Atomic Spectra

As noted in Quantization of Energy , the energies of some small systems are quantized. Atomic and molecular emission and absorption spectra have been known for over a century to be discrete (or quantized). (See [Figure 2].) Maxwell and others had realized that there must be a connection between the spectrum of an atom and its structure, something like the resonant frequencies of musical instruments. But, in spite of years of efforts by many great minds, no one had a workable theory. (It was a running joke that any theory of atomic and molecular spectra could be destroyed by throwing a book of data at it, so complex were the spectra.) Following Einstein's proposal of photons with quantized energies directly proportional to their wavelengths, it became even more evident that electrons in atoms can exist only in discrete orbits.



Part (a) shows, from left to right, a discharge tube, slit, and diffraction grating producing a line spectrum. Part (b) shows the emission line spectrum for iron. The discrete lines imply quantized energy states for the atoms that produce them. The line spectrum for each element is unique, providing a powerful and much used analytical tool, and many line spectra were well known for many years before they could be explained with physics. (credit for (b): Yttrium91, Wikimedia Commons)

In some cases, it had been possible to devise formulas that described the emission spectra. As you might expect, the simplest atom—hydrogen, with its single electron—has a relatively simple spectrum. The hydrogen spectrum had been observed in the infrared (IR), visible, and ultraviolet (UV), and several series of spectral lines had been observed. (See [Figure 3].) These series are named after early researchers who studied them in particular depth.

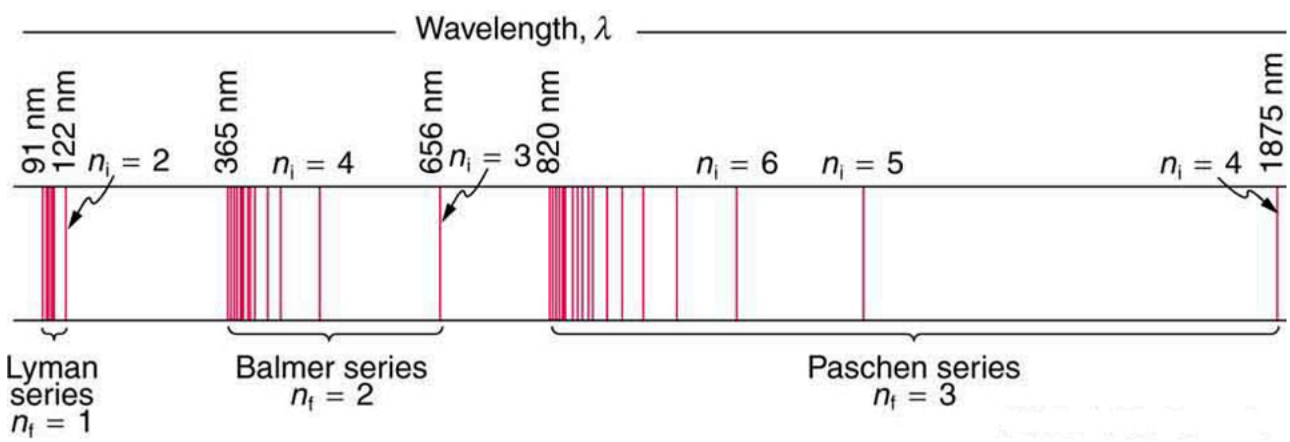
The observed **hydrogen-spectrum wavelengths** can be calculated using the following formula:

$$\frac{1}{\lambda}=R\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right),$$

where λ is the wavelength of the emitted EM radiation and R is the **Rydberg constant**, determined by the experiment to be

$$R=1.097\times10^7/\text{m} \ (\text{or m}^{-1}).$$

The constant n_f is a positive integer associated with a specific series. For the Lyman series, $n_f=1$; for the Balmer series, $n_f=2$; for the Paschen series, $n_f=3$; and so on. The Lyman series is entirely in the UV, while part of the Balmer series is visible with the remainder UV. The Paschen series and all the rest are entirely IR. There are apparently an unlimited number of series, although they lie progressively farther into the infrared and become difficult to observe as n_f increases. The constant n_i is a positive integer, but it must be greater than n_f . Thus, for the Balmer series, $n_f=2$ and $n_i=3,4,5,6,...$. Note that n_i can approach infinity. While the formula in the wavelengths equation was just a recipe designed to fit data and was not based on physical principles, it did imply a deeper meaning. Balmer first devised the formula for his series alone, and it was later found to describe all the other series by using different values of n_f . Bohr was the first to comprehend the deeper meaning. Again, we see the interplay between experiment and theory in physics. Experimentally, the spectra were well established, an equation was found to fit the experimental data, but the theoretical foundation was missing.



A schematic of the hydrogen spectrum shows several series named for those who contributed most to their determination. Part of the Balmer series is in the visible spectrum, while the Lyman series is entirely in the UV, and the Paschen series and others are in the IR. Values of n_{f} and n_{f} are shown for some of the lines.

Calculating Wave Interference of a Hydrogen Line

What is the distance between the slits of a grating that produces a first-order maximum for the second Balmer line at an angle of 15° ?

Strategy and Concept

For an Integrated Concept problem, we must first identify the physical principles involved. In this example, we need to know (a) the wavelength of light as well as (b) conditions for an interference maximum for the pattern from a double slit. Part (a) deals with a topic of the present chapter, while part (b) considers the wave interference material of [Wave Optics].

Solution for (a)

Hydrogen spectrum wavelength. The Balmer series requires that $n_{\text{f}} = 2$. The first line in the series is taken to be for $n_{\text{i}} = 3$, and so the second would have $n_{\text{i}} = 4$. The calculation is a straightforward application of the wavelength equation. Entering the determined values for n_{f} and n_{i} yields

$$1\lambda \quad = \quad R(1n_{2f}-1n_{2i}) \quad = \quad (1.097\times10^{7}\mathrm{m}^{-1})(12^{2}-14^{2}) \quad = \quad 2.057\times10^{6}\mathrm{m}^{-1}.$$

Inverting to find λ gives

$$\lambda \quad = \quad 12.057\times10^{6}\mathrm{m}^{-1}=486\times10^{-9}\mathrm{m} \quad = \quad 486\mathrm{nm}.$$

Discussion for (a)

This is indeed the experimentally observed wavelength, corresponding to the second (blue-green) line in the Balmer series. More impressive is the fact that the same simple recipe predicts *all* of the hydrogen spectrum lines, including new ones observed in subsequent experiments. What is nature telling us?

Solution for (b)

Double-slit interference ([Wave Optics]). To obtain constructive interference for a double slit, the path length difference from two slits must be an integral multiple of the wavelength. This condition was expressed by the equation

$$d\sin\theta=m\lambda,$$

where d is the distance between slits and θ is the angle from the original direction of the beam. The number m is the order of the interference; $m = 1$ in this example. Solving for d and entering known values yields

$$d=(1)(486\mathrm{nm})\sin15^{\circ}=1.88\times10^{-6}\mathrm{m}.$$

Discussion for (b)

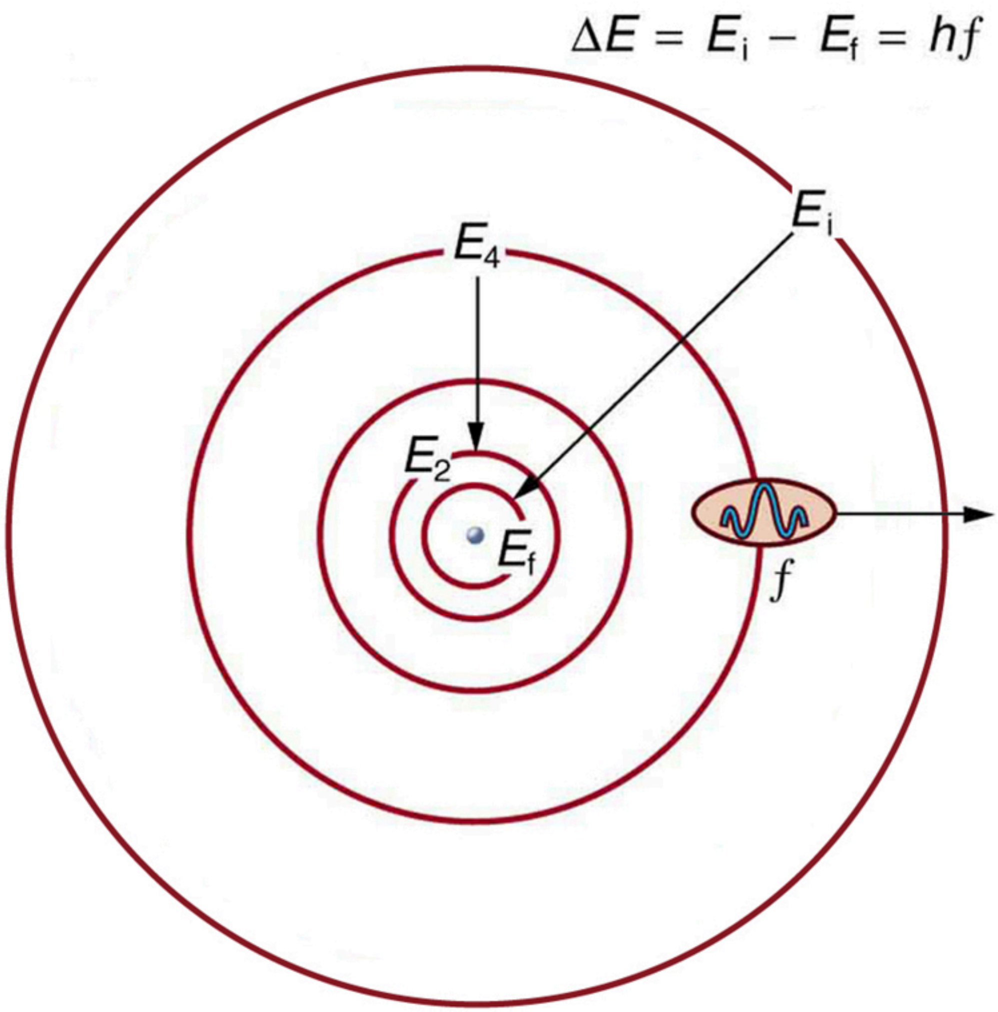
This number is similar to those used in the interference examples of [Introduction to Quantum Physics] (and is close to the spacing between slits in commonly used diffraction glasses).

Bohr’s Solution for Hydrogen

Bohr was able to derive the formula for the hydrogen spectrum using basic physics, the planetary model of the atom, and some very important new proposals. His first proposal is that only certain orbits are allowed: we say that *the orbits of electrons in atoms are quantized*. Each orbit has a different energy, and electrons can move to a higher orbit by absorbing energy and drop to a lower orbit by emitting energy. If the orbits are quantized, the amount of energy absorbed or emitted is also quantized, producing discrete spectra. Photon absorption and emission are among the primary methods of transferring energy into and out of atoms. The energies of the photons are quantized, and their energy is explained as being equal to the change in energy of the electron when it moves from one orbit to another. In equation form, this is

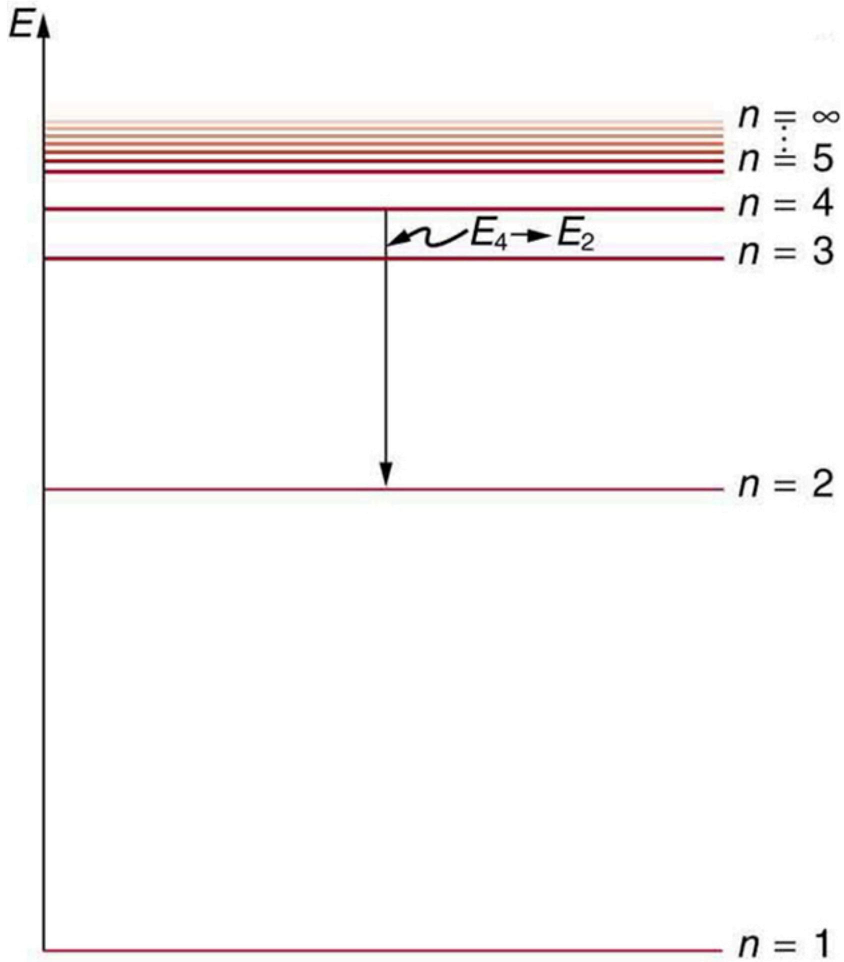
$$\Delta E = hf = E_i - E_f.$$

Here, ΔE is the change in energy between the initial and final orbits, and hf is the energy of the absorbed or emitted photon. It is quite logical (that is, expected from our everyday experience) that energy is involved in changing orbits. A blast of energy is required for the space shuttle, for example, to climb to a higher orbit. What is not expected is that atomic orbits should be quantized. This is not observed for satellites or planets, which can have any orbit given the proper energy. (See [Figure 4].)



The planetary model of the atom, as modified by Bohr, has the orbits of the electrons quantized. Only certain orbits are allowed, explaining why atomic spectra are discrete (quantized). The energy carried away from an atom by a photon comes from the electron dropping from one allowed orbit to another and is thus quantized. This is likewise true for atomic absorption of photons.

[Figure 5] shows an **energy-level diagram**, a convenient way to display energy states. In the present discussion, we take these to be the allowed energy levels of the electron. Energy is plotted vertically with the lowest or ground state at the bottom and with excited states above. Given the energies of the lines in an atomic spectrum, it is possible (although sometimes very difficult) to determine the energy levels of an atom. Energy-level diagrams are used for many systems, including molecules and nuclei. A theory of the atom or any other system must predict its energies based on the physics of the system.



An energy-level diagram plots energy vertically and is useful in visualizing the energy states of a system and the transitions between them. This diagram is for the hydrogen-atom electrons, showing a transition between two orbits having energies E_{4} and E_{2} .

Bohr was clever enough to find a way to calculate the electron orbital energies in hydrogen. This was an important first step that has been improved upon, but it is well worth repeating here, because it does correctly describe many characteristics of hydrogen. Assuming circular orbits, Bohr proposed that the **angular momentum L of an electron in its orbit is quantized**, that is, it has only specific, discrete values. The value for L is given by the formula

$$L=m_e v r_n=n\frac{h}{2\pi}(n=1,2,3,\ldots),$$

where L is the angular momentum, m_e is the electron's mass, r_n is the radius of the n th orbit, and h is Planck's constant. Note that angular momentum is $L=I\omega$. For a small object at a radius $r, I=mr^2$ and $\omega=v/r$, so that $L=(mr^2)(v/r)=mvr$. Quantization says that this value of mvr can only be equal to $h/2, 2h/2, 3h/2$, etc. At the time, Bohr himself did not know why angular momentum should be quantized, but using this assumption he was able to calculate the energies in the hydrogen spectrum, something no one else had done at the time.

From Bohr's assumptions, we will now derive a number of important properties of the hydrogen atom from the classical physics we have covered in the text. We start by noting the centripetal force causing the electron to follow a circular path is supplied by the Coulomb force. To be more general, we note that this analysis is valid for any single-electron atom. So, if a nucleus has Z protons ($Z=1$ for hydrogen, 2 for helium, etc.) and only one electron, that atom is called a **hydrogen-like atom**. The spectra of hydrogen-like ions are similar to hydrogen, but shifted to higher energy by the greater attractive force between the electron and nucleus. The magnitude of the centripetal force is $m_e v^2/r_n$, while the Coulomb force is $k(Zq_e)(q_e)/r_n^2$. The tacit assumption here is that the nucleus is more massive than the stationary electron, and the electron orbits about it. This is consistent with the planetary model of the atom. Equating these,

$$\frac{kZq_e^2}{r_n^2}=\frac{m_e v^2}{r_n}(Coulomb=centripetal).$$

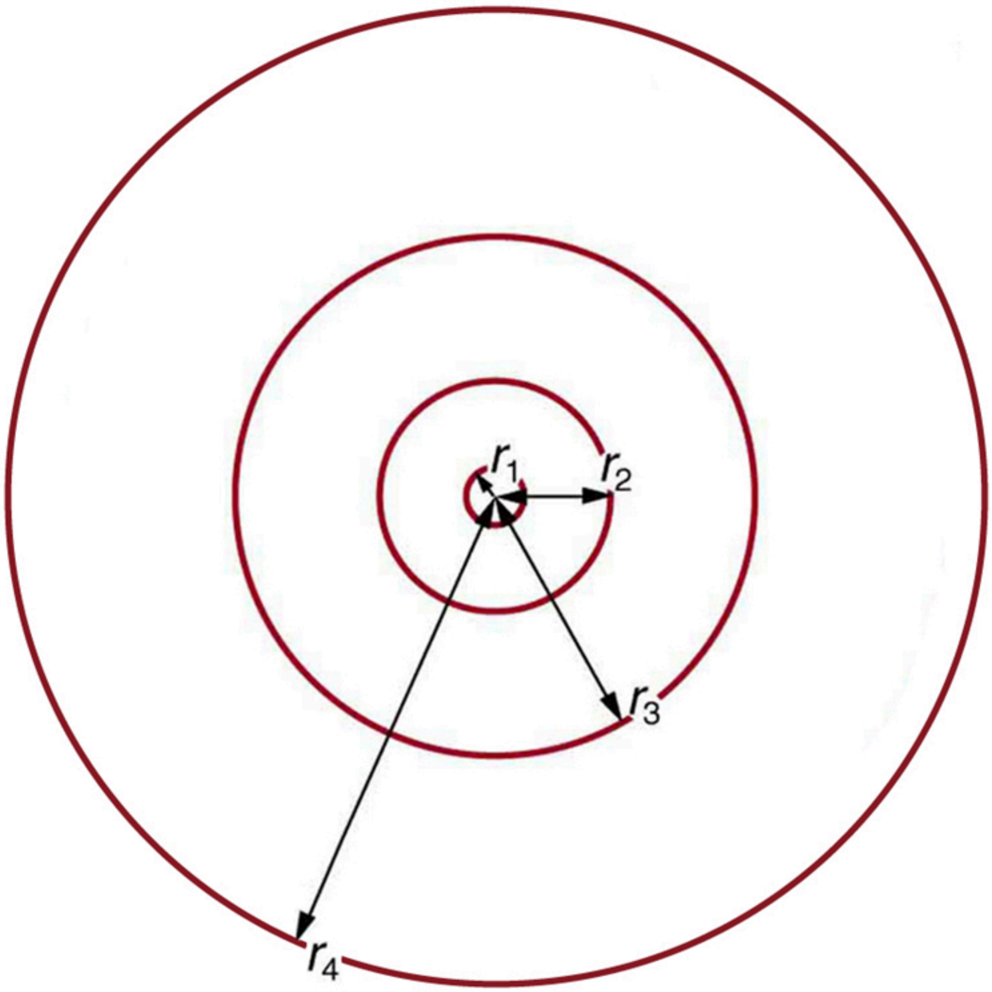
Angular momentum quantization is stated in an earlier equation. We solve that equation for v , substitute it into the above, and rearrange the expression to obtain the radius of the orbit. This yields:

$$r_n=\frac{n^2}{Z}a_{\mathrm{B}}, \text{ for allowed orbits}(n=1,2,3,\ldots),$$

where a_B is defined to be the **Bohr radius**, since for the lowest orbit $(n=1)$ and for hydrogen $(Z=1)$, $r_1 = a_B$. It is left for this chapter's Problems and Exercises to show that the Bohr radius is

$$a_B = h^2 4\pi^2 m_e k q_{2e} = 0.529\times10^{-10}\text{m.}$$

These last two equations can be used to calculate the **radii of the allowed (quantized) electron orbits in any hydrogen-like atom**. It is impressive that the formula gives the correct size of hydrogen, which is measured experimentally to be very close to the Bohr radius. The earlier equation also tells us that the orbital radius is proportional to n^2, as illustrated in [Figure 6].



The allowed electron orbits in hydrogen have the radii shown. These radii were first calculated by Bohr and are given by the equation \(r_n=n^2/Z a_B \). The lowest orbit has the experimentally verified diameter of a hydrogen atom.

To get the electron orbital energies, we start by noting that the electron energy is the sum of its kinetic and potential energy:

$$E_n = \text{KE} + \text{PE}.$$

Kinetic energy is the familiar $\text{KE} = (1/2) m_e v^2$, assuming the electron is not moving at relativistic speeds. Potential energy for the electron is electrical, or $\text{PE} = q_e V$, where V is the potential due to the nucleus, which looks like a point charge. The nucleus has a positive charge $Z q_e$; thus, $V = k Z q_e / r_n$, recalling an earlier equation for the potential due to a point charge. Since the electron's charge is negative, we see that $\text{PE} = -k Z q_e / r_n$. Entering the expressions for KE and PE, we find

$$E_n = 12 m_e v^2 - k Z q_{2e} r_n.$$

Now we substitute r_n and v from earlier equations into the above expression for energy. Algebraic manipulation yields

$$E_n = -Z^2 n^2 E_0 (n=1, 2, 3, \ldots)$$

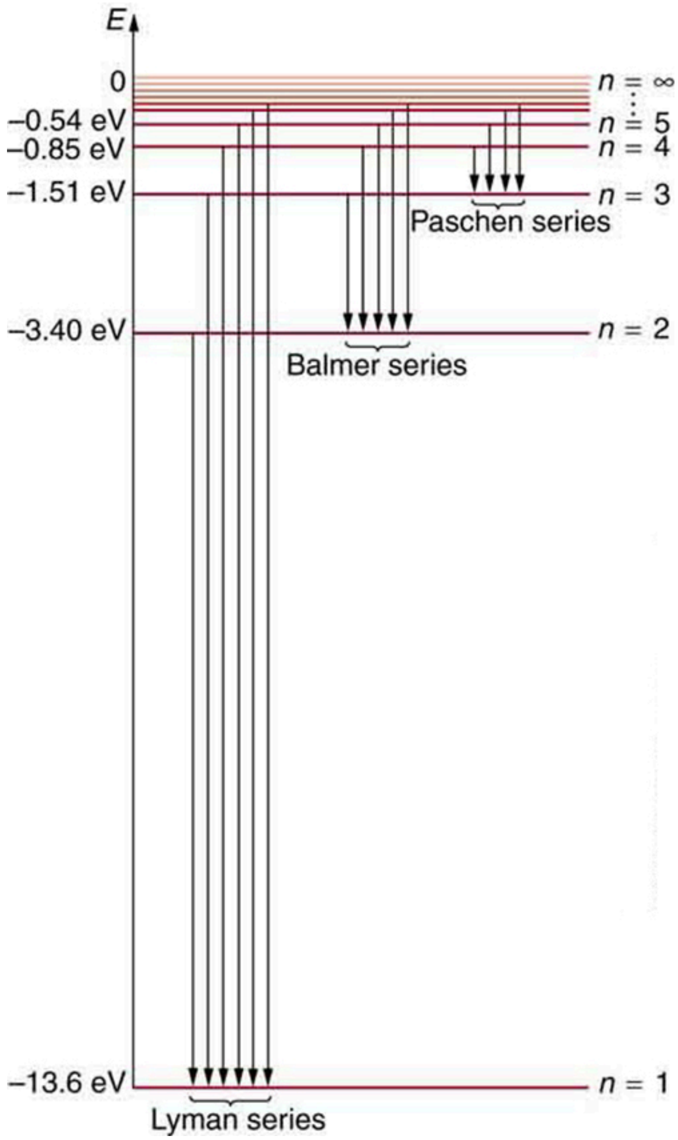
for the orbital **energies of hydrogen-like atoms**. Here, E_0 is the **ground-state energy** $(n=1)$ for hydrogen $(Z=1)$ and is given by

$$E_0 = \frac{2\pi^2 q_e^4 m_e k^2}{h^2} = 13.6\text{eV}.$$

Thus, for hydrogen,

$$E_n = -\frac{13.6\text{eV}}{n^2} \quad (n = 1, 2, 3, ...).$$

[Figure 7] shows an energy-level diagram for hydrogen that also illustrates how the various spectral series for hydrogen are related to transitions between energy levels.



Energy-level diagram for hydrogen showing the Lyman, Balmer, and Paschen series of transitions. The orbital energies are calculated using the above equation, first derived by Bohr.

Electron total energies are negative, since the electron is bound to the nucleus, analogous to being in a hole without enough kinetic energy to escape. As n approaches infinity, the total energy becomes zero. This corresponds to a free electron with no kinetic energy, since r_n gets very large for large n , and the electric potential energy thus becomes zero. Thus, 13.6 eV is needed to ionize hydrogen (to go from –13.6 eV to 0, or unbound), an experimentally verified number. Given more energy, the electron becomes unbound with some kinetic energy. For example, giving 15.0 eV to an electron in the ground state of hydrogen strips it from the atom and leaves it with 1.4 eV of kinetic energy.

Finally, let us consider the energy of a photon emitted in a downward transition, given by the equation to be

$$\Delta E = hf = E_\text{i} - E_\text{f}.$$

Substituting $E_n = (-13.6\text{eV}/n^2)$, we see that

$$hf = (13.6\text{eV})\left(\frac{1}{n_\text{f}^2} - \frac{1}{n_\text{i}^2}\right).$$

Dividing both sides of this equation by hc gives an expression for $1/\lambda$:

$$\frac{hf}{hc}=\frac{f}{c}=\frac{1}{\lambda}=\frac{(13.6\,\text{eV})}{hc}\left(\frac{1}{n_{\text{f}}^2}-\frac{1}{n_{\text{i}}^2}\right).$$

It can be shown that

$$\left(\frac{13.6\,\text{eV}}{hc}\right)=\frac{(13.6\,\text{eV})(1.602\times10^{-19}\,\text{J/eV})}{(6.626\times10^{-34}\,\text{J}\cdot\text{s})(2.998\times10^{8}\,\text{m/s})}=1.097\times10^{7}\,\text{m}^{-1}=R$$

is the **Rydberg constant**. Thus, we have used Bohr’s assumptions to derive the formula first proposed by Balmer years earlier as a recipe to fit experimental data.

$$\frac{1}{\lambda}=R\left(\frac{1}{n_{\text{f}}^2}-\frac{1}{n_{\text{i}}^2}\right)$$

We see that Bohr’s theory of the hydrogen atom answers the question as to why this previously known formula describes the hydrogen spectrum. It is because the energy levels are proportional to $1/n^2$, where n is a non-negative integer. A downward transition releases energy, and so n_i must be greater than n_{f} . The various series are those where the transitions end on a certain level. For the Lyman series, $n_{\text{f}}=1$ — that is, all the transitions end in the ground state (see also [Figure 7]). For the Balmer series, $n_{\text{f}}=2$, or all the transitions end in the first excited state; and so on. What was once a recipe is now based in physics, and something new is emerging—angular momentum is quantized.

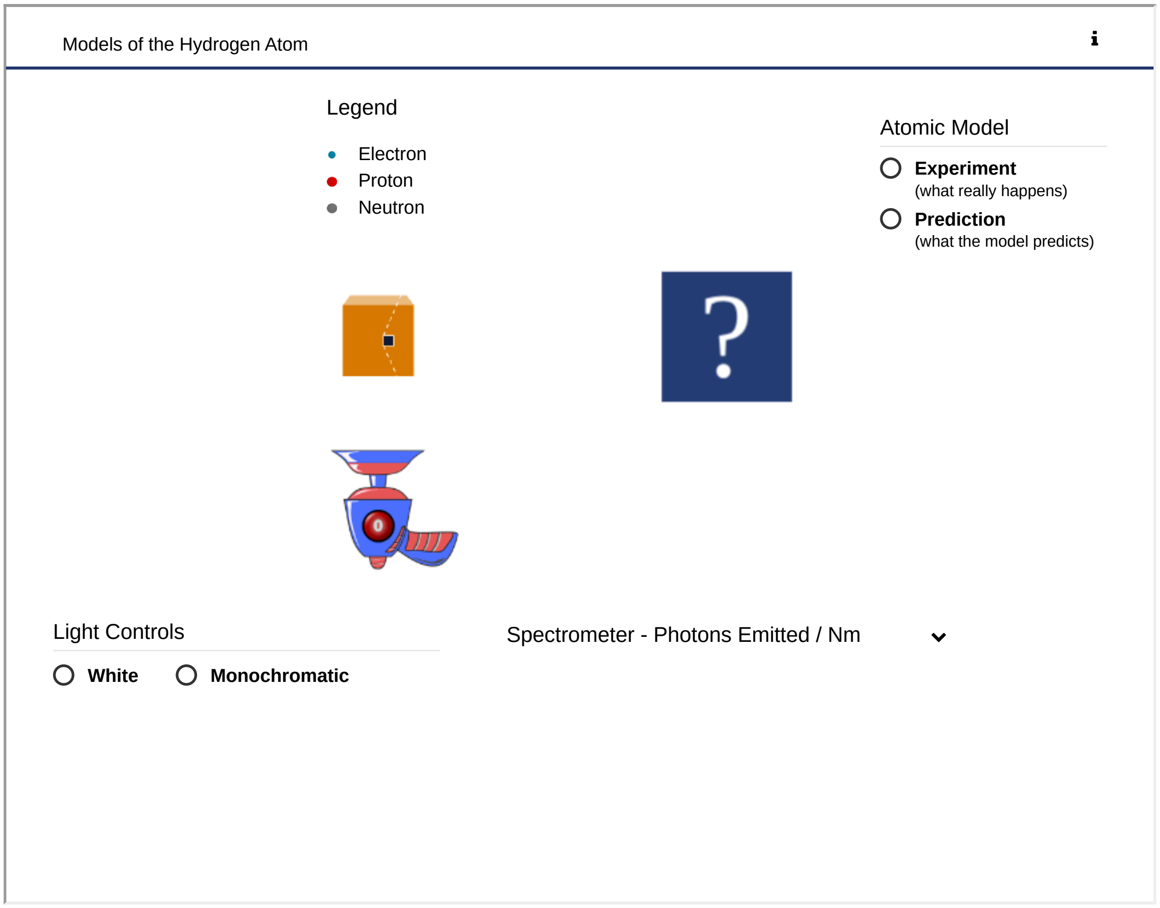
Triumphs and Limits of the Bohr Theory

Bohr did what no one had been able to do before. Not only did he explain the spectrum of hydrogen, he correctly calculated the size of the atom from basic physics. Some of his ideas are broadly applicable. Electron orbital energies are quantized in all atoms and molecules. Angular momentum is quantized. The electrons do not spiral into the nucleus, as expected classically (accelerated charges radiate, so that the electron orbits classically would decay quickly, and the electrons would sit on the nucleus—matter would collapse). These are major triumphs.

But there are limits to Bohr’s theory. It cannot be applied to multielectron atoms, even one as simple as a two-electron helium atom. Bohr’s model is what we call *semiclassical*. The orbits are quantized (nonclassical) but are assumed to be simple circular paths (classical). As quantum mechanics was developed, it became clear that there are no well-defined orbits; rather, there are clouds of probability. Bohr’s theory also did not explain that some spectral lines are doublets (split into two) when examined closely. We shall examine many of these aspects of quantum mechanics in more detail, but it should be kept in mind that Bohr did not fail. Rather, he made very important steps along the path to greater knowledge and laid the foundation for all of atomic physics that has since evolved.

PhET Explorations: Models of the Hydrogen Atom

How did scientists figure out the structure of atoms without looking at them? Try out different models by shooting light at the atom. Check how the prediction of the model matches the experimental results.



Section Summary

- The planetary model of the atom pictures electrons orbiting the nucleus in the way that planets orbit the sun. Bohr used the planetary model to develop the first reasonable theory of hydrogen, the simplest atom. Atomic and molecular spectra are quantized, with hydrogen spectrum wavelengths given by the formula

  $$\frac{1}{\lambda}=R\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right),$$

  where λ is the wavelength of the emitted EM radiation and R is the Rydberg constant, which has the value

  $$R=1.097\times10^{7}\,\mathrm{m}^{-1}.$$

- The constants n_i and n_f are positive integers, and n_i must be greater than n_f .
- Bohr correctly proposed that the energy and radii of the orbits of electrons in atoms are quantized, with energy for transitions between orbits given by

  $$\Delta E=hf=E_i-E_f,$$

  where ΔE is the change in energy between the initial and final orbits and hf is the energy of an absorbed or emitted photon. It is useful to plot orbital energies on a vertical graph called an energy-level diagram.
- Bohr proposed that the allowed orbits are circular and must have quantized orbital angular momentum given by

  $$L=m_e v r_n=n\frac{h}{2\pi}\text{ where }(n=1,2,3\ldots),$$

  where L is the angular momentum, r_n is the radius of the nth orbit, and h is Planck's constant. For all one-electron (hydrogen-like) atoms, the radius of an orbit is given by

  $$r_n=\frac{n^2}{Z}a_B\text{ where (allowed orbits }n=1,2,3,...),$$

  Z is the atomic number of an element (the number of electrons is has when neutral) and a_B is defined to be the Bohr radius, which is

  $$a_B=\frac{h^2}{4\pi^2 m_e k q_e^2}=0.529\times10^{-10}\,\mathrm{m}.$$
- Furthermore, the energies of hydrogen-like atoms are given by

$$E_n = -\frac{Z^2}{n^2}E_0 \text{ where } (n = 1, 2, 3 ...),$$

where E_0 is the ground-state energy and is given by

$$E_0 = \frac{2\pi^2 q_e^4 m_e k^2}{h^2} = 13.6\,\text{eV}.$$

Thus, for hydrogen,

$$E_n = -\frac{13.6\,\text{eV}}{n^2} \text{ where } (n = 1, 2, 3 ...).$$

- The Bohr Theory gives accurate values for the energy levels in hydrogen-like atoms, but it has been improved upon in several respects.

Conceptual Questions

How do the allowed orbits for electrons in atoms differ from the allowed orbits for planets around the sun? Explain how the correspondence principle applies here.

Explain how Bohr's rule for the quantization of electron orbital angular momentum differs from the actual rule.

What is a hydrogen-like atom, and how are the energies and radii of its electron orbits related to those in hydrogen?

Problems & Exercises

By calculating its wavelength, show that the first line in the Lyman series is UV radiation.

Show Solution

Strategy

The Lyman series corresponds to transitions ending at the ground state, so $n_f = 1$. The first line in the series corresponds to a transition from $n_i = 2$ to $n_f = 1$. We'll use the Rydberg formula to calculate the wavelength and determine if it falls in the UV region (wavelengths less than 400 nm).

Solution

Using the Rydberg formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

With $n_f = 1$ and $n_i = 2$:

$$\frac{1}{\lambda} = (1.097 \times 10^7\ \text{m}^{-1})\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = (1.097 \times 10^7\ \text{m}^{-1})\left(1 - \frac{1}{4}\right)$$
$$\frac{1}{\lambda} = (1.097 \times 10^7\ \text{m}^{-1})\left(\frac{3}{4}\right) = 8.228 \times 10^6\ \text{m}^{-1}$$
$$\lambda = \frac{1}{8.228 \times 10^6\ \text{m}^{-1}} = 1.22 \times 10^{-7}\ \text{m} = 122\ \text{nm}$$

Since 122 nm is much less than 400 nm (the lower limit of visible light), this is indeed **UV radiation**.

Discussion

The entire Lyman series lies in the ultraviolet region because these transitions all end at the ground state (n = 1), involving relatively large energy changes. The first line at 122 nm (also called Lyman-alpha) is the longest wavelength and lowest energy transition in this series. As we move to higher lines in the series (from n = 3, 4, 5, … to n = 1), the wavelengths get progressively shorter, approaching the series limit at 91.2 nm as n approaches infinity. This UV radiation from hydrogen is important in astronomy—Lyman-alpha is one of the most studied spectral lines in astrophysics, used to probe distant galaxies, intergalactic medium, and the early universe. However, because Earth's atmosphere absorbs UV radiation, these lines must be observed from space-based telescopes.

Find the wavelength of the third line in the Lyman series, and identify the type of EM radiation.

Show Solution

Strategy

The Lyman series corresponds to transitions ending at the ground state, so $n_f = 1$. The first line in the series corresponds to a transition from $n_i = 2$ to $n_f = 1$, the second line from $n_i = 3$ to $n_f = 1$, and the third line from $n_i = 4$ to $n_f = 1$. We'll use the Rydberg formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

where $R = 1.097 \times 10^7\ \text{m}^{-1}$ is the Rydberg constant.

Solution

For the third line in the Lyman series, we have $n_{\mathrm{f}} = 1$ and $n_i = 4$:

$$1\lambda=(1.097\times10^{7}\ \mathrm{m}^{-1})(11^{2}-14^{2})$$
$$1\lambda=(1.097\times10^{7}\ \mathrm{m}^{-1})(1-116)=(1.097\times10^{7}\ \mathrm{m}^{-1})(1516)$$
$$1\lambda=1.028\times10^{7}\ \mathrm{m}^{-1}$$
$$\lambda=11.028\times10^{7}\ \mathrm{m}^{-1}=9.72\times10^{-8}\ \mathrm{m}=97.2\ \mathrm{nm}$$

This wavelength of 97.2 nm is in the **ultraviolet (UV)** region of the electromagnetic spectrum.

Discussion

The entire Lyman series lies in the UV region, which makes sense because these transitions all end at the ground state (n = 1), which means they involve relatively large energy changes. As we move to higher lines in the series (from the first to the second to the third, etc.), the wavelength increases (the photon energy decreases) because the energy difference between successive levels decreases. The third line at 97.2 nm is still well into the UV (UV extends from about 10 nm to 400 nm), though it's longer than the first line (121 nm) and second line (103 nm). The series limit (as $n_i \to \infty$) approaches 91.2 nm, which represents the minimum energy needed to ionize hydrogen from its ground state.

Look up the values of the quantities in $a_{\mathrm{B}} = h^2 4\pi^2 m_e k q_{2e}$, and verify that the Bohr radius a_{B} is $0.529 \times 10^{-10}\mathrm{m}$.

Show Solution

Strategy

We'll substitute the known values of fundamental constants into the equation $a_{\mathrm{B}} = h^2 4\pi^2 m_e k q_{2e}$ and verify that the result is 0.529×10^{-10} m . The constants we need are:

- Planck's constant: $h = 6.626 \times 10^{-34}\ \mathrm{J\cdot s}$
- Electron mass: $m_e = 9.109 \times 10^{-31}\ \mathrm{kg}$
- Coulomb's constant: $k = 8.988 \times 10^{9}\ \mathrm{N\cdot m^2/C^2}$
- Elementary charge: $q_e = 1.602 \times 10^{-19}\ \mathrm{C}$

Solution

Substituting these values into the Bohr radius equation:

$$a_{\mathrm{B}}=h^2 4\pi^2 m_e k q_{2e}=(6.626\times10^{-34}\ \mathrm{J\cdot s})^2 4\pi^2(9.109\times10^{-31}\ \mathrm{kg})(8.988\times10^{9}\ \mathrm{N\cdot m^2/C^2})(1.602\times10^{-19}\ \mathrm{C})^2$$
$$a_{\mathrm{B}}=4.390\times10^{-67}\ \mathrm{J^2\cdot s^2}(39.478)(9.109\times10^{-31})(8.988\times10^{9})(2.566\times10^{-38})\ \mathrm{kg\cdot N\cdot m^2/C^2\cdot C^2}$$
$$a_{\mathrm{B}}=4.390\times10^{-67}8.298\times10^{-57}\ \mathrm{m}=0.529\times10^{-10}\ \mathrm{m}=5.29\times10^{-11}\ \mathrm{m}$$

This verifies that the Bohr radius is **0.529×10^{-10} m or 0.529 Å** (angstroms).

Discussion

The Bohr radius represents the most probable distance of the electron from the nucleus in the ground state of the hydrogen atom. This fundamental length scale emerges naturally from the combination of quantum mechanics (Planck's constant h), the properties of the electron (its mass m_e and charge q_e), and the electromagnetic force (Coulomb's constant k). The fact that this theoretical prediction matches experimental measurements of the hydrogen atom's size was one of the great triumphs of Bohr's theory. The Bohr radius serves as a natural unit for expressing atomic distances—for example, in hydrogen-like atoms, the radius of the n-th orbit is $r_n = n^2 a_{\mathrm{B}}/Z$. The small value (about 0.05 nm or half an angstrom) explains why atoms are invisible to the naked eye and why we need powerful microscopes or indirect methods to study atomic structure. It's worth noting that in modern quantum mechanics, we no longer think of the electron as orbiting at a fixed radius; instead, the Bohr radius represents the peak of the radial probability distribution for finding the electron in the ground state.

Verify that the ground state energy E_0 is 13.6 eV by using $E_0 = 2\pi^2 q_{4e} m_e k^2 h^2$.

Show Solution

Strategy

We'll substitute the known fundamental constants into the given equation and calculate the result. The constants we need are:

- Electron charge: $q_e = 1.602 \times 10^{-19}$ C

- Electron mass: $m_e = 9.109 \times 10^{-31}$ kg
- Coulomb's constant: $k = 8.988 \times 10^{9}$ N·m^2/C^2
- Planck's constant: $h = 6.626 \times 10^{-34}$ J·s

Solution

Substituting these values into the equation:

$$E_0 = \frac{2\pi^2 q_e^4 m_e k^2}{h^2}$$
$$E_0 = \frac{2\pi^2 (1.602\times10^{-19}\ \text{C})^4 (9.109\times10^{-31}\ \text{kg})(8.988\times10^{9}\ \text{N·m}^2/\text{C}^2)^2}{(6.626\times10^{-34}\ \text{J·s})^2}$$

Working through the calculation step by step:

$$E_0 = \frac{2(9.870)(6.595\times10^{-76})(9.109\times10^{-31})(8.079\times10^{19})}{4.390\times10^{-67}}$$
$$E_0 = \frac{9.521\times10^{-86}}{4.390\times10^{-67}} = 2.169\times10^{-18}\ \text{J}$$

Converting to electron volts (dividing by $q_e = 1.602 \times 10^{-19}$ J/eV):

$$E_0 = \frac{2.169\times10^{-18}\ \text{J}}{1.602\times10^{-19}\ \text{J/eV}} = 13.6\ \text{eV}$$

This verifies that the ground state energy of hydrogen is indeed **13.6 eV**.

Discussion

This result is fundamental to atomic physics. The ground state energy of –13.6 eV (negative because it represents a bound state) means that 13.6 eV of energy is required to ionize a hydrogen atom from its ground state. This value emerges directly from the fundamental constants of nature—the electron's charge and mass, Planck's constant, and Coulomb's constant. The fact that Bohr could derive this experimentally known value from first principles was a major triumph of his theory and provided strong support for the quantum model of the atom. The energy levels for excited states are given by $E_n = E_0/n^2$, so all hydrogen energy levels can be calculated from this single fundamental value.

If a hydrogen atom has its electron in the $n = 4$ state, how much energy in eV is needed to ionize it?

Show Solution

Strategy

The energy of an electron in the n-th energy level of hydrogen is given by $E_n = \frac{-13.6\ \text{eV}}{n^2}$. To ionize the atom from the $n = 4$ state means bringing the electron from E_4 to zero energy (the ionization threshold). The ionization energy is the magnitude of E_4 .

Solution

For $n = 4$:

$$E_4 = \frac{-13.6\ \text{eV}}{4^2} = \frac{-13.6\ \text{eV}}{16} = -0.850\ \text{eV}$$

The ionization energy is:

$$\text{Ionization energy} = 0 - E_4 = 0 - (-0.850\ \text{eV}) = 0.850\ \text{eV}$$

Therefore, **0.850 eV** of energy is needed to ionize a hydrogen atom from the $n = 4$ state.

Discussion

This energy is much less than the 13.6 eV needed to ionize hydrogen from its ground state ($n = 1$). The higher the energy level, the less tightly the electron is bound to the nucleus, and therefore the less energy is required to ionize it. This makes physical sense because the electron in the $n = 4$ state is farther from the nucleus (the orbital radius increases as n^2) and experiences a weaker attractive force. The ionization energy decreases as $1/n^2$: for $n = 2$ it's 3.4 eV, for $n = 3$ it's 1.51 eV, and for $n = 4$ it's 0.850 eV. As $n \to \infty$, the ionization energy approaches zero, meaning the electron is barely bound. This is why highly excited atoms (high n) are easily ionized by collisions, electric fields, or photons, and why such "Rydberg atoms" are important in studying quantum-to-classical transitions and in applications like quantum computing.

A hydrogen atom in an excited state can be ionized with less energy than when it is in its ground state. What is n for a hydrogen atom if 0.850 eV of energy can ionize it?

Show Solution

Strategy

The energy of an electron in the n-th energy level of hydrogen is given by $E_n = -\frac{13.6 \text{ eV}}{n^2}$. The energy is negative because the electron is in a bound state. To ionize the atom from level n, we must provide enough energy to bring the electron from E_n to zero energy (the ionization threshold). Therefore, the ionization energy is $$ E_n = \frac{13.6 \text{ eV}}{n^2}$. We can solve for n $$.

Solution

Setting the ionization energy equal to 0.850 eV:

$$\frac{13.6 \text{ eV}}{n^2} = 0.850 \text{ eV}$$

Solving for n^2:

$$n^2 = \frac{13.6 \text{ eV}}{0.850 \text{ eV}} = 16.0$$
$$n = \sqrt{16.0} = 4$$

Therefore, the hydrogen atom is in the **n = 4** energy level.

Discussion

This result makes physical sense. The higher the energy level (larger n), the less tightly bound the electron is to the nucleus, and therefore the less energy is required to ionize it. For the ground state (n = 1), we need 13.6 eV to ionize hydrogen. For n = 2, we need 13.6/4 = 3.4 eV. For n = 3, we need 13.6/9 = 1.51 eV. And for n = 4, we need 13.6/16 = 0.850 eV, which matches the given ionization energy. As n approaches infinity, the ionization energy approaches zero, which makes sense because the electron is already nearly free at very high quantum numbers. This problem demonstrates why excited atoms are more easily ionized than ground-state atoms—a fact that's important in understanding phenomena like flame tests, gas discharge tubes, and ionized gases in stars.

Find the radius of a hydrogen atom in the $n = 2$ state according to Bohr's theory.

Show Solution

Strategy

According to Bohr's theory, the radius of the n-th orbit in hydrogen is given by $r_n = n^2 a_B$, where $a_B = 0.529 \times 10^{-10}$ m is the Bohr radius. For the $n = 2$ state, we simply substitute $n = 2$ into this formula.

Solution

Using $r_n = n^2 a_B$ with $n = 2$:

$$r_2 = (2)^2 a_B = 4 \times (0.529 \times 10^{-10} \text{ m}) = 2.12 \times 10^{-10} \text{ m}$$

The radius of the $n = 2$ state is **2.12×10^{-10} m or 2.12 Å**.

Discussion

The $n = 2$ state has a radius exactly 4 times larger than the ground state ($n = 1$), demonstrating the n^2 scaling of orbital radius in Bohr's model. This is the first excited state of hydrogen. When an atom transitions from $n = 2$ to $n = 1$, it emits a photon at 121.6 nm (Lyman-alpha), the first line of the Lyman series. In modern quantum mechanics, we interpret this radius as the location of the peak in the radial probability distribution for the 2s orbital, or the average radius for 2p orbitals. The larger radius of excited states explains why excited atoms are physically bigger and more easily perturbed by external fields than ground-state atoms. The n^2 scaling also explains why very high-n "Rydberg atoms" can be macroscopically large—for example, atoms with $n = 100$ would have radii of about 0.5 micrometers, large enough to be affected by room-temperature thermal radiation.

Show that $(13.6 \text{eV})/hc = 1.097 \times 10^7 \text{m} = R$ (Rydberg's constant), as discussed in the text.

Show Solution

Strategy

We need to calculate $\frac{13.6 \text{ eV}}{hc}$ and show it equals the Rydberg constant $R = 1.097 \times 10^7 \text{ m}^{-1}$. We'll use the convenient relationship $hc = 1240 \text{ eV} \cdot \text{nm}$ to simplify the calculation.

Solution

Using the relationship $hc = 1240\ \text{eV}\cdot\text{nm}$:

$$\frac{13.6\ \text{eV}}{hc} = \frac{13.6\ \text{eV}}{1240\ \text{eV}\cdot\text{nm}}$$
$$= \frac{13.6}{1240}\ \text{nm}^{-1} = 0.01097\ \text{nm}^{-1}$$

Converting to m^{-1} (since 1 nm = 10^{-9} m, so 1 nm^{-1} = 10^{9} m^{-1}):

$$= 0.01097 \times 10^{9}\ \text{m}^{-1} = 1.097 \times 10^{7}\ \text{m}^{-1}$$

This is exactly equal to the Rydberg constant $R = 1.097 \times 10^{7}\ \text{m}^{-1}$.

Discussion

This result beautifully demonstrates the connection between Bohr's atomic theory and the empirically discovered Rydberg formula. The Rydberg constant, originally determined from experimental spectroscopic data, can now be understood as arising from the fundamental ground-state binding energy of hydrogen (13.6 eV) divided by hc . This relationship appears in the formula for the wavelengths of spectral lines:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

The fact that Bohr's theory could not only explain the existence of discrete spectral lines but also derive the previously mysterious Rydberg constant from first principles was one of the theory's greatest triumphs. It showed that quantum mechanics could make precise quantitative predictions, not just qualitative explanations, and established the validity of the quantum approach to atomic structure.

What is the smallest-wavelength line in the Balmer series? Is it in the visible part of the spectrum?

Show Solution

Strategy

The Balmer series corresponds to transitions ending at $n_f = 2$. The smallest wavelength (highest energy) occurs when the initial state has the highest possible n_i , which approaches infinity. We'll use the Rydberg formula with $n_f = 2$ and $n_i \to \infty$ to find the series limit.

Solution

Using the Rydberg formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For the series limit, $n_f = 2$ and $n_i \to \infty$, so $\frac{1}{n_i^2} \to 0$:

$$\frac{1}{\lambda_{\text{min}}} = R\left(\frac{1}{2^2} - 0\right) = \frac{R}{4} = \frac{1.097 \times 10^{7}\ \text{m}^{-1}}{4} = 2.743 \times 10^{6}\ \text{m}^{-1}$$
$$\lambda_{\text{min}} = \frac{1}{2.743 \times 10^{6}\ \text{m}^{-1}} = 3.65 \times 10^{-7}\ \text{m} = 365\ \text{nm}$$

The smallest wavelength in the Balmer series is **365 nm**, which is in the **ultraviolet** region (just below the 400 nm threshold for visible light).

Discussion

The Balmer series limit at 365 nm represents the minimum energy photon that can ionize a hydrogen atom from the $n = 2$ state. All transitions in the Balmer series that we observe have wavelengths longer than 365 nm. The first four lines of the Balmer series are in the visible region (H-alpha at 656 nm is red, H-beta at 486 nm is blue-green, H-gamma at 434 nm is violet, and H-delta at 410 nm is deep violet), but as we move to higher transitions (higher initial n values), the wavelengths get shorter and eventually cross into the UV at 365 nm. This series limit is important historically—Balmer discovered the empirical formula for hydrogen spectral lines by studying these visible lines, and his work laid the foundation for Bohr's quantum theory of the atom. The fact that the series has a well-defined limit (rather than continuing indefinitely) was one of the key clues that atomic energy levels are quantized.

Show that the entire Paschen series is in the infrared part of the spectrum. To do this, you only need to calculate the shortest wavelength in the series.

Show Solution

Strategy

The Paschen series corresponds to transitions that end at $n_\text{f} = 3$. The shortest wavelength (highest energy transition) in any series occurs when $n_i \to \infty$, because this represents the largest possible energy difference for that series. If we can show that even this shortest wavelength is in the infrared ($\lambda > 700$ nm), then all other lines in the Paschen series must also be in the infrared, since they have longer wavelengths. We'll use the Rydberg formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_\text{f}^2} - \frac{1}{n_i^2}\right)$$

Solution

For the Paschen series, $n_\text{f} = 3$. As $n_i \to \infty$, the term $\frac{1}{n_i^2} \to 0$, so:

$$1\lambda_{\text{min}}=R(13^2-0)=R9$$

Substituting $R = 1.097 \times 10^7\ \text{m}^{-1}$:

$$1\lambda_{\text{min}}=1.097\times10^7\ \text{m}^{-1}9=1.219\times10^6\ \text{m}^{-1}$$
$$\lambda_{\text{min}}=11.219\times10^6\ \text{m}^{-1}=8.20\times10^{-7}\ \text{m}=820\ \text{nm}$$

Since the infrared region begins at wavelengths greater than about 700 nm, and the shortest wavelength in the Paschen series is 820 nm, the entire Paschen series lies in the infrared region.

Discussion

This result demonstrates why the Paschen series was discovered later than the Balmer series—it's not visible to the human eye. The visible spectrum extends from approximately 400 nm (violet) to 700 nm (red), while infrared radiation has wavelengths longer than 700 nm. Since 820 nm is the *shortest* wavelength in the Paschen series, all other lines in the series (corresponding to transitions from $n_i = 4,5,6,...$ to $n_f = 3$) must have even longer wavelengths and therefore are also in the infrared. For example, the first line of the Paschen series (from $n_i = 4$ to $n_f = 3$) has a wavelength of 1875 nm, which is well into the infrared. This series is important in astronomy for studying stellar atmospheres and in understanding the complete hydrogen spectrum.

Do the Balmer and Lyman series overlap? To answer this, calculate the shortest-wavelength Balmer line and the longest-wavelength Lyman line.

Show Solution

Strategy

To determine if the Balmer and Lyman series overlap, we need to compare the shortest wavelength of the Balmer series (its series limit) with the longest wavelength of the Lyman series (its first line). If the Balmer minimum is shorter than or equal to the Lyman maximum, the series would overlap. We'll calculate both wavelengths using the Rydberg formula.

Solution

Shortest Balmer wavelength (series limit, $n_f = 2$, $n_i \to \infty$):

$$1\lambda_{\text{B,min}}=R(12^2-0)=R4$$
$$\lambda_{\text{B,min}}=4R=41.097\times10^7\ \text{m}^{-1}=365\ \text{nm}$$

Longest Lyman wavelength (first line, $n_f = 1$, $n_i = 2$):

$$1\lambda_{\text{L,max}}=R(11^2-12^2)=R(34)$$
$$\lambda_{\text{L,max}}=43R=43\times1.097\times10^7\ \text{m}^{-1}=122\ \text{nm}$$

Since **365 nm > 122 nm**, the Balmer and Lyman series **do not overlap**.

Discussion

The Balmer series spans from 365 nm (series limit) to 656 nm (H-alpha line), while the Lyman series spans from 91.2 nm (series limit) to 122 nm (Lyman-alpha). There's a clear gap of 243 nm (from 122 nm to 365 nm) between these two series. This gap is significant—it's nearly twice the width of the Lyman series itself. This separation makes sense because the Balmer series involves transitions to $n = 2$ while Lyman involves transitions to $n = 1$, and the energy difference between these final states is substantial (10.2 eV). The non-overlapping nature of hydrogen's spectral series made it easier for early spectroscopists to identify and categorize the different series. In contrast, for heavier hydrogen-like ions with larger Z values, the energy levels scale as Z^2 , which can cause different series to overlap in wavelength even though they correspond to different transitions.

(a) Which line in the Balmer series is the first one in the UV part of the spectrum?

(b) How many Balmer series lines are in the visible part of the spectrum?

(c) How many are in the UV?

Show Solution

Strategy

The Balmer series corresponds to transitions ending at $n_{\text{f}} = 2$, with initial states $n_i = 3,4,5,6,...$. The visible spectrum extends from approximately 400 nm (violet) to 700 nm (red), while ultraviolet (UV) radiation has wavelengths shorter than 400 nm. We'll calculate the wavelength for each Balmer line using the Rydberg formula until we find the transition into the UV region. Since wavelength decreases as n_i increases, we need to find which value of n_i produces the first wavelength below 400 nm.

$$1\lambda=R(1n_{2\mathrm{f}}-1n_{2i})$$

Solution

For the Balmer series, $n_\mathrm{f}=2$. Let's calculate wavelengths for successive values of n_i :

For $n_i=3$ (H-alpha line):

$$1\lambda=(1.097\times10^{7}\ \mathrm{m}^{-1})(14-19)=(1.097\times10^{7}\ \mathrm{m}^{-1})(536)=1.524\times10^{6}\ \mathrm{m}^{-1}$$
$$\lambda=656\ \text{nm (red, visible)}$$

For $n_i=4$ (H-beta line):

$$1\lambda=(1.097\times10^{7}\ \mathrm{m}^{-1})(14-116)=(1.097\times10^{7}\ \mathrm{m}^{-1})(316)=2.057\times10^{6}\ \mathrm{m}^{-1}$$
$$\lambda=486\ \text{nm (blue-green, visible)}$$

For $n_i=5$ (H-gamma line):

$$1\lambda=(1.097\times10^{7}\ \mathrm{m}^{-1})(14-125)=(1.097\times10^{7}\ \mathrm{m}^{-1})(21100)=2.304\times10^{6}\ \mathrm{m}^{-1}$$
$$\lambda=434\ \text{nm (violet, visible)}$$

For $n_i=6$ (H-delta line):

$$1\lambda=(1.097\times10^{7}\ \mathrm{m}^{-1})(14-136)=(1.097\times10^{7}\ \mathrm{m}^{-1})(836)=2.438\times10^{6}\ \mathrm{m}^{-1}$$
$$\lambda=410\ \text{nm (violet, visible)}$$

For $n_i=7$ (H-epsilon line):

$$1\lambda=(1.097\times10^{7}\ \mathrm{m}^{-1})(14-149)=(1.097\times10^{7}\ \mathrm{m}^{-1})(45196)=2.519\times10^{6}\ \mathrm{m}^{-1}$$
$$\lambda=397\ \text{nm (UV)}$$

(a) The line corresponding to the transition from $n_i=7$ to $n_f=2$ (H-epsilon line) is the first one in the UV part of the spectrum, with $\lambda=397\ \mathrm{nm}$.

(b) Four Balmer series lines are in the visible part of the spectrum (corresponding to $n_i=3,4,5,6$).

(c) Infinitely many Balmer lines are in the UV. As n_i increases from 7 to infinity, all wavelengths become progressively shorter, remaining in the UV and approaching the series limit of 365 nm.

Discussion

The Balmer series is special because it's the only hydrogen spectral series with lines in the visible region—this is why it was discovered first and studied most extensively in early spectroscopy. The four visible lines (H-alpha at 656 nm appearing red, H-beta at 486 nm appearing blue-green, H-gamma at 434 nm appearing violet, and H-delta at 410 nm appearing violet) can be seen with the naked eye in hydrogen discharge tubes and in astronomical observations of nebulae and stars. The fact that only four lines are visible while infinitely many are in the UV illustrates how the energy levels become more closely spaced as n increases. The series limit (as $n_i\to\infty$) is at 365 nm, representing the energy needed to ionize hydrogen from the $n=2$ state. This wavelength marks the shortest possible Balmer line and separates discrete spectral lines from the continuum that appears beyond the series limit.

A wavelength of $4.653\mu\mathrm{m}$ is observed in a hydrogen spectrum for a transition that ends in the $n_\mathrm{f}=5$ level. What was n_i for the initial level of the electron?

Show Solution

Strategy

We're given the wavelength $\lambda=4.653\ \mu\mathrm{m}$ and the final level $n_f=5$. We need to find the initial level n_i . Using the Rydberg formula, we can solve for n_i : $1\lambda=R(1n_{2f}-1n_{2i})$.

Solution

First, convert the wavelength to meters:

$$\lambda=4.653\ \mu\mathrm{m}=4.653\times10^{-6}\ \mathrm{m}$$

Using the Rydberg formula:

$$1\lambda=R(1n_{2f}-1n_{2i})$$

$14.653\times10^{-6}=(1.097\times10^{7})(125-1n_{2i})$
$2.149\times10^{5}=(1.097\times10^{7})(125-1n_{2i})$
$2.149\times10^{5}1.097\times10^{7}=125-1n_{2i}$
$0.0196=0.0400-1n_{2i}$
$1n_{2i}=0.0400-0.0196=0.0204$
$n_{2i}=10.0204=49.0$
$n_{i}=\sqrt{49.0}=7$

The electron was initially in the **n = 7** level.

Discussion

This transition from $n=7$ to $n=5$ produces infrared radiation at 4.653 μm, which is well beyond the visible range. This is part of what we might call an extended Pfund series (transitions ending at $n=5$). The Pfund series includes several lines in the mid-infrared, all invisible to the human eye but detectable with infrared spectrometers. The long wavelength (low energy) of this transition makes sense because both initial and final states are highly excited, so the energy difference is relatively small. Transitions between high-lying states are important in astrophysics for studying the conditions in stellar atmospheres and interstellar gas clouds, where hydrogen atoms can be excited to these high levels by the intense radiation fields. These infrared hydrogen lines are also used to study the kinematics of gas in galaxies through Doppler shifts.

A singly ionized helium ion has only one electron and is denoted He^{+} . What is the ion's radius in the ground state compared to the Bohr radius of hydrogen atom?

Show Solution

Strategy

A singly ionized helium ion (He^{+}) is a hydrogen-like atom with one electron orbiting a nucleus containing two protons (so $Z=2$). We can use the general formula for the radius of hydrogen-like atoms:

$$r_{n}=n^{2}Za_{\mathrm{B}}$$

where $a_{\mathrm{B}}=0.529\times10^{-10}\ \mathrm{m}$ is the Bohr radius. For the ground state, $n=1$.

Solution

For He^{+} in the ground state, we have $n=1$ and $Z=2$:

$$r_{1}=1^{2}2a_{\mathrm{B}}=a_{\mathrm{B}}2=0.5a_{\mathrm{B}}$$

Therefore, the radius of He^{+} in its ground state is **half the Bohr radius** of hydrogen, or:

$$r_{1}=0.5\times(0.529\times10^{-10}\ \mathrm{m})=0.265\times10^{-10}\ \mathrm{m}=0.265\ \text{Å}$$

Discussion

This result makes physical sense: the helium nucleus has twice the positive charge of the hydrogen nucleus, so it exerts a stronger attractive force on the electron. This stronger force pulls the electron into a tighter orbit, reducing the orbital radius by a factor of 2. More generally, for any hydrogen-like ion in the ground state, the radius is inversely proportional to the atomic number: $r_{1}=a_{\mathrm{B}}/Z$. This means that highly ionized atoms (large Z) have very small radii. For example, Li^{2+} (lithium with two electrons removed, $Z=3$) would have a ground-state radius of $a_{\mathrm{B}}/3$. The same principle applies to energy levels: the binding energy of He^{+} is proportional to $Z^{2}=4$, making it four times more tightly bound than hydrogen (54.4 eV compared to 13.6 eV for hydrogen).

A beryllium ion with a single electron (denoted Be^{3+}) is in an excited state with radius the same as that of the ground state of hydrogen.

(a) What is n for the Be^{3+} ion?

(b) How much energy in eV is needed to ionize the ion from this excited state?

Show Solution

Strategy

For hydrogen-like ions (single electron), the orbital radius is given by $r_n = n^2 Z a_B$ and the energy is $E_n = -Z^2 n^2 (13.6\text{ eV})$, where Z is the atomic number. For Be^{3+}, $Z = 4$. (a) We'll find which n gives the same radius as hydrogen's ground state ($r = a_B$). (b) We'll calculate the ionization energy from that state.

Solution

(a) The ground state of hydrogen has radius $r_1 = a_B$. For Be^{3+} with $Z = 4$:

$$r_n = n^2 Z a_B = n^2 4 a_B$$

Setting this equal to a_B:

$$n^2 4 a_B = a_B$$
$$n^2 = 4$$
$$n = 2$$

The Be^{3+} ion is in the **n = 2** state.

(b) The energy of the $n = 2$ state for Be^{3+} is:

$$E_2 = -Z^2 n^2 (13.6\text{ eV}) = -(4)^2 (2)^2 (13.6\text{ eV}) = -164 (13.6\text{ eV}) = -4 \times 13.6\text{ eV}$$
$$E_2 = -54.4\text{ eV}$$

The ionization energy is the energy needed to bring the electron from $E_2 = -54.4\text{ eV}$ to $E = 0$:

$$\text{Ionization energy} = 0 - (-54.4\text{ eV}) = 54.4\text{ eV}$$

The ionization energy is **54.4 eV**.

Discussion

This problem beautifully illustrates how the Z^2 scaling affects both orbital radii and energies in hydrogen-like ions. The Be^{3+} ion has $Z = 4$, so its orbitals are compressed by a factor of $Z = 4$ (radius scales as $1/Z$) compared to hydrogen at the same n. To match hydrogen's ground state radius, the Be^{3+} electron must be in $n = 2$, which is its first excited state. However, despite having the same radius as hydrogen's ground state, the Be^{3+} ion in $n = 2$ is much more tightly bound—it requires 54.4 eV to ionize, exactly 4 times the 13.6 eV needed for hydrogen's ground state (energy scales as Z^2). This shows that nuclear charge dominates the binding energy. Be^{3+} is created in very hot environments like stellar interiors or in high-energy laboratory plasmas. The high ionization energy of 54.4 eV (even from an excited state!) means that only very energetic photons (in the soft X-ray region, around 23 nm wavelength) or very energetic collisions can ionize it. For comparison, the ground state ($n = 1$) of Be^{3+} would require 217.6 eV to ionize, placing it in the extreme UV to soft X-ray spectral region.

Atoms can be ionized by thermal collisions, such as at the high temperatures found in the solar corona. One such ion is C^{+5}, a carbon atom with only a single electron.

(a) By what factor are the energies of its hydrogen-like levels greater than those of hydrogen?

(b) What is the wavelength of the first line in this ion's Paschen series?

(c) What type of EM radiation is this?

Show Solution

Strategy

The ion C^{+5} is a carbon atom with 5 electrons removed, leaving only one electron. Since carbon has 6 protons, $Z = 6$ for this hydrogen-like ion. For hydrogen-like atoms, the energy levels are given by:

$$E_n = -Z^2 n^2 E_0$$

where $E_0 = 13.6\text{ eV}$ is the ground-state energy of hydrogen. The energy of a photon emitted during a transition is $\Delta E = E_i - E_f$, which can be related to wavelength using $E = hc/\lambda$.

Solution

(a) The energy levels of C^{+5} are greater than those of hydrogen by a factor of Z^2:

$\text{Factor} = Z^2 = 6^2 = 36$

The energies are **36 times greater** than those of hydrogen.

(b) For the Paschen series, $n_f = 3$. The first line corresponds to a transition from $n_i = 4$ to $n_f = 3$. The energy of the emitted photon is:

$$\Delta E = E_4 - E_3 = -\frac{Z^2 E_0}{n_i^2} - \left(-\frac{Z^2 E_0}{n_f^2}\right) = Z^2 E_0 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$
$$\Delta E = (36)(13.6\ \text{eV})\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = (489.6\ \text{eV})\left(\frac{1}{9} - \frac{1}{16}\right)$$
$$\Delta E = (489.6\ \text{eV})\left(\frac{16-9}{144}\right) = (489.6\ \text{eV})\left(\frac{7}{144}\right) = 23.8\ \text{eV}$$

Converting to wavelength using $\lambda = \frac{hc}{\Delta E}$ and $hc = 1240\ \text{eV}\cdot\text{nm}$:

$$\lambda = \frac{1240\ \text{eV}\cdot\text{nm}}{23.8\ \text{eV}} = 52.1\ \text{nm}$$

(c) A wavelength of 52.1 nm is in the **ultraviolet (UV)** region of the electromagnetic spectrum, specifically in the extreme ultraviolet (EUV) or far-UV range.

Discussion

This problem illustrates how hydrogen-like ions with high atomic numbers produce much higher energy transitions than hydrogen itself. The factor of 36 increase in energy (due to $Z^2 = 36$) means that what would be an infrared transition in hydrogen (the Paschen series) becomes an extreme ultraviolet transition in C^{+5} . For comparison, the first line of the Paschen series in hydrogen has a wavelength of 1875 nm (infrared), while the same transition in C^{+5} has a wavelength 36 times shorter: 52.1 nm (extreme UV). This is exactly what we expect since $\lambda \propto 1/(\Delta E) \propto 1/Z^2$. Such highly ionized carbon atoms are found in the solar corona and other high-temperature astrophysical plasmas where temperatures exceed one million Kelvin. The extreme UV radiation from these ions is important for understanding solar physics and space weather, though it is absorbed by Earth's atmosphere and must be observed from space-based telescopes.

Verify Equations $r_n = \frac{n^2}{Z} a_B$ and $a_B = \frac{h^2}{4\pi^2 m_e k q_e^2} = 0.529 \times 10^{-10}\,\text{m}$ using the approach stated in the text. That is, equate the Coulomb and centripetal forces and then insert an expression for velocity from the condition for angular momentum quantization.

Show Solution

Strategy

We'll follow Bohr's approach by: (1) equating the electrostatic Coulomb force to the centripetal force needed to keep the electron in circular orbit, and (2) using the quantization condition for angular momentum $L = m_e v r_n = \frac{nh}{2\pi}$ to eliminate the velocity. This will yield expressions for r_n in terms of fundamental constants and quantum number n .

Solution

Starting with the force balance equation, the Coulomb attractive force equals the centripetal force:

$$\frac{kZq_e^2}{r_n^2} = \frac{m_e v^2}{r_n}$$

Multiplying both sides by r_n and rearranging:

$$kZq_e^2 = m_e v^2 r_n$$
$$r_n = \frac{kZq_e^2}{m_e v^2}$$

From Bohr's quantization condition:

$$m_e v r_n = \frac{nh}{2\pi}$$

Solving for v :

$$v = \frac{nh}{2\pi m_e r_n}$$

Substituting this expression for v into our equation for r_n :

$$r_n = \frac{kZq_e^2}{m_e} \cdot \frac{1}{v^2} = \frac{kZq_e^2}{m_e} \cdot \frac{4\pi^2 m_e^2 r_n^2}{n^2 h^2}$$

Simplifying:

$$r_n = \frac{kZq_e^2 \cdot 4\pi^2 m_e r_n^2}{m_e n^2 h^2}$$

$$1=\frac{kZq_e^2\cdot 4\pi^2 m_e r_n}{n^2h^2}$$
$$r_n=\frac{n^2h^2}{4\pi^2 m_e kZq_e^2}=\frac{n^2}{Z}\cdot\frac{h^2}{4\pi^2 m_e k q_e^2}$$

Defining the Bohr radius as:

$$a_{\mathrm{B}}=\frac{h^2}{4\pi^2 m_e k q_e^2}=0.529\times10^{-10}\text{ m}$$

We obtain the general formula:

$$r_n=\frac{n^2}{Z}a_{\mathrm{B}}$$

This verifies both equations.

Discussion

This derivation is at the heart of Bohr's atomic model and represents a brilliant synthesis of classical and quantum ideas. The force balance equation (Coulomb force = centripetal force) is pure classical mechanics, while the quantization condition $L=\frac{nh}{2\pi}$ introduces quantum mechanics. The resulting formula $r_n=\frac{n^2}{Z}a_{\mathrm{B}}$ has profound implications: (1) Orbital radii are quantized and scale as n^2, explaining why higher energy states are much larger; (2) For hydrogen-like ions, the radius scales as $1/Z$, meaning higher nuclear charge squeezes the electron orbits inward; (3) The Bohr radius a_{B} emerges as a natural length scale built from fundamental constants, representing the ground state size of hydrogen. At the time (1913), this theory successfully explained the Rydberg formula for hydrogen spectral lines and predicted the radii of excited states. Although superseded by full quantum mechanics, Bohr's model remains pedagogically valuable and gives surprisingly accurate results for hydrogen and hydrogen-like ions. The approach also introduced the revolutionary idea that quantum numbers (like n) determine discrete atomic properties—a concept that became central to all of quantum physics.

The wavelength of the four Balmer series lines for hydrogen are found to be 410.3, 434.2, 486.3, and 656.5 nm. What average percentage difference is found between these wavelength numbers and those predicted by $\frac{1}{\lambda}=R\left(\frac{1}{n_{\mathrm{f}}^2}-\frac{1}{n_{\mathrm{i}}^2}\right)$? It is amazing how well a simple formula (disconnected originally from theory) could duplicate this phenomenon.

Show Solution

Strategy

The Balmer series has $n_{\mathrm{f}}=2$, and the four lines correspond to transitions from $n_i=6,5,4,3$ (in order of the given experimental wavelengths from shortest to longest). We'll calculate the theoretical wavelength for each transition using the Rydberg formula, compare with the experimental values, calculate the percentage difference for each line, and then find the average.

$$\text{Percentage difference}=\left|\frac{\lambda_{\text{exp}}-\lambda_{\text{theory}}}{\lambda_{\text{theory}}}\right|\times100\%$$

Solution

For the Balmer series with $R=1.097\times10^7\text{ m}^{-1}$ and $n_f=2$:

Line 1: $n_i=6$ (experimental: 410.3 nm)

$$\frac{1}{\lambda}=(1.097\times10^7\text{ m}^{-1})\left(\frac{1}{4}-\frac{1}{36}\right)=(1.097\times10^7\text{ m}^{-1})\left(\frac{8}{36}\right)=2.438\times10^6\text{ m}^{-1}$$
$$\lambda_{\text{theory}}=410.2\text{ nm}$$
$$\text{Percentage difference}=\left|\frac{410.3-410.2}{410.2}\right|\times100\%=0.024\%$$

Line 2: $n_i=5$ (experimental: 434.2 nm)

$$\frac{1}{\lambda}=(1.097\times10^7\text{ m}^{-1})\left(\frac{1}{4}-\frac{1}{25}\right)=(1.097\times10^7\text{ m}^{-1})\left(\frac{21}{100}\right)=2.304\times10^6\text{ m}^{-1}$$
$$\lambda_{\text{theory}}=434.1\text{ nm}$$
$$\text{Percentage difference}=\left|\frac{434.2-434.1}{434.1}\right|\times100\%=0.023\%$$

Line 3: $n_i=4$ (experimental: 486.3 nm)

$$\frac{1}{\lambda}=(1.097\times10^7\text{ m}^{-1})\left(\frac{1}{4}-\frac{1}{16}\right)=(1.097\times10^7\text{ m}^{-1})\left(\frac{3}{16}\right)=2.057\times10^6\text{ m}^{-1}$$
$$\lambda_{\text{theory}}=486.1\text{ nm}$$
$$\text{Percentage difference}=\left|\frac{486.3-486.1}{486.1}\right|\times100\%=0.041\%$$

Line 4: $n_i=3$ (experimental: 656.5 nm)

$$\frac{1}{\lambda}=(1.097\times10^{7}\ \text{m}^{-1})\left(\frac{1}{4}-\frac{1}{9}\right)=(1.097\times10^{7}\ \text{m}^{-1})\left(\frac{5}{36}\right)=1.524\times10^{6}\ \text{m}^{-1}$$
$$\lambda_{\text{theory}}=656.3\ \text{nm}$$
$$\text{Percentage difference}=\left|\frac{656.5-656.3}{656.3}\right|\times100\%=0.030\%$$

Average percentage difference:

$$\text{Average}=\frac{0.024\%+0.023\%+0.041\%+0.030\%}{4}=\frac{0.118\%}{4}=0.030\%$$

The average percentage difference is approximately **0.03%** or about **0.3 parts per thousand**.

Discussion

This remarkably small average percentage difference (only about 0.03%) demonstrates the extraordinary accuracy of the Rydberg formula in predicting hydrogen spectral lines. The fact that such a simple empirical formula—originally developed by Balmer and Rydberg before the development of quantum mechanics—could predict wavelengths to within a few hundredths of a percent is truly remarkable. This agreement provided strong evidence that there was a fundamental physical principle underlying atomic spectra, even before Bohr developed his quantum theory of the hydrogen atom. The small discrepancies that do exist (all less than 0.05%) can be attributed to several factors: experimental uncertainties in measuring wavelengths, the finite mass of the nucleus (which we assumed to be infinite in the simple Bohr model), and relativistic corrections for the electron's motion. When Bohr showed that his quantum theory could derive the Rydberg formula from first principles and explain *why* it worked so well, it was a watershed moment in physics that validated the quantum approach to atomic structure.

Glossary

hydrogen spectrum wavelengths
: the wavelengths of visible light from hydrogen; can be calculated by $\frac{1}{\lambda}=R\left(\frac{1}{n_{\text{f}}^{2}}-\frac{1}{n_{\text{i}}^{2}}\right)$

Rydberg constant
: a physical constant related to the atomic spectra with an established value of $1.097\times10^{7}\,\text{m}^{-1}$

double-slit interference
: an experiment in which waves or particles from a single source impinge upon two slits so that the resulting interference pattern may be observed

energy-level diagram
: a diagram used to analyze the energy level of electrons in the orbits of an atom

Bohr radius
: the mean radius of the orbit of an electron around the nucleus of a hydrogen atom in its ground state

hydrogen-like atom
: any atom with only a single electron

energies of hydrogen-like atoms
: Bohr formula for energies of electron states in hydrogen-like atoms:

$$E_{n}=-\frac{Z^{2}}{n^{2}}E_{0}\ (n=1,\ 2,\ 3,\ldots)$$

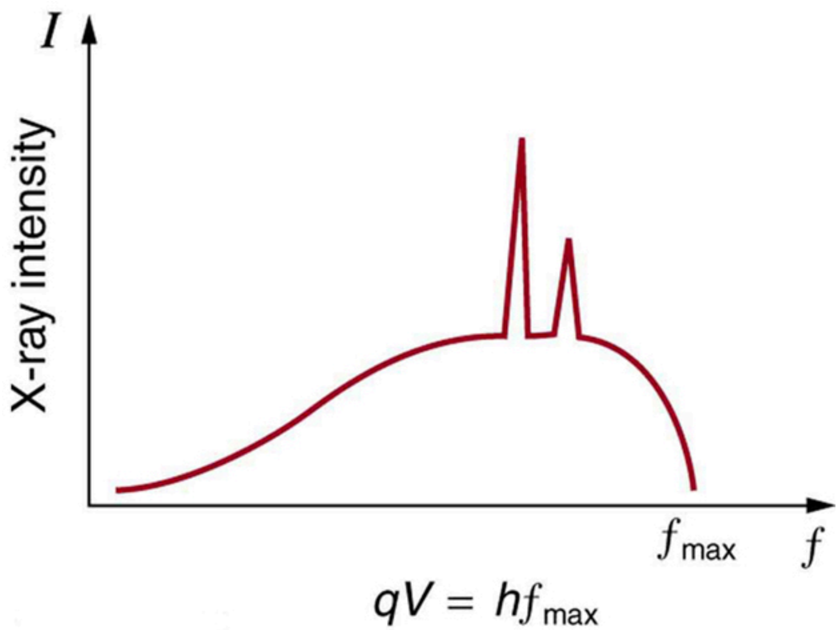
X Rays: Atomic Origins and Applications

- Define x-ray tube and its spectrum.
- Show the x-ray characteristic energy.
- Specify the use of X-rays in medical observations.
- Explain the use of X-rays in CT scanners in diagnostics.

Each type of atom (or element) has its own characteristic electromagnetic spectrum. **X-rays** lie at the high-frequency end of an atom's spectrum and are characteristic of the atom as well. In this section, we explore characteristic X-rays and some of their important applications.

We have previously discussed X-rays as a part of the electromagnetic spectrum in Photon Energies and the Electromagnetic Spectrum. That module illustrated how an x-ray tube (a specialized CRT) produces X-rays. Electrons emitted from a hot filament are accelerated with a high voltage, gaining significant kinetic energy and striking the anode.

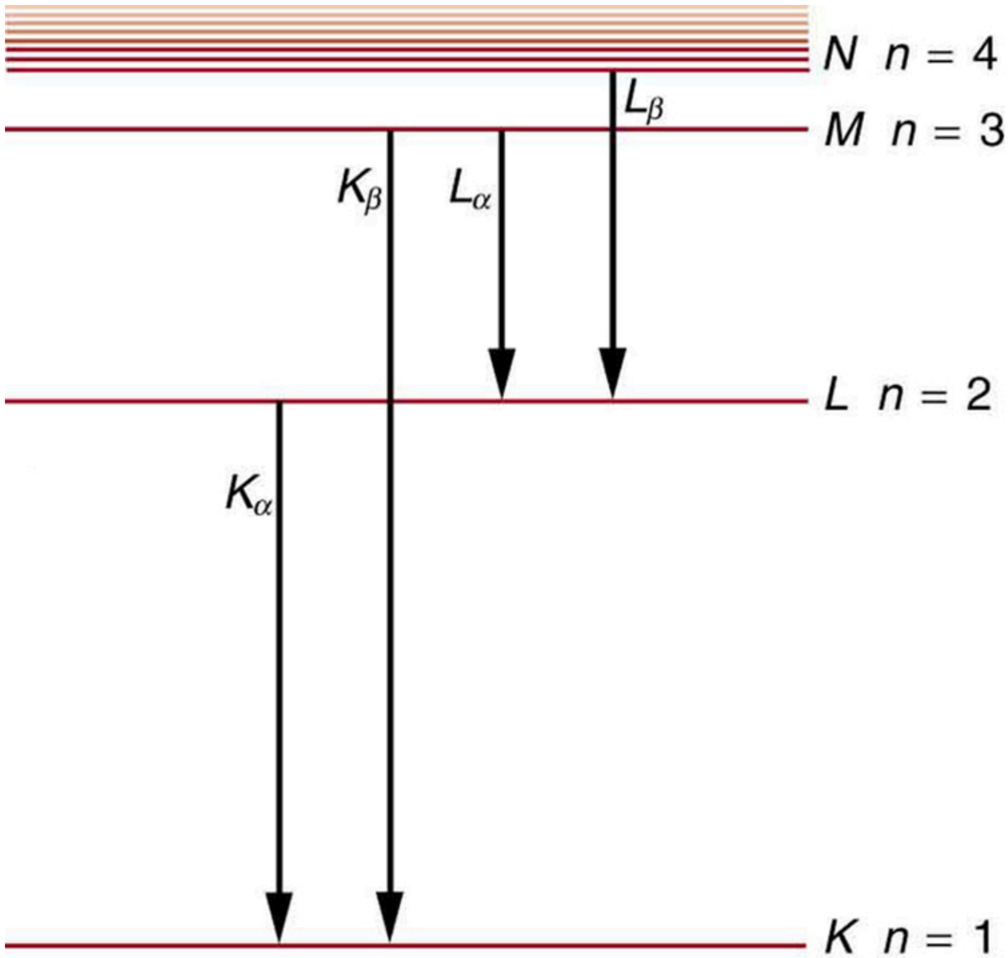
There are two processes by which X-rays are produced in the anode of an x-ray tube. In one process, the deceleration of electrons produces X-rays, and these x rays are called *bremsstrahlung*, or braking radiation. The second process is atomic in nature and produces *characteristic X-rays*, so called because they are characteristic of the anode material. The x-ray spectrum in [Figure 1] is typical of what is produced by an x-ray tube, showing a broad curve of bremsstrahlung radiation with characteristic x-ray peaks on it.



X-ray spectrum obtained when energetic electrons strike a material, such as in the anode of a CRT. The smooth part of the spectrum is bremsstrahlung radiation, while the peaks are characteristic of the anode material. A different anode material would have characteristic x-ray peaks at different frequencies.

The spectrum in [Figure 1] is collected over a period of time in which many electrons strike the anode, with a variety of possible outcomes for each hit. The broad range of x-ray energies in the bremsstrahlung radiation indicates that an incident electron's energy is not usually converted entirely into photon energy. The highest-energy X-ray produced is one for which all of the electron's energy was converted to photon energy. Thus the accelerating voltage and the maximum x-ray energy are related by conservation of energy. Electric potential energy is converted to kinetic energy and then to photon energy, so that $E_{\text{max}} = hf_{\text{max}} = q_e V$. Units of electron volts are convenient. For example, a 100-kV accelerating voltage produces x-ray photons with a maximum energy of 100 keV.

Some electrons excite atoms in the anode. Part of the energy that they deposit by collision with an atom results in one or more of the atom's inner electrons being knocked into a higher orbit or the atom being ionized. When the anode's atoms de-excite, they emit characteristic electromagnetic radiation. The most energetic of these are produced when an inner-shell vacancy is filled—that is, when an $n = 1$ or $n = 2$ shell electron has been excited to a higher level, and another electron falls into the vacant spot. A *characteristic X-ray* (see Photon Energies and the Electromagnetic Spectrum) is electromagnetic (EM) radiation emitted by an atom when an inner-shell vacancy is filled. [Figure 2] shows a representative energy-level diagram that illustrates the labeling of characteristic X-rays. X-rays created when an electron falls into an $n = 1$ shell vacancy are called K_α when they come from the next higher level; that is, an $n = 2$ to $n = 1$ transition. The labels K,L,M,... come from the older alphabetical labeling of shells starting with K rather than using the principal quantum numbers 1, 2, 3, …. A more energetic K_β X-ray is produced when an electron falls into an $n = 1$ shell vacancy from the $n = 3$ shell; that is, an $n = 3$ to $n = 1$ transition. Similarly, when an electron falls into the $n = 2$ shell from the $n = 3$ shell, an L_α X-ray is created. The energies of these X-rays depend on the energies of electron states in the particular atom and, thus, are characteristic of that element: every element has its own set of x-ray energies. This property can be used to identify elements, for example, to find trace (small) amounts of an element in an environmental or biological sample.



A characteristic X-ray is emitted when an electron fills an inner-shell vacancy, as shown for several transitions in this approximate energy level diagram for a multiple-electron atom. Characteristic X-rays are labeled according to the shell that had the vacancy and the shell from which the electron came. A K_α X-ray, for example, is produced when an electron coming from the $n=2$ shell fills the $n=1$ shell vacancy.

Characteristic X-Ray Energy

Calculate the approximate energy of a K_α X-ray from a tungsten anode in an x-ray tube.

Strategy

How do we calculate energies in a multiple-electron atom? In the case of characteristic X-rays, the following approximate calculation is reasonable. Characteristic X-rays are produced when an inner-shell vacancy is filled. Inner-shell electrons are nearer the nucleus than others in an atom and thus feel little net effect from the others. This is similar to what happens inside a charged conductor, where its excess charge is distributed over the surface so that it produces no electric field inside. It is reasonable to assume the inner-shell electrons have hydrogen-like energies, as given by $E_n = -Z^2 n^2 E_0 (n=1,2,3,...)$. As noted, a K_α X-ray is produced by an $n=2$ ** to $n=1$ ** transition. Since there are two electrons in a filled K *_ shell, a vacancy would leave one electron, so that the effective charge would be $Z-1$ rather than Z _.* For tungsten, $Z=74$, so that the effective charge is 73.

Solution

$E_n = -Z^2 n^2 E_0 (n=1,2,3,...)$ gives the orbital energies for hydrogen-like atoms to be $E_n = -(Z^2/n^2)E_0$, where $E_0 = 13.6\text{eV}$. As noted, the effective Z is 73. Now the K_α x-ray energy is given by

$$E_{K_\alpha} = \Delta E = E_\text{i} - E_\text{f} = E_2 - E_1,$$

where

$$E_1 = -Z^2 1^2 E_0 = -73^2 1 (13.6\text{eV}) = -72.5\text{keV}$$

and

$E_2 = -Z^2 2^2 E_0 = -73^2 4 (13.6\text{eV}) = -18.1\text{keV}.$

Thus,

$E_{K_\alpha} = -18.1\ \text{keV} - (-72.5\text{keV}) = 54.4 keV.$

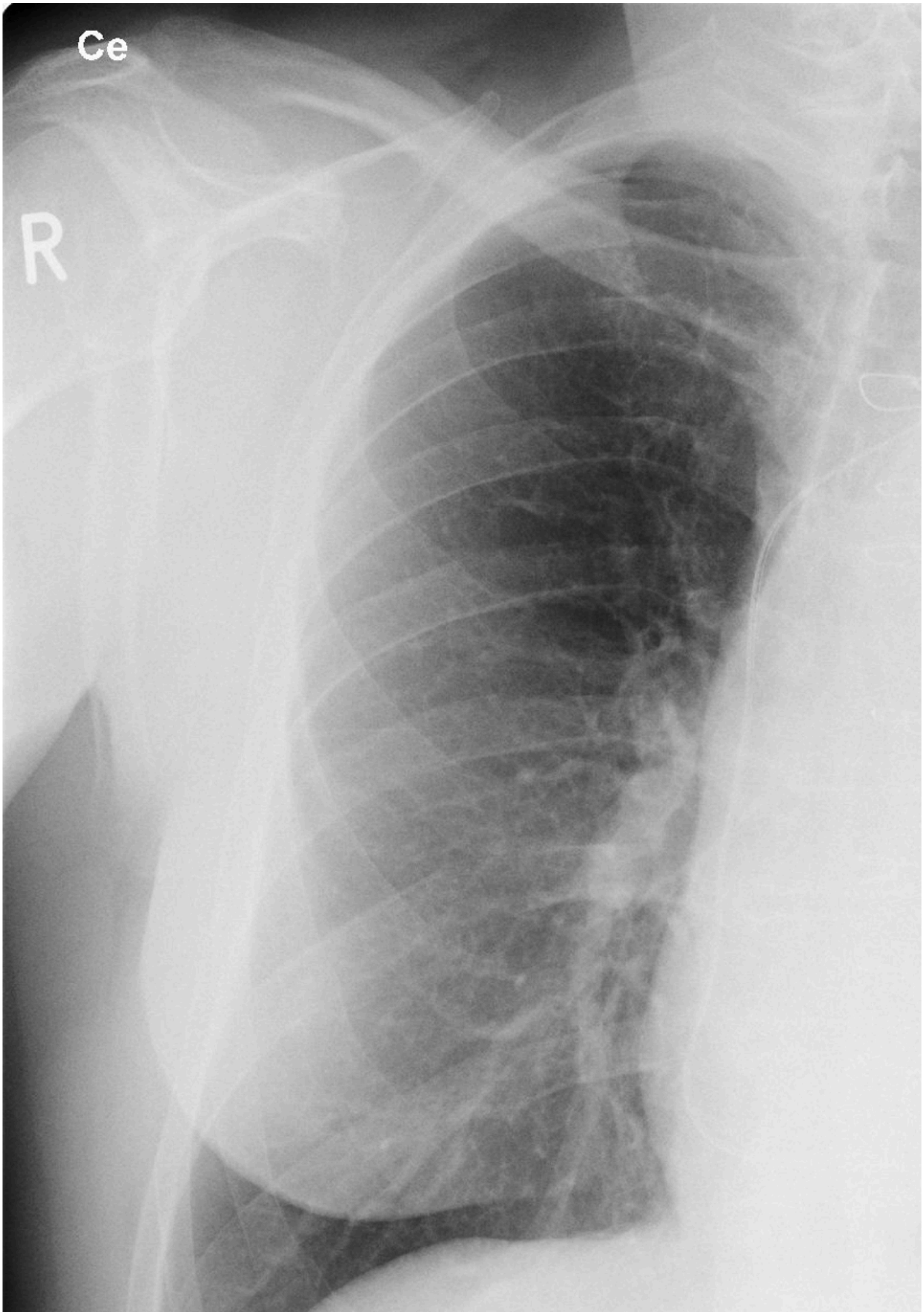
Discussion

This large photon energy is typical of characteristic X-rays from heavy elements. It is large compared with other atomic emissions because it is produced when an inner-shell vacancy is filled, and inner-shell electrons are tightly bound. Characteristic X-ray energies become progressively larger for heavier elements because their energy increases approximately as Z^2. Significant accelerating voltage is needed to create these inner-shell vacancies. In the case of tungsten, at least 72.5 kV is needed, because other shells are filled and you cannot simply bump one electron to a higher filled shell. Tungsten is a common anode material in x-ray tubes; so much of the energy of the impinging electrons is absorbed, raising its temperature, that a high-melting-point material like tungsten is required.

Medical and Other Diagnostic Uses of X-rays

All of us can identify diagnostic uses of x-ray photons. Among these are the universal dental and medical X-rays that have become an essential part of medical diagnostics. (See [Figure 4] and [Figure 5].) X rays are also used to inspect our luggage at airports, as shown in [Figure 3], and for early detection of cracks in crucial aircraft components. An X-ray is not only a noun meaning high-energy photon, it also is an image produced by X-rays, and it has been made into a familiar verb—to be x-rayed.

An x-ray image reveals fillings in a person's teeth. (credit: Dmitry G, Wikimedia Commons)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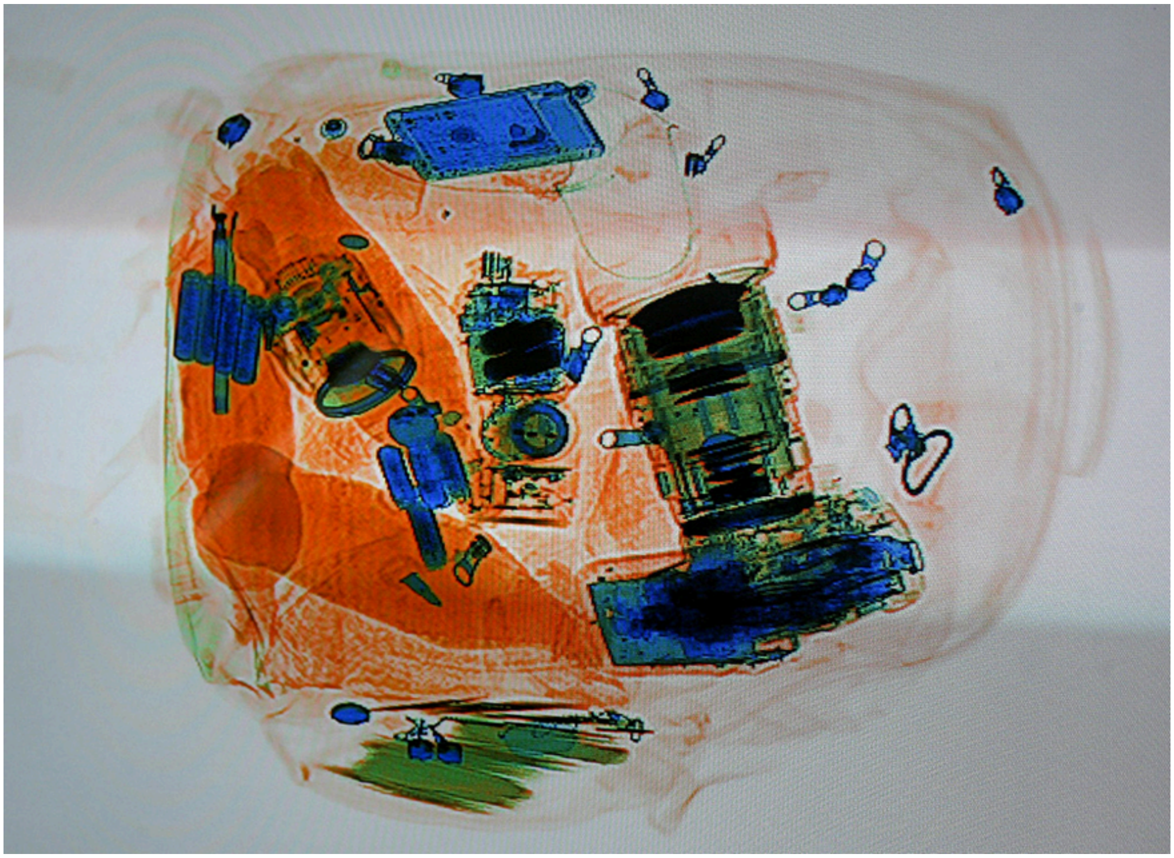
This x-ray image of a person’s chest shows many details, including an artificial pacemaker. (credit: Sunzi99, Wikimedia Commons)



This x-ray image shows the contents of a piece of luggage. The denser the material, the darker the shadow. (credit: IDuke, Wikimedia Commons)

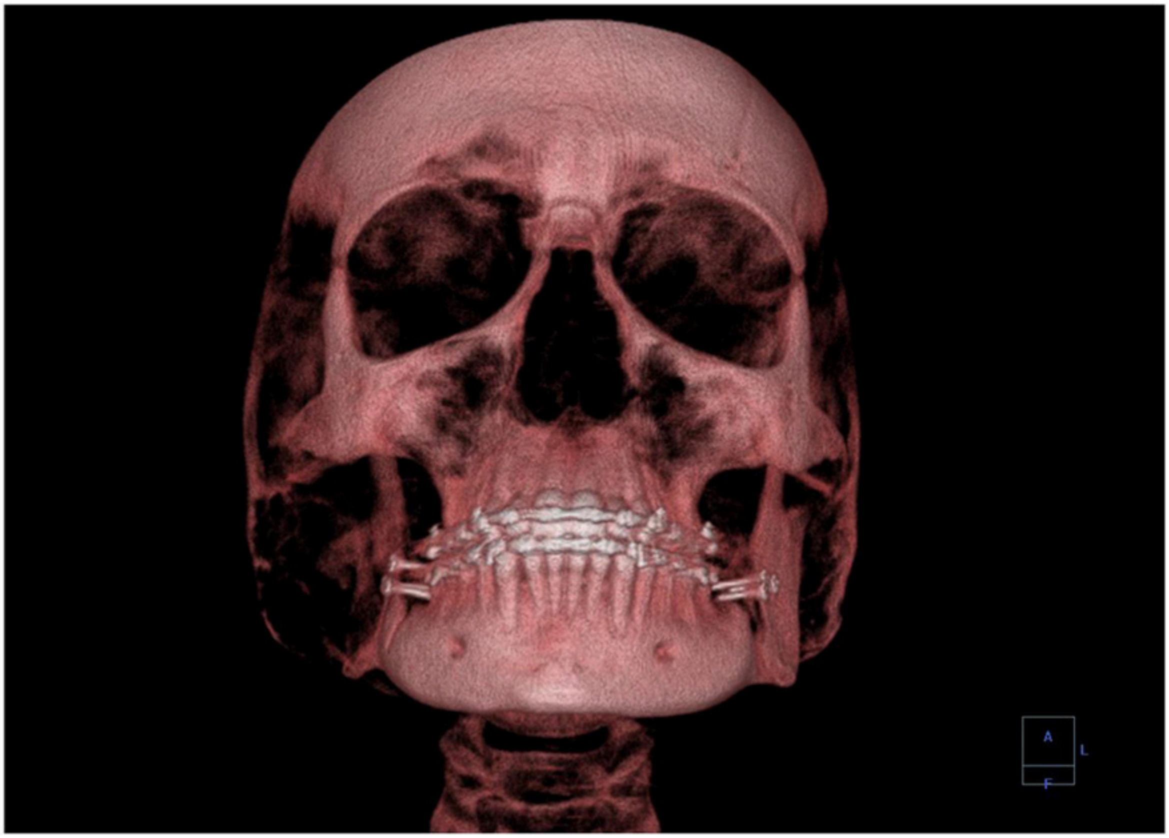
The most common x-ray images are simple shadows. Since x-ray photons have high energies, they penetrate materials that are opaque to visible light. The more energy an x-ray photon has, the more material it will penetrate. So an x-ray tube may be operated at 50.0 kV for a chest X-ray, whereas it may need to be operated at 100 kV to examine a broken leg in a cast. The depth of penetration is related to the density of the material as well as to the energy of the photon. The denser the material, the fewer x-ray photons get through and the darker the shadow. Thus X-rays excel at detecting breaks in bones and in imaging other physiological structures, such as some tumors, that differ in density from surrounding material. Because of their high photon energy, X-rays produce significant ionization in materials and damage cells in biological organisms. Modern uses minimize exposure to the patient and eliminate exposure to others. Biological effects of X-rays will be explored in the next chapter along with other types of ionizing radiation such as those produced by nuclei.

As the x-ray energy increases, the Compton effect (see [Photon Momentum]) becomes more important in the attenuation of the X-rays. Here, the X-ray scatters from an outer electron shell of the atom, giving the ejected electron some kinetic energy while losing energy itself. The probability for attenuation of the X-rays depends upon the number of electrons present (the material’s density) as well as the thickness of the material. Chemical composition of the medium, as characterized by its atomic number Z , is not important here. Low-energy X-rays provide better contrast (sharper images). However, due to greater attenuation and less scattering, they are more absorbed by thicker materials. Greater contrast can be achieved by injecting a substance with a large atomic number, such as barium or iodine. The structure of the part of the body that contains the substance (e.g., the gastro-intestinal tract or the abdomen) can easily be seen this way.

Breast cancer is the second-leading cause of death among women worldwide. Early detection can be very effective, hence the importance of x-ray diagnostics. A mammogram cannot diagnose a malignant tumor, only give evidence of a lump or region of increased density within the breast. X-ray absorption by different types of soft tissue is very similar, so contrast is difficult; this is especially true for younger women, who typically have denser breasts. For older women who are at greater risk of developing breast cancer, the presence of more fat in the breast gives the lump or tumor more contrast. MRI (Magnetic resonance imaging) has recently been used as a supplement to conventional X-rays to improve detection and eliminate false positives. The subject’s radiation dose from X-rays will be treated in a later chapter.

A standard X-ray gives only a two-dimensional view of the object. Dense bones might hide images of soft tissue or organs. If you took another X-ray from the side of the person (the first one being from the front), you would gain additional information. While shadow images are sufficient in many applications, far more sophisticated images can be produced with modern technology. [Figure 6] shows the use of a computed tomography (CT) scanner, also called computed axial tomography (CAT) scanner. X-rays are passed through a narrow section (called a slice) of the patient's body (or body part) over a range of directions. An array of many detectors on the other side of the patient registers the X-rays. The system is then rotated around the patient and another image is taken, and so on. The x-ray tube and detector array are mechanically attached and so rotate together. Complex computer image processing of the relative absorption of the X-rays along different directions produces a highly-detailed image. Different slices are taken as the patient moves through the scanner on a table. Multiple images of different slices can also be computer analyzed to produce three-dimensional information, sometimes enhancing specific types of tissue, as shown in [Figure 7]. G. Hounsfield (UK) and A. Cormack (US) won the Nobel Prize in Medicine in 1979 for their development of computed tomography.

A patient being positioned in a CT scanner aboard the hospital ship USNS Mercy. The CT scanner passes X-rays through slices of the patient's body (or body part) over a range of directions. The relative absorption of the X-rays along different directions is computer analyzed to produce highly detailed images. Three-dimensional information can be obtained from multiple slices. (credit: Rebecca Moat, U.S. Navy)

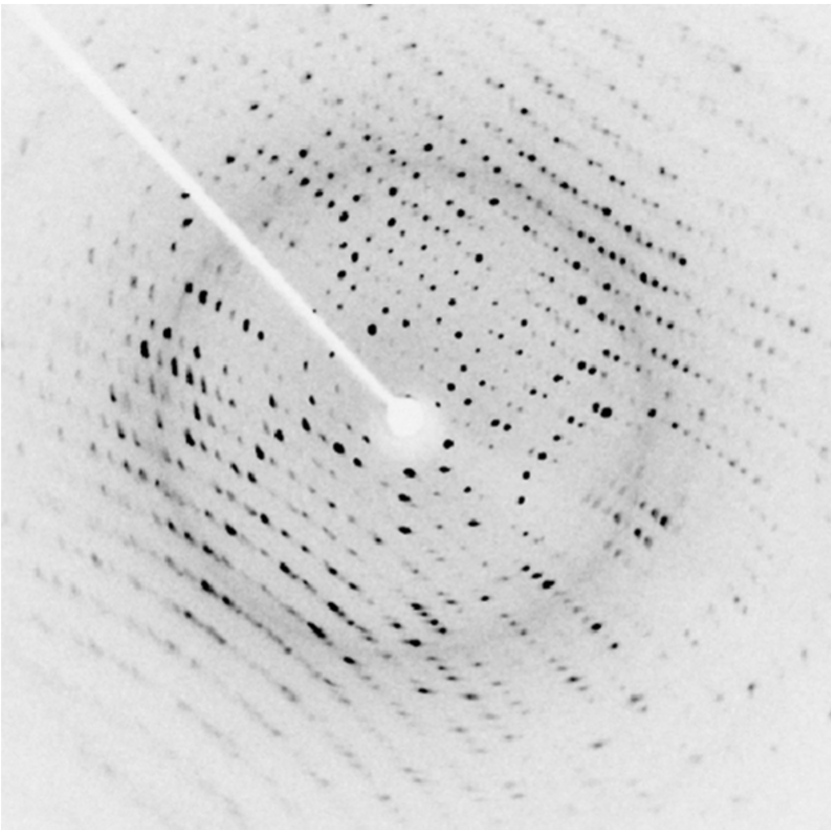


This three-dimensional image of a skull was produced by computed tomography, involving analysis of several x-ray slices of the head. (credit: Emailshankar, Wikimedia Commons)

X-Ray Diffraction and Crystallography

Since x-ray photons are very energetic, they have relatively short wavelengths. For example, the 54.4-keV K_α X-ray of [Example 1] has a wavelength $\lambda = hc/E = 0.0228 \text{nm}$. Thus, typical x-ray photons act like rays when they encounter macroscopic objects, like teeth, and produce sharp shadows; however, since atoms are on the order of 0.1 nm in size, X-rays can be used to detect the location, shape, and size of atoms and molecules. The process is called **x-ray diffraction**, because it involves the diffraction and interference of X-rays to produce patterns that can be analyzed for information about the structures that scattered the X-rays. Perhaps the most famous example of x-ray diffraction is the discovery of the double-helix structure of DNA in 1953 by an international team of scientists working at the Cavendish Laboratory—American James Watson, Englishman Francis Crick, and New Zealand–born Maurice Wilkins. Using x-ray diffraction data produced by Rosalind Franklin, they were the first to discern the structure of DNA that is so crucial to life. For this, Watson, Crick, and Wilkins were awarded the 1962 Nobel Prize in Physiology or Medicine. There is much debate and controversy over the issue that Rosalind Franklin was not included in the prize.

[Figure 8] shows a diffraction pattern produced by the scattering of X-rays from a crystal. This process is known as x-ray crystallography because of the information it can yield about crystal structure, and it was the type of data Rosalind Franklin supplied to Watson and Crick for DNA. Not only do X-rays confirm the size and shape of atoms, they give information on the atomic arrangements in materials. For example, current research in high-temperature superconductors involves complex materials whose lattice arrangements are crucial to obtaining a superconducting material. These can be studied using x-ray crystallography.



X-ray diffraction from the crystal of a protein, hen egg lysozyme, produced this interference pattern. Analysis of the pattern yields information about the structure of the protein. (credit: Del45, Wikimedia Commons)

Historically, the scattering of X-rays from crystals was used to prove that x rays are energetic EM waves. This was suspected from the time of the discovery of X-rays in 1895, but it was not until 1912 that the German Max von Laue (1879–1960) convinced two of his colleagues to scatter X-rays from crystals. If a diffraction pattern is obtained, he reasoned, then the X-rays must be waves, and their wavelength could be determined. (The spacing of atoms in various crystals was reasonably well known at the time, based on good values for Avogadro's number.) The experiments were convincing, and the 1914 Nobel Prize in Physics was given to von Laue for his suggestion leading to the proof that X-rays are EM waves. In 1915, the unique father-and-son team of Sir William Henry Bragg and his son Sir William Lawrence Bragg were awarded a joint Nobel Prize for inventing the x-ray spectrometer and the then-new science of x-ray analysis. The elder Bragg had migrated to Australia from England just after graduating in mathematics. He learned physics and chemistry during his career at the University of Adelaide. The younger Bragg was born in Adelaide but went back to the Cavendish Laboratories in England to a career in x-ray and neutron crystallography; he provided support for Watson, Crick, and Wilkins for their work on unraveling the mysteries of DNA and to Max Perutz for his 1962 Nobel Prize-winning work on the structure of hemoglobin. Here again, we witness the enabling nature of physics—establishing instruments and designing experiments as well as solving mysteries in the biomedical sciences.

Certain other uses for X-rays will be studied in later chapters. X-rays are useful in the treatment of cancer because of the inhibiting effect they have on cell reproduction. X-rays observed coming from outer space are useful in determining the nature of their sources, such as neutron stars and possibly black holes. Created in nuclear bomb explosions, X-rays can also be used to detect clandestine atmospheric tests of these weapons. X-rays can cause excitations of atoms, which then fluoresce (emitting characteristic EM radiation), making x-ray-induced fluorescence a valuable analytical tool in a range of fields from art to archaeology.

Section Summary

- X-rays are relatively high-frequency EM radiation. They are produced by transitions between inner-shell electron levels, which produce X-rays characteristic of the atomic element, or by decelerating electrons.
- X-rays have many uses, including medical diagnostics and x-ray diffraction.

Conceptual Questions

Explain why characteristic X-rays are the most energetic in the EM emission spectrum of a given element.

Why does the energy of characteristic X-rays become increasingly greater for heavier atoms?

Observers at a safe distance from an atmospheric test of a nuclear bomb feel its heat but receive none of its copious X-rays. Why is air opaque to X-rays but transparent to infrared?

Lasers are used to burn and read CDs. Explain why a laser that emits blue light would be capable of burning and reading more information than one that emits infrared.

Crystal lattices can be examined with X-rays but not UV. Why?

CT scanners do not detect details smaller than about 0.5 mm. Is this limitation due to the wavelength of X-rays? Explain.

Problem Exercises

(a) What is the shortest-wavelength x-ray radiation that can be generated in an x-ray tube with an applied voltage of 50.0 kV? (b) Calculate the photon energy in eV. (c) Explain the relationship of the photon energy to the applied voltage.

Show Solution

Strategy

(a) The shortest-wavelength X-ray is produced when all of an electron's kinetic energy is converted into a single photon. When an electron is accelerated through a potential difference V, it gains kinetic energy $KE = eV$. The maximum photon energy is $E_{max} = hf_{max} = hc\lambda_{min}$. Setting $eV = hc\lambda_{min}$ and solving for λ_{min}, we can use the convenient relationship $hc = 1240 \text{ eV·nm}$.

(b) The maximum photon energy equals the kinetic energy gained by the electron.

(c) We'll examine the relationship between the accelerating voltage and the photon energy.

Solution

(a) The maximum photon energy equals the electron's kinetic energy:

$$E_{max}=eV=50.0 \text{ keV}=50{,}000 \text{ eV}$$

Using $E = hc\lambda$ and $hc = 1240 \text{ eV·nm}$:

$$\lambda_{min}=hcE_{max}=1240 \text{ eV·nm}50{,}000 \text{ eV}=0.0248 \text{ nm}$$
$$\lambda_{min}=0.0248 \text{ nm}=0.248\times10^{-10} \text{ m}=2.48\times10^{-11} \text{ m}$$

(b) The maximum photon energy is:

$$E_{max}=50.0 \text{ keV}=50{,}000 \text{ eV}$$

(c) The relationship between photon energy and applied voltage is direct and simple:

$$E_{photon}(\text{in eV})=e\times V(\text{in volts})$$

Since the elementary charge $e = 1$ electron charge, the numerical value of the energy in electron volts equals the numerical value of the voltage in volts. For example, 50.0 kV produces photons with a maximum energy of 50.0 keV. This makes electron volts a particularly convenient unit for X-ray work.

Discussion

This problem illustrates the production of bremsstrahlung (braking radiation) X-rays. When high-speed electrons strike the anode of an X-ray tube, they decelerate rapidly, and some convert all their kinetic energy into a single photon. The wavelength of 0.0248 nm is about 20,000 times shorter than visible light (which has wavelengths of 400-700 nm), placing it well into the X-ray region of the electromagnetic spectrum.

The direct relationship between accelerating voltage and maximum photon energy ($E = eV$) is one of the most straightforward examples of energy conversion in physics. It reflects conservation of energy: electrical potential energy is converted to kinetic energy of the electron, which is then converted to photon energy. This relationship has important practical implications: (1) To produce higher-energy (more penetrating) X-rays, simply increase the tube voltage; (2) The voltage sets a sharp upper limit on photon energy—no X-ray can have more energy than eV ; (3) Most X-rays produced have lower energies because electrons typically don't convert all their energy to a single photon.

In actual X-ray spectra, the continuous bremsstrahlung spectrum shows all energies from near zero up to this maximum, with characteristic X-ray peaks superimposed. The 50 kV accelerating voltage used here is typical for medical chest X-rays, providing enough penetration to image through soft tissue while still being absorbed differently by bones. Higher voltages (100-150 kV) are used for thicker body parts or to examine denser materials.

A color television tube also generates some X-rays when its electron beam strikes the screen. What is the shortest wavelength of these X-rays, if a 30.0-kV potential is used to accelerate the electrons? (Note that TVs have shielding to prevent these X-rays from exposing viewers.)

Show Solution

Strategy

The shortest-wavelength X-ray is produced when all of the electron's kinetic energy is converted into a single photon. The electron gains kinetic energy equal to $KE = q_eV$ when accelerated through a potential difference V. This energy equals the maximum photon energy: $E_{max} = hf_{max} = hc\lambda_{min}$. We can use the convenient relationship $hc = 1240 \text{ eV·nm}$ to find the wavelength.

Solution

The maximum photon energy equals the kinetic energy gained by the electron:

$$E_{max}=q_eV=30.0\text{ keV}=30{,}000\text{ eV}$$

Using $E=\frac{hc}{\lambda}$ and $hc=1240\text{ eV}\cdot\text{nm}$:

$$\lambda_{min}=\frac{hc}{E_{max}}=\frac{1240\text{ eV}\cdot\text{nm}}{30{,}000\text{ eV}}=0.0413\text{ nm}=4.13\times10^{-11}\text{ m}$$

The shortest wavelength is **0.0413 nm** or **4.13 × 10^{-11} m**.

Discussion

This wavelength is characteristic of the bremsstrahlung (braking radiation) X-rays produced when electrons are rapidly decelerated upon striking the screen material. The 30 kV accelerating voltage in old CRT televisions was sufficient to produce X-rays, which is why TV manufacturers included lead shielding in the glass screen and around the tube. The X-ray energy of 30 keV is relatively low compared to medical X-rays (which typically use 50-150 kV), but prolonged exposure could still pose a health risk, hence the shielding requirements. Modern flat-panel displays (LCD, LED, OLED) do not produce X-rays because they don't use high-voltage electron beams. The inverse relationship between accelerating voltage and minimum wavelength ($\lambda_{min}=\frac{hc}{eV}$) is fundamental to X-ray production: higher voltages produce shorter wavelengths (more penetrating X-rays). This wavelength of 0.0413 nm is about 1000 times shorter than visible light wavelengths (400-700 nm), placing it firmly in the X-ray region of the electromagnetic spectrum. The fact that we can calculate this precisely from just the accelerating voltage demonstrates the direct connection between electrical energy and photon energy in quantum mechanics.

An X-ray tube has an applied voltage of 100 kV. (a) What is the most energetic x-ray photon it can produce? Express your answer in electron volts and joules. (b) Find the wavelength of such an X–ray.

Show Solution

Strategy

(a) The most energetic X-ray photon is produced when an electron converts all its kinetic energy (gained from the applied voltage) into a single photon. The energy in eV equals the accelerating voltage, and we can convert to joules using $1\text{ eV}=1.60\times10^{-19}\text{ J}$.

(b) We'll use the relationship $E=\frac{hc}{\lambda}$ to find the wavelength, using either $hc=1240\text{ eV}\cdot\text{nm}$ or SI units with $h=6.626\times10^{-34}\text{ J}\cdot\text{s}$ and $c=3.00\times10^{8}\text{ m/s}$.

Solution

(a) The maximum photon energy equals the electron's kinetic energy after being accelerated through 100 kV:

$$E_{max}=eV=100\text{ keV}=100\times10^{3}\text{ eV}$$

Converting to joules:

$$E_{max}=(100\times10^{3}\text{ eV})\times(1.60\times10^{-19}\text{ J/eV})=1.60\times10^{-14}\text{ J}$$

The maximum energy is **100 keV** or **1.60 × 10^{-14} J**.

(b) Using $E=\frac{hc}{\lambda}$ with $hc=1240\text{ eV}\cdot\text{nm}$:

$$\lambda=\frac{hc}{E}=\frac{1240\text{ eV}\cdot\text{nm}}{100\times10^{3}\text{ eV}}=\frac{1240}{100{,}000}\text{ nm}=0.0124\text{ nm}$$
$$\lambda=0.0124\text{ nm}=1.24\times10^{-11}\text{ m}=0.124\times10^{-10}\text{ m}$$

The wavelength is **0.0124 nm** or **1.24 × 10^{-11} m**.

Discussion

The 100 kV accelerating voltage is typical for medical diagnostic X-rays, particularly for imaging denser body parts (like examining a broken leg in a cast, as mentioned in the chapter) or for industrial radiography. The resulting X-rays with wavelength 0.0124 nm are quite penetrating—about half the wavelength of the 50 kV X-rays calculated in the previous problem, which makes them more energetic and able to penetrate through denser materials.

To put this wavelength in perspective: it's about 40,000 times shorter than visible light, about 10 times smaller than the size of a typical atom (0.1 nm), and comparable to the spacing between atoms in a crystal. This last fact is crucial for X-ray crystallography—X-rays with wavelengths comparable to atomic spacings can be diffracted by crystals, revealing atomic arrangements.

The energy of 100 keV (or $1.60\times10^{-14}\text{ J}$) might seem tiny in everyday terms, but it's enormous on the atomic scale. This is about 7 million times the ionization energy of hydrogen (13.6 eV). Such energetic photons can easily ionize atoms by knocking out inner-shell electrons, which is why X-rays are classified as ionizing radiation and why they can damage biological tissue. The biological effects of X-rays depend on both the dose (total energy deposited) and the photon energy—higher energy photons penetrate deeper but are less likely to be absorbed per unit thickness. Modern medical imaging carefully balances the need for good image quality (requiring sufficient X-ray penetration and dose) against minimizing radiation exposure to the patient. This optimization has led to dramatic reductions in medical radiation doses while improving image quality through better detectors and image processing.

The maximum characteristic x-ray photon energy comes from the capture of a free electron into a K shell vacancy. What is this photon energy in keV for tungsten, assuming the free electron has no initial kinetic energy?

Show Solution

Strategy

When a free electron (with zero initial kinetic energy) is captured into a K shell vacancy, it transitions from essentially $n=\infty$ (unbound state with $E=0$) to $n=1$ (K shell). The energy of the emitted photon equals the magnitude of the K shell binding energy. For hydrogen-like atoms, the K shell energy is $E_1 = -\frac{Z^2}{1^2}E_0$, where $E_0 = 13.6\text{ eV}$. However, for inner-shell electrons in heavy atoms, we must account for screening by using an effective charge $Z_{eff} = Z - 1$ because one electron remains in the K shell. For tungsten, $Z = 74$.

Solution

For tungsten with $Z = 74$, the effective nuclear charge for a K-shell electron is approximately $Z_{eff} = Z - 1 = 73$ (accounting for the other K-shell electron’s screening).

The K-shell binding energy is:

$$E_K = -\frac{Z_{eff}^2}{n^2}E_0 = -\frac{(73)^2}{1^2}(13.6\text{ eV}) = -(5329)(13.6\text{ eV}) = -72{,}474\text{ eV}$$

Converting to keV:

$$E_K = -72.5\text{ keV}$$

When a free electron (at $E = 0$) is captured into this K-shell state, the energy of the emitted photon is:

$$E_{photon} = E_{initial} - E_{final} = 0 - (-72.5\text{ keV}) = 72.5\text{ keV}$$

The maximum characteristic X-ray photon energy for tungsten is **72.5 keV**.

Discussion

This energy represents the absolute maximum characteristic X-ray that tungsten can emit—it occurs when a completely free electron (from outside the atom) falls directly into a K-shell vacancy. This is slightly higher than the K_α X-ray energy of 54.4 keV calculated in Example 1 of this chapter, which corresponds to an $n = 2$ to $n = 1$ transition (an electron from the L shell filling a K shell vacancy). The difference of about 18 keV represents the binding energy of the L shell. This 72.5 keV result matches the K-edge absorption energy for tungsten—the minimum photon energy needed to knock a K-shell electron completely out of the atom. The high energy of this X-ray (72.5 keV) explains why tungsten is used as the anode material in X-ray tubes: it can produce very penetrating X-rays suitable for medical imaging. The calculation also explains why an X-ray tube operating at less than 72.5 kV cannot produce K-shell characteristic X-rays from tungsten—the accelerating voltage must be at least this high to create K-shell vacancies in the first place. The Z^2 dependence of the binding energy (and thus the characteristic X-ray energy) means that heavier elements produce much more energetic characteristic X-rays: for example, lead ($Z = 82$) produces K-shell X-rays near 88 keV, while copper ($Z = 29$) produces them near 9 keV.

What are the approximate energies of the K_α and K_β X-rays for copper?

Show Solution

Strategy

Characteristic X-rays are produced when inner-shell vacancies are filled. A K_α X-ray results from an $n = 2$ to $n = 1$ transition (L to K shell), while a K_β X-ray comes from an $n = 3$ to $n = 1$ transition (M to K shell). We can use the hydrogen-like atom approximation with an effective nuclear charge $Z_{eff} = Z - 1$ to account for the screening by the remaining K-shell electron. For copper, $Z = 29$. The energy levels are $E_n = -\frac{Z_{eff}^2}{n^2}E_0$ where $E_0 = 13.6\text{ eV}$, and the photon energy is $E_{photon} = E_{initial} - E_{final}$.

Solution

For copper with $Z = 29$, the effective nuclear charge experienced by a K-shell electron is approximately:

$$Z_{eff} = Z - 1 = 29 - 1 = 28$$

The energy levels are:

$$E_n = -\frac{Z_{eff}^2}{n^2}E_0 = -\frac{(28)^2}{n^2}(13.6\text{ eV}) = -\frac{784}{n^2}(13.6\text{ eV})$$

For $n = 1$ (K shell):

$$E_1 = -\frac{784}{1}(13.6\text{ eV}) = -10{,}662\text{ eV} = -10.7\text{ keV}$$

For n =2 (L shell):

$$E_2=-7844(13.6\text{ eV})=-2{,}666\text{ eV}=-2.67\text{ keV}$$

For n =3 (M shell):

$$E_3=-7849(13.6\text{ eV})=-1{,}185\text{ eV}=-1.19\text{ keV}$$

For K-alpha (n =2 → n =1):

$$E_{K\alpha}=E_2-E_1=(-2.67)-(-10.7)=8.03\text{ keV}$$

For K-beta (n =3 → n =1):

$$E_{K\beta}=E_3-E_1=(-1.19)-(-10.7)=9.51\text{ keV}$$

Rounding to three significant figures:

- **K-alpha: 8.00 keV**
- **K-beta: 9.48 keV**

Discussion

These calculated values for copper's characteristic X-rays are quite close to the measured values (K_α =8.05 keV and K_β =8.91 keV experimentally). The slight discrepancy comes from our simplified model using $Z_{\text{eff}}=Z-1$, which doesn't account for the different screening experienced by electrons in different shells. More sophisticated calculations would use different effective charges for each shell.

Copper is commonly used as an anode material in X-ray diffraction studies because its K_α X-rays (8.05 keV, wavelength 0.154 nm) have a wavelength well-suited for studying crystal structures. The wavelength is comparable to interatomic spacings in crystals, making copper K-alpha radiation ideal for X-ray crystallography. The characteristic X-ray energy depends approximately on Z^2 (since $E \propto Z_{2\text{eff}}$), so heavier elements produce much more energetic characteristic X-rays. For comparison:

- Aluminum (Z =13): $K_\alpha \approx 1.5$ keV
- Copper (Z =29): $K_\alpha \approx 8.0$ keV
- Molybdenum (Z =42): $K_\alpha \approx 17.5$ keV
- Tungsten (Z =74): $K_\alpha \approx 59$ keV

The K_β X-ray is always more energetic than the K_α because the electron falls from a higher shell (M vs L), releasing more energy. However, K_α X-rays are typically more intense than K_β X-rays because transitions from the L shell are more probable than from the M shell. In X-ray spectroscopy and crystallography, filters or monochromators are often used to select predominantly the K_α radiation, which provides sharper, more well-defined wavelengths for precision measurements.

Glossary

X-rays
: a form of electromagnetic radiation

x-ray diffraction
: a technique that provides the detailed information about crystallographic structure of natural and manufactured materials

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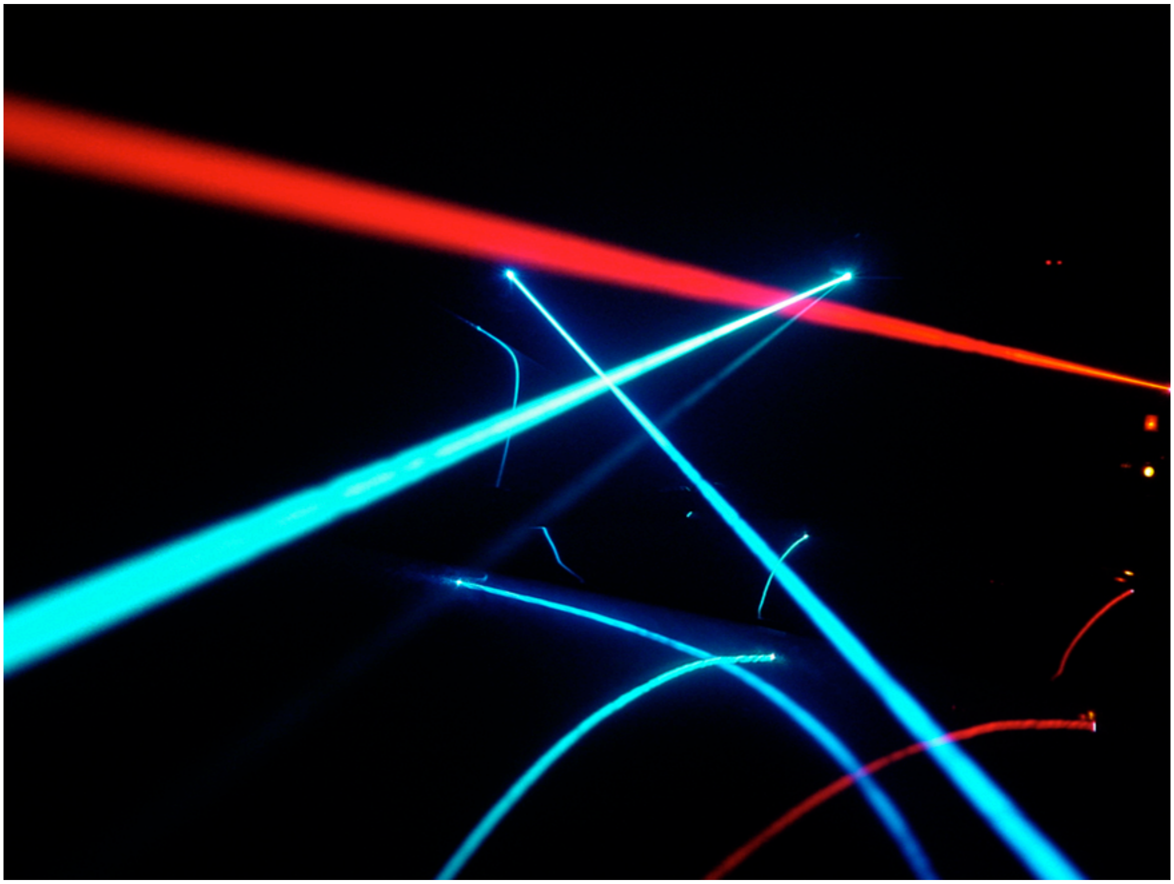
Applications of Atomic Excitations and De-Excitations

- Define and discuss fluorescence.
- Define metastable.
- Describe how laser emission is produced.
- Explain population inversion.
- Define and discuss holography.

Many properties of matter and phenomena in nature are directly related to atomic energy levels and their associated excitations and de-excitations. The color of a rose, the output of a laser, and the transparency of air are but a few examples. (See [Figure 1].) While it may not appear that glow-in-the-dark pajamas and lasers have much in common, they are in fact different applications of similar atomic de-excitations.

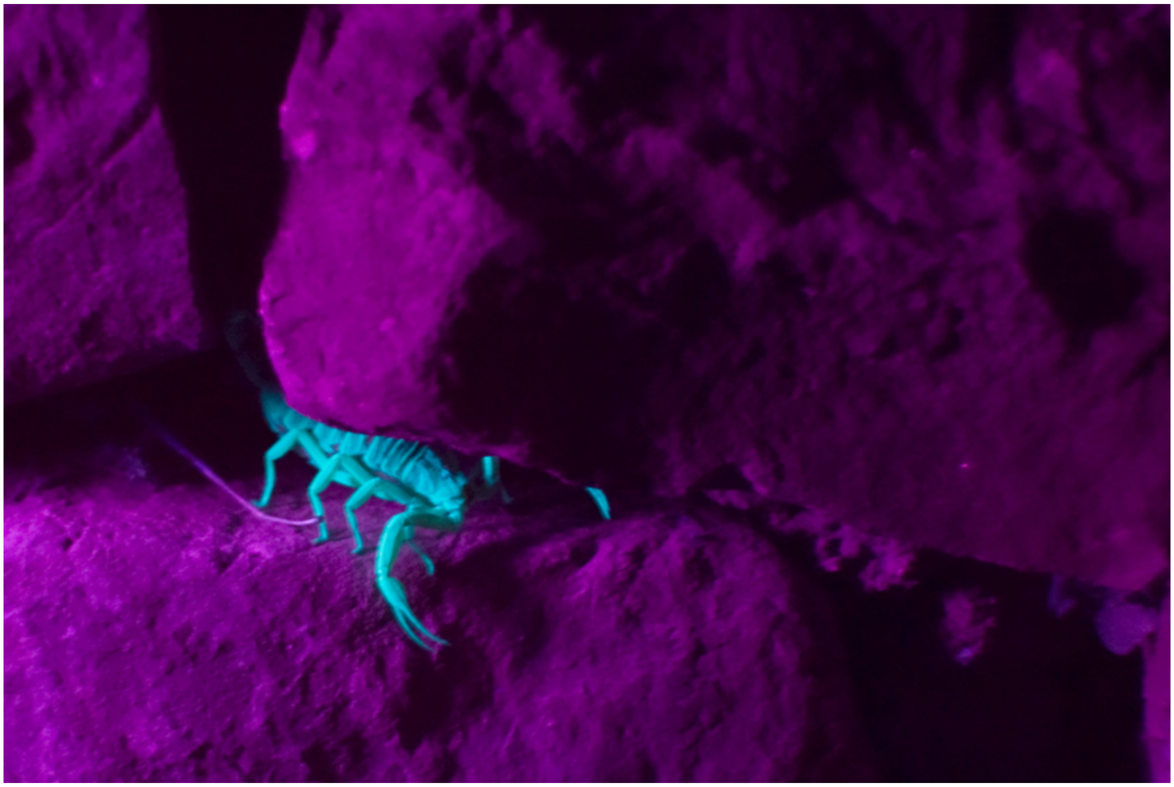


Light from a laser is based on a particular type of atomic de-excitation. (credit: Jeff Keyzer)

The color of a material is due to the ability of its atoms to absorb certain wavelengths while reflecting or reemitting others. A simple red material, for example a tomato, absorbs all visible wavelengths except red. This is because the atoms of its hydrocarbon pigment (lycopene) have levels separated by a variety of energies corresponding to all visible photon energies except red. Air is another interesting example. It is transparent to visible light, because there are few energy levels that visible photons can excite in air molecules and atoms. Visible light, thus, cannot be absorbed. Furthermore, visible light is only weakly scattered by air, because visible wavelengths are so much greater than the sizes of the air molecules and atoms. Light must pass through kilometers of air to scatter enough to cause red sunsets and blue skies.

Fluorescence and Phosphorescence

The ability of a material to emit various wavelengths of light is similarly related to its atomic energy levels. [Figure 2] shows a scorpion illuminated by a UV lamp, sometimes called a black light. Some rocks also glow in black light, the particular colors being a function of the rock’s mineral composition. Black lights are also used to make certain posters glow.



Objects glow in the visible spectrum when illuminated by an ultraviolet (black) light. Emissions are characteristic of the mineral involved, since they are related to its energy levels. In the case of scorpions, proteins near the surface of their skin give off the characteristic blue glow. This is a colorful example of fluorescence in which excitation is induced by UV radiation while de-excitation occurs in the form of visible light. (credit: Ken Bosma, Flickr)

In the fluorescence process, an atom is excited to a level several steps above its ground state by the absorption of a relatively high-energy UV photon. This is called **atomic excitation**. Once it is excited, the atom can de-excite in several ways, one of which is to re-emit a photon of the same energy as excited it, a single step back to the ground state. This is called **atomic de-excitation**. All other paths of de-excitation involve smaller steps, in which lower-energy (longer wavelength) photons are emitted. Some of these may be in the visible range, such as for the scorpion in [Figure 2]. **Fluorescence** is defined to be any process in which an atom or molecule, excited by a photon of a given energy, and de-excites by emission of a lower-energy photon.

Fluorescence can be induced by many types of energy input. Fluorescent paint, dyes, and even soap residues in clothes make colors seem brighter in sunlight by converting some UV into visible light. X-rays can induce fluorescence, as is done in x-ray fluoroscopy to make brighter visible images. Electric discharges can induce fluorescence, as in so-called neon lights and in gas-discharge tubes that produce atomic and molecular spectra. Common fluorescent lights use an electric discharge in mercury vapor to cause atomic emissions from mercury atoms. The inside of a fluorescent light is coated with a fluorescent material that emits visible light over a broad spectrum of wavelengths. By choosing an appropriate coating, fluorescent lights can be made more like sunlight or like the reddish glow of candlelight, depending on needs. Fluorescent lights are more efficient in converting electrical energy into visible light than incandescent filaments (about four times as efficient), the blackbody radiation of which is primarily in the infrared due to temperature limitations.

This atom is excited to one of its higher levels by absorbing a UV photon. It can de-excite in a single step, re-emitting a photon of the same energy, or in several steps. The process is called fluorescence if the atom de-excites in smaller steps, emitting energy different from that which excited it. Fluorescence can be induced by a variety of energy inputs, such as UV, X-rays, and electrical discharge.

The spectacular Waitomo caves on North Island in New Zealand provide a natural habitat for glow-worms. The glow-worms hang up to 70 silk threads of about 30 or 40 cm each to trap prey that fly towards them in the dark. The fluorescence process is very efficient, with nearly 100% of the energy input turning into light. (In comparison, fluorescent lights are about 20% efficient.)

Fluorescence has many uses in biology and medicine. It is commonly used to label and follow a molecule within a cell. Such tagging allows one to study the structure of DNA and proteins. Fluorescent dyes and antibodies are usually used to tag the molecules, which are then illuminated with UV light and their emission of visible light is observed. Since the fluorescence of each element is characteristic, identification of elements within a sample can be done this way.

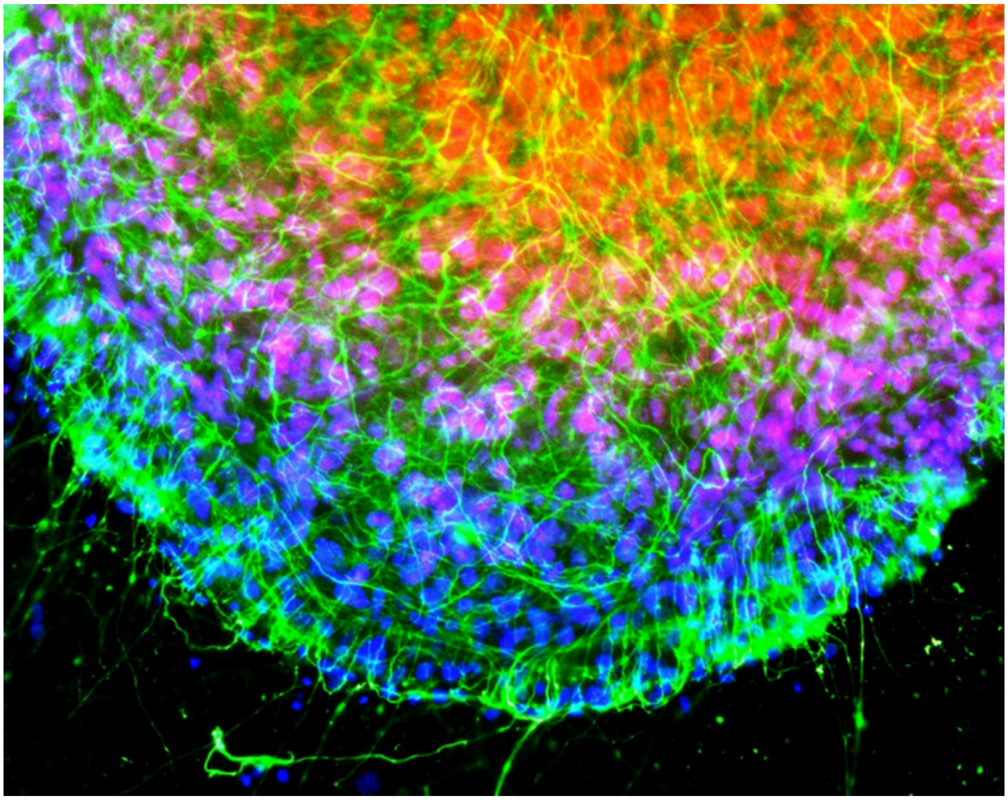
[Figure 3] shows a commonly used fluorescent dye called fluorescein. Below that, [Figure 4] reveals the diffusion of a fluorescent dye in water by observing it under UV light.

Fluorescein, shown here in powder form, is used to dye laboratory samples. (credit: Benjah-bmm27, Wikimedia Commons)

Here, fluorescent powder is added to a beaker of water. The mixture gives off a bright glow under ultraviolet light. (credit: Bricksnite, Wikimedia Commons)

Nano-Crystals

Recently, a new class of fluorescent materials has appeared—“nano-crystals.” These are single-crystal molecules less than 100 nm in size. The smallest of these are called “quantum dots.” These semiconductor indicators are very small (2–6 nm) and provide improved brightness. They also have the advantage that all colors can be excited with the same incident wavelength. They are brighter and more stable than organic dyes and have a longer lifetime than conventional phosphors. They have become an excellent tool for long-term studies of cells, including migration and morphology. ([Figure 5].)



Microscopic image of chicken cells using nano-crystals of a fluorescent dye. Cell nuclei exhibit blue fluorescence while neurofilaments exhibit green. (credit: Weerapong Prasongchean, Wikimedia Commons)

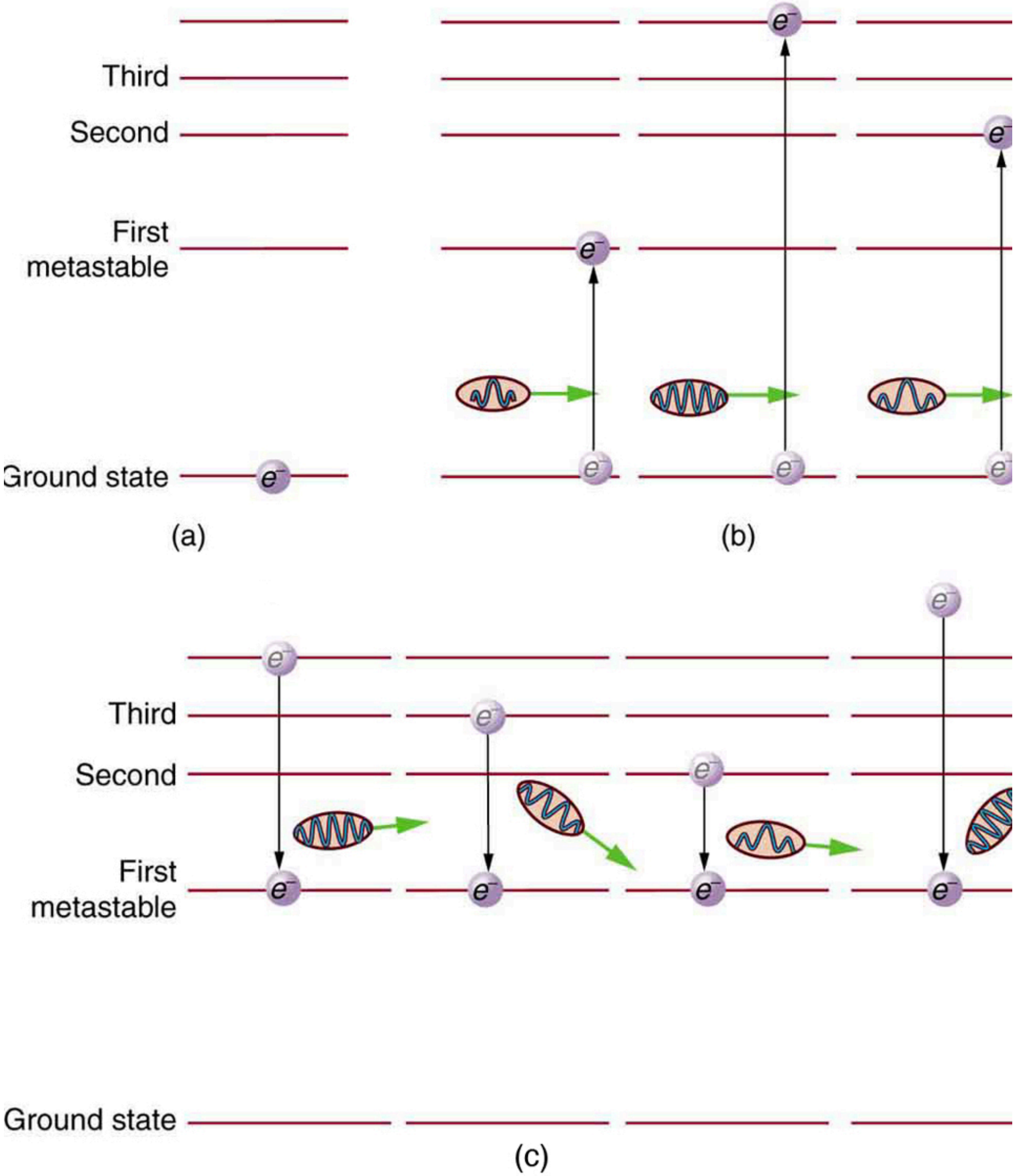
Once excited, an atom or molecule will usually spontaneously de-excite quickly. (The electrons raised to higher levels are attracted to lower ones by the positive charge of the nucleus.) Spontaneous de-excitation has a very short mean lifetime of typically about 10^{-8}s . However, some levels have significantly longer lifetimes, ranging up to milliseconds to minutes or even hours. These energy levels are inhibited and are slow in de-exciting because their quantum numbers differ greatly from those of available lower levels. Although these level lifetimes are short in human terms, they are many orders of magnitude longer than is typical and, thus, are said to be **metastable**, meaning relatively stable. **Phosphorescence** is the de-excitation of a metastable state. Glow-in-the-dark materials, such as luminous dials on some watches and clocks and on children’s toys and pajamas, are made of phosphorescent substances. Visible light excites the atoms or molecules to metastable states that decay slowly, releasing the stored excitation energy partially as visible light. In some ceramics, atomic excitation energy can be frozen in after the ceramic has cooled from its firing. It is very slowly released, but the ceramic can be induced to phosphoresce by heating—a process called “thermoluminescence.” Since the release is slow, thermoluminescence can be used to date antiquities. The less light emitted, the older the ceramic. (See [Figure 6].)

Atoms frozen in an excited state when this Chinese ceramic figure was fired can be stimulated to de-excite and emit EM radiation by heating a sample of the ceramic—a process called thermoluminescence. Since the states slowly de-excite over centuries, the amount of thermoluminescence decreases with age, making it possible to use this effect to date and authenticate antiquities. This figure dates from the 11th century. (credit: Vassil, Wikimedia Commons)

Lasers

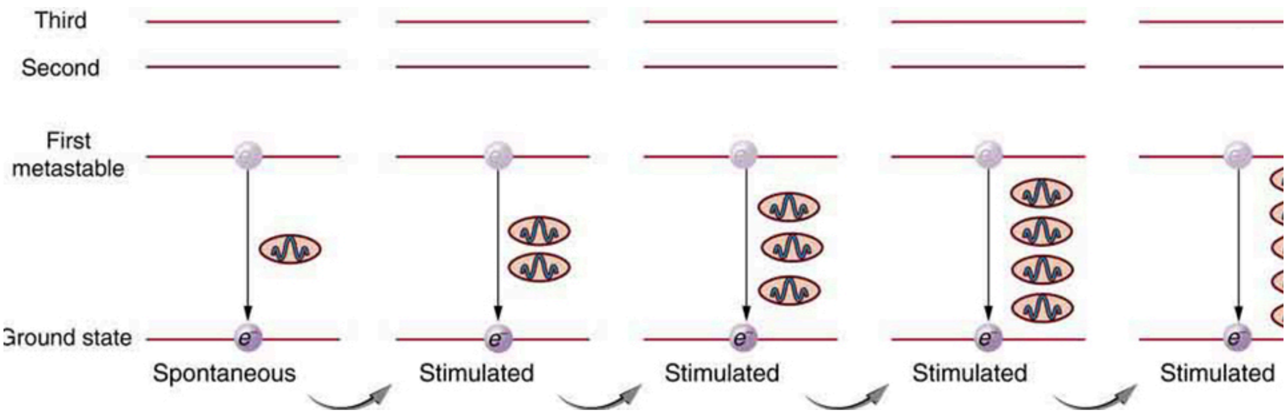
Lasers today are commonplace. Lasers are used to read bar codes at stores and in libraries, laser shows are staged for entertainment, laser printers produce high-quality images at relatively low cost, and lasers send prodigious numbers of telephone messages through optical fibers. Among other things, lasers are also employed in surveying, weapons guidance, tumor eradication, retinal welding, and for reading music CDs and computer CD-ROMs.

Why do lasers have so many varied applications? The answer is that lasers produce single-wavelength EM radiation that is also very coherent—that is, the emitted photons are in phase. Laser output can, thus, be more precisely manipulated than incoherent mixed-wavelength EM radiation from other sources. The reason laser output is so pure and coherent is based on how it is produced, which in turn depends on a metastable state in the lasing material. Suppose a material had the energy levels shown in [Figure 7]. When energy is put into a large collection of these atoms, electrons are raised to all possible levels. Most return to the ground state in less than about 10^{-8}s , but those in the metastable state linger. This includes those electrons originally excited to the metastable state and those that fell into it from above. It is possible to get a majority of the atoms into the metastable state, a condition called a **population inversion**.



(a) Energy-level diagram for an atom showing the first few states, one of which is metastable. (b) Massive energy input excites atoms to a variety of states. (c) Most states decay quickly, leaving electrons only in the metastable and ground state. If a majority of electrons are in the metastable state, a population inversion has been achieved.

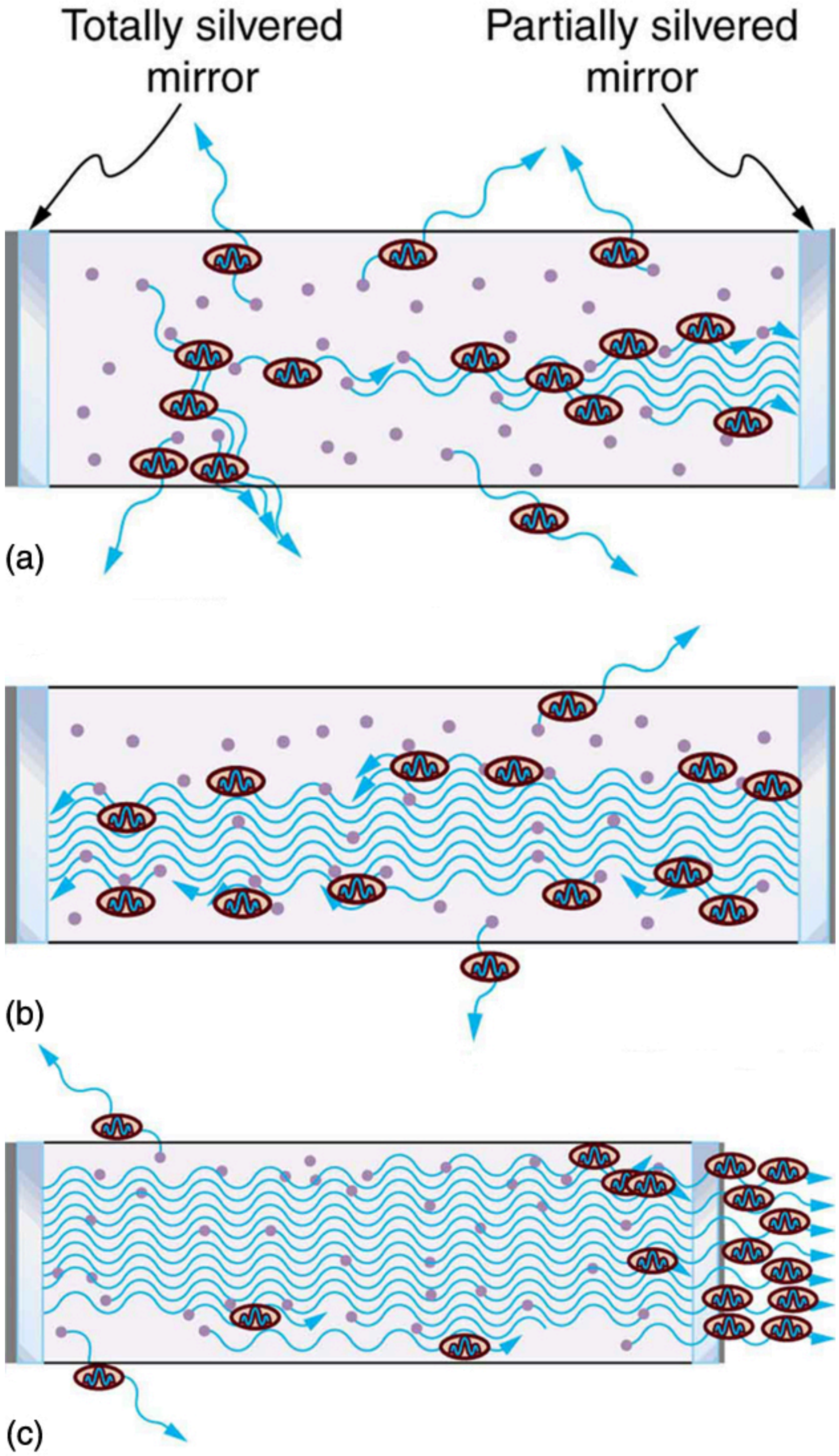
Once a population inversion is achieved, a very interesting thing can happen, as shown in [Figure 8]. An electron spontaneously falls from the metastable state, emitting a photon. This photon finds another atom in the metastable state and stimulates it to decay, emitting a second photon of *the same wavelength and in phase* with the first, and so on. **Stimulated emission** is the emission of electromagnetic radiation in the form of photons of a given frequency, triggered by photons of the same frequency. For example, an excited atom, with an electron in an energy orbit higher than normal, releases a photon of a specific frequency when the electron drops back to a lower energy orbit. If this photon then strikes another electron in the same high-energy orbit in another atom, another photon of the same frequency is released. The emitted photons and the triggering photons are always in phase, have the same polarization, and travel in the same direction. The probability of absorption of a photon is the same as the probability of stimulated emission, and so a majority of atoms must be in the metastable state to produce energy. Einstein (again Einstein, and back in 1917!) was one of the important contributors to the understanding of stimulated emission of radiation. Among other things, Einstein was the first to realize that stimulated emission and absorption are equally probable. The laser acts as a temporary energy storage device that subsequently produces a massive energy output of single-wavelength, in-phase photons.



One atom in the metastable state spontaneously decays to a lower level, producing a photon that goes on to stimulate another atom to de-excite. The second photon has exactly the same energy and wavelength as the first and is in phase with it. Both go on to stimulate the emission of other photons. A population inversion is necessary for there to be a net production rather than a net absorption of the photons.

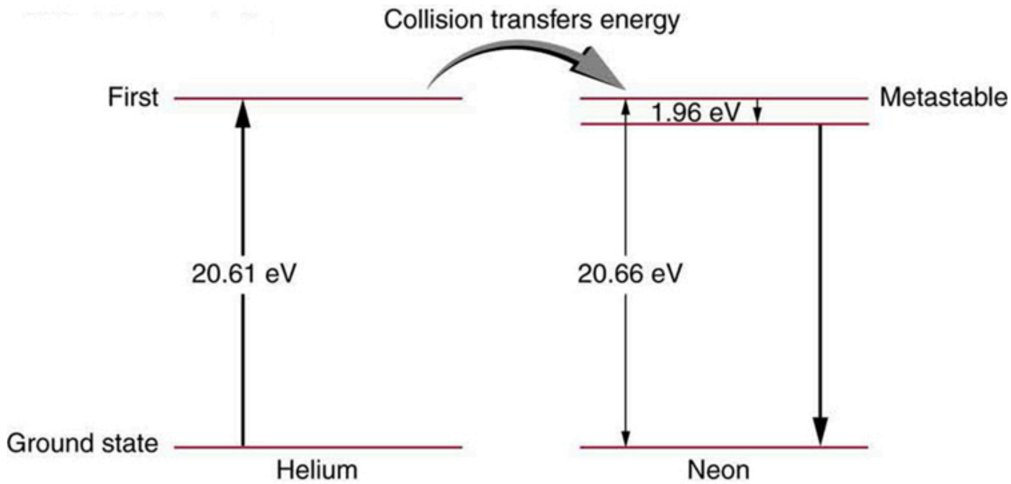
The name **laser** is an acronym for light amplification by stimulated emission of radiation, the process just described. The process was proposed and developed following the advances in quantum physics. A joint Nobel Prize was awarded in 1964 to American Charles Townes (1915–), and Nikolay Basov (1922–2001) and Aleksandr Prokhorov (1916–2002), from the Soviet Union, for the development of lasers. The Nobel Prize in 1981 went to Arthur Schawlow (1921-1999) for pioneering laser applications. The original devices were called masers, because they produced microwaves. The first working laser was created in 1960 at Hughes Research labs (CA) by T. Maiman. It used a pulsed high-powered flash lamp and a ruby rod to produce red light. Today the name laser is used for all such devices developed to produce a variety of wavelengths, including microwave, infrared, visible, and ultraviolet radiation. [Figure 9] shows how a laser can be constructed to enhance the stimulated emission of radiation. Energy input can be from a flash tube, electrical discharge, or other sources, in a process sometimes called optical pumping. A large percentage of the original pumping energy is dissipated in other forms, but a population inversion must be achieved. Mirrors can be used to enhance stimulated emission by multiple passes of the radiation back and forth through the lasing material. One of the mirrors is semitransparent to allow some of the light to pass through. The laser output from a laser is a mere 1% of the light passing back and forth in a laser.

As described earlier in the section on laser vision correction, Donna Strickland and Gérard Mourou, working at University of Rochester, developed a method to greatly increase the power of lasers, while also enabling them to be miniaturized. By passing the light over a specific type of grating, their method segments (or chirps) the delivery of the beam components in a matter that generates little heat at the source. Chirped pulse amplification is now used in some of the world's most powerful lasers as well as those commonly used to make precise microcuts or burns in medical applications. Strickland and Mourou were awarded the Nobel Prize in 2018.



Typical laser construction has a method of pumping energy into the lasing material to produce a population inversion. (a) Spontaneous emission begins with some photons escaping and others stimulating further emissions. (b) and (c) Mirrors are used to enhance the probability of stimulated emission by passing photons through the material several times.

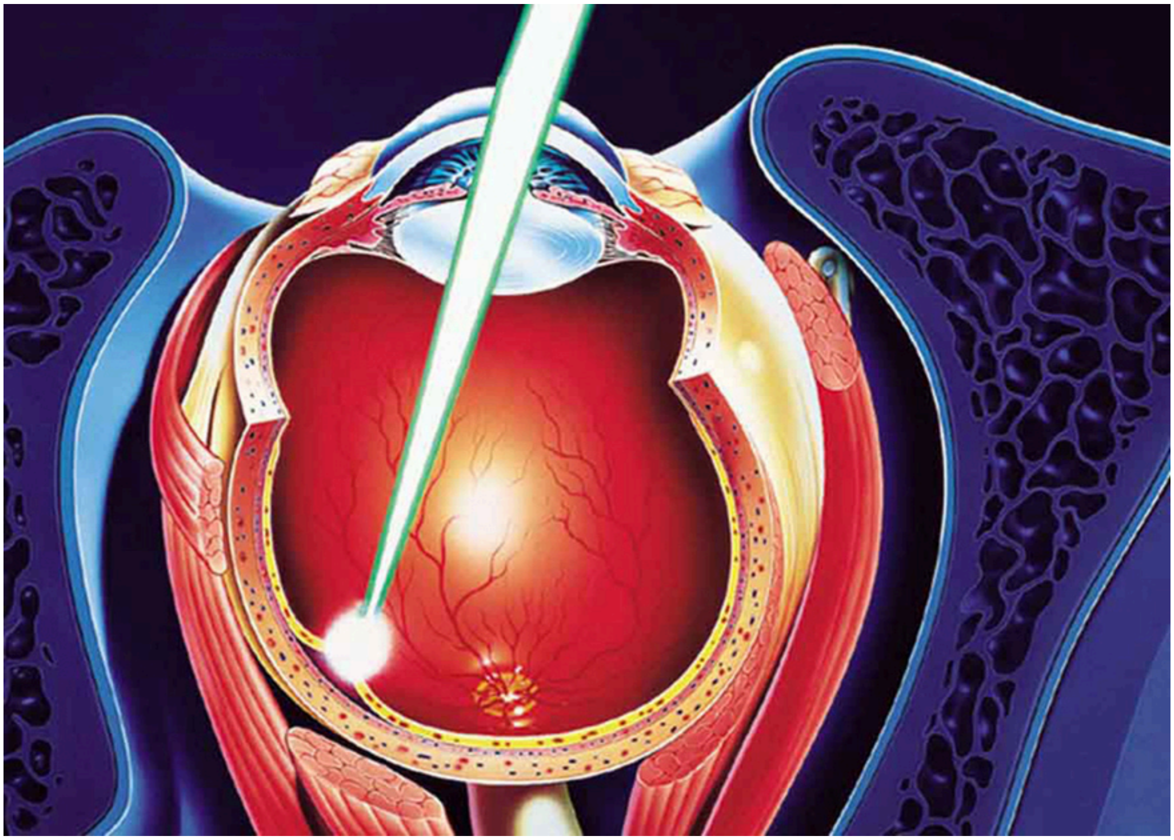
Lasers are constructed from many types of lasing materials, including gases, liquids, solids, and semiconductors. But all lasers are based on the existence of a metastable state or a phosphorescent material. Some lasers produce continuous output; others are pulsed in bursts as brief as 10^{-14}s . Some laser outputs are fantastically powerful—some greater than 10^{12}W —but the more common, everyday lasers produce something on the order of 10^{-3}W . The helium-neon laser that produces a familiar red light is very common. [Figure 10] shows the energy levels of helium and neon, a pair of noble gases that work well together. An electrical discharge is passed through a helium-neon gas mixture in which the number of atoms of helium is ten times that of neon. The first excited state of helium is metastable and, thus, stores energy. This energy is easily transferred by collision to neon atoms, because they have an excited state at nearly the same energy as that in helium. That state in neon is also metastable, and this is the one that produces the laser output. (The most likely transition is to the nearby state, producing 1.96 eV photons, which have a wavelength of 633 nm and appear red.) A population inversion can be produced in neon, because there are so many more helium atoms and these put energy into the neon. Helium-neon lasers often have continuous output, because the population inversion can be maintained even while lasing occurs. Probably the most common lasers in use today, including the common laser pointer, are semiconductor or diode lasers, made of silicon. Here, energy is pumped into the material by passing a current in the device to excite the electrons. Special coatings on the ends and fine cleavings of the semiconductor material allow light to bounce back and forth and a tiny fraction to emerge as laser light. Diode lasers can usually run continually and produce outputs in the milliwatt range.



Energy levels in helium and neon. In the common helium-neon laser, an electrical discharge pumps energy into the metastable states of both atoms. The gas mixture has about ten times more helium atoms than neon atoms. Excited helium atoms easily de-excite by transferring energy to neon in a collision. A population inversion in neon is achieved, allowing lasing by the neon to occur.

There are many medical applications of lasers. Lasers have the advantage that they can be focused to a small spot. They also have a well-defined wavelength. Many types of lasers are available today that provide wavelengths from the ultraviolet to the infrared. This is important, as one needs to be able to select a wavelength that will be preferentially absorbed by the material of interest. Objects appear a certain color because they absorb all other visible colors incident upon them. What wavelengths are absorbed depends upon the energy spacing between electron orbitals in that molecule. Unlike the hydrogen atom, biological molecules are complex and have a variety of absorption wavelengths or lines. But these can be determined and used in the selection of a laser with the appropriate wavelength. Water is transparent to the visible spectrum but will absorb light in the UV and IR regions. Blood (hemoglobin) strongly reflects red but absorbs most strongly in the UV.

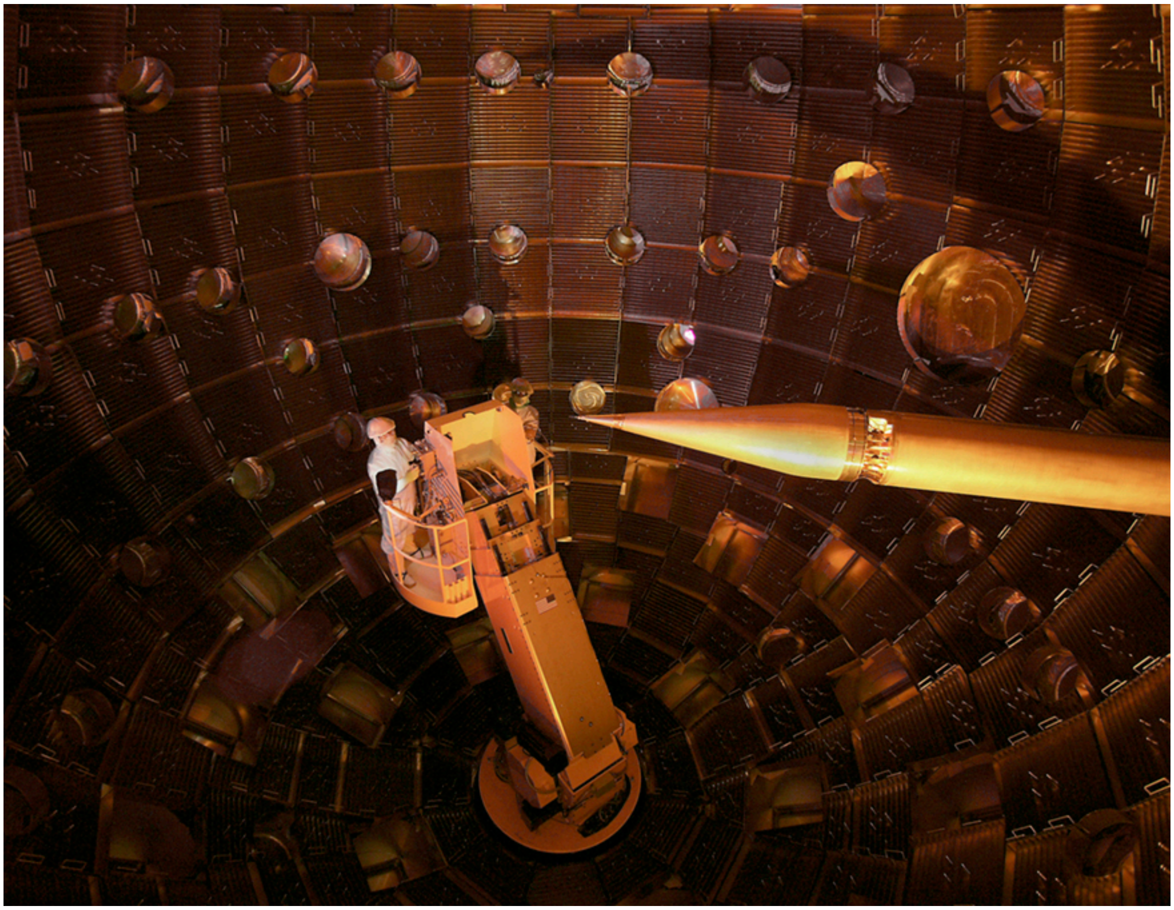
Laser surgery uses a wavelength that is strongly absorbed by the tissue it is focused upon. One example of a medical application of lasers is shown in [Figure 11]. A detached retina can result in total loss of vision. Burns made by a laser focused to a small spot on the retina form scar tissue that can hold the retina in place, salvaging the patient’s vision. Other light sources cannot be focused as precisely as a laser due to refractive dispersion of different wavelengths. Similarly, laser surgery in the form of cutting or burning away tissue is made more accurate because laser output can be very precisely focused and is preferentially absorbed because of its single wavelength. Depending upon what part or layer of the retina needs repairing, the appropriate type of laser can be selected. For the repair of tears in the retina, a green argon laser is generally used. This light is absorbed well by tissues containing blood, so coagulation or “welding” of the tear can be done.



A detached retina is burned by a laser designed to focus on a small spot on the retina, the resulting scar tissue holding it in place. The lens of the eye is used to focus the light, as is the device bringing the laser output to the eye.

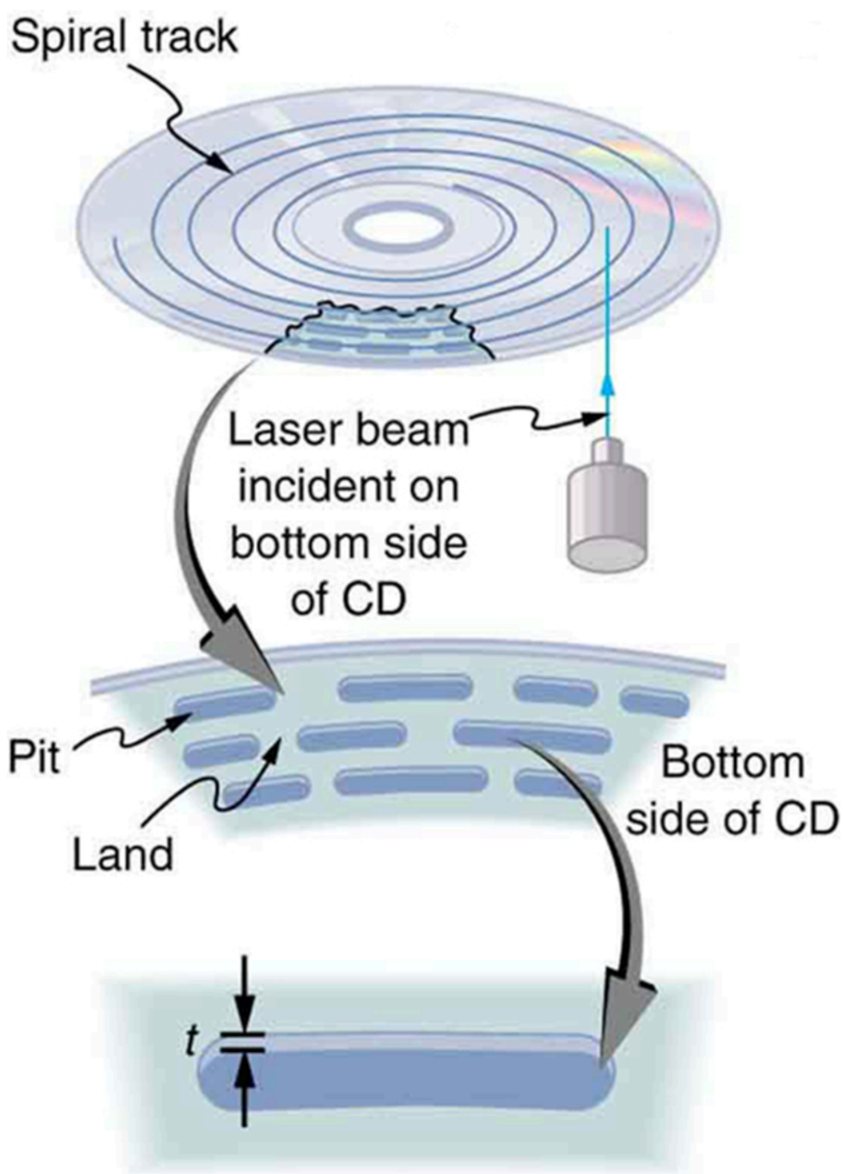
In dentistry, the use of lasers is rising. Lasers are most commonly used for surgery on the soft tissue of the mouth. They can be used to remove ulcers, stop bleeding, and reshape gum tissue. Their use in cutting into bones and teeth is not quite so common; here the erbium YAG (yttrium aluminum garnet) laser is used.

The massive combination of lasers shown in [Figure 12] can be used to induce nuclear fusion, the energy source of the sun and hydrogen bombs. Since lasers can produce very high power in very brief pulses, they can be used to focus an enormous amount of energy on a small glass sphere containing fusion fuel. Not only does the incident energy increase the fuel temperature significantly so that fusion can occur, it also compresses the fuel to great density, enhancing the probability of fusion. The compression or implosion is caused by the momentum of the impinging laser photons.



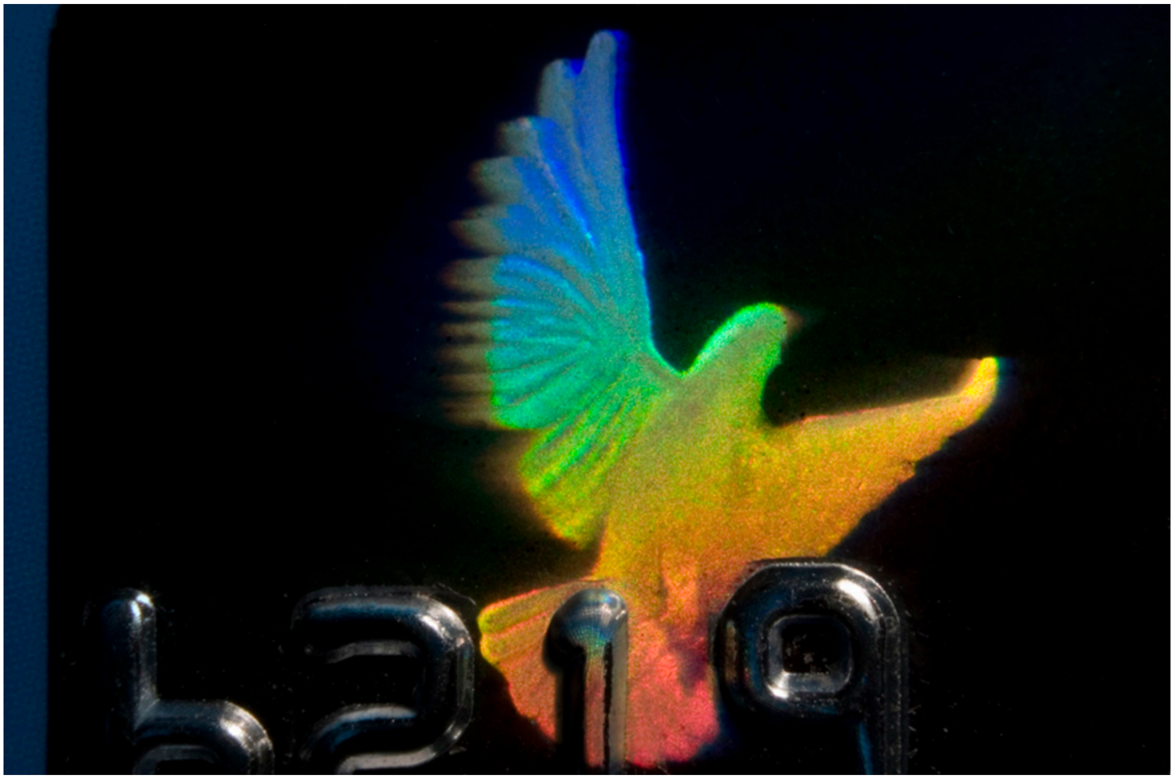
This system of lasers at Lawrence Livermore Laboratory is used to ignite nuclear fusion. A tremendous burst of energy is focused on a small fuel pellet, which is imploded to the high density and temperature needed to make the fusion reaction proceed. (credit: Lawrence Livermore National Laboratory, Lawrence Livermore National Security, LLC, and the Department of Energy)

Music CDs are now so common that vinyl records are quaint antiquities. CDs (and DVDs) store information digitally and have a much larger information-storage capacity than vinyl records. An entire encyclopedia can be stored on a single CD. [Figure 13] illustrates how the information is stored and read from the CD. Pits made in the CD by a laser can be tiny and very accurately spaced to record digital information. These are read by having an inexpensive solid-state infrared laser beam scatter from pits as the CD spins, revealing their digital pattern and the information encoded upon them.



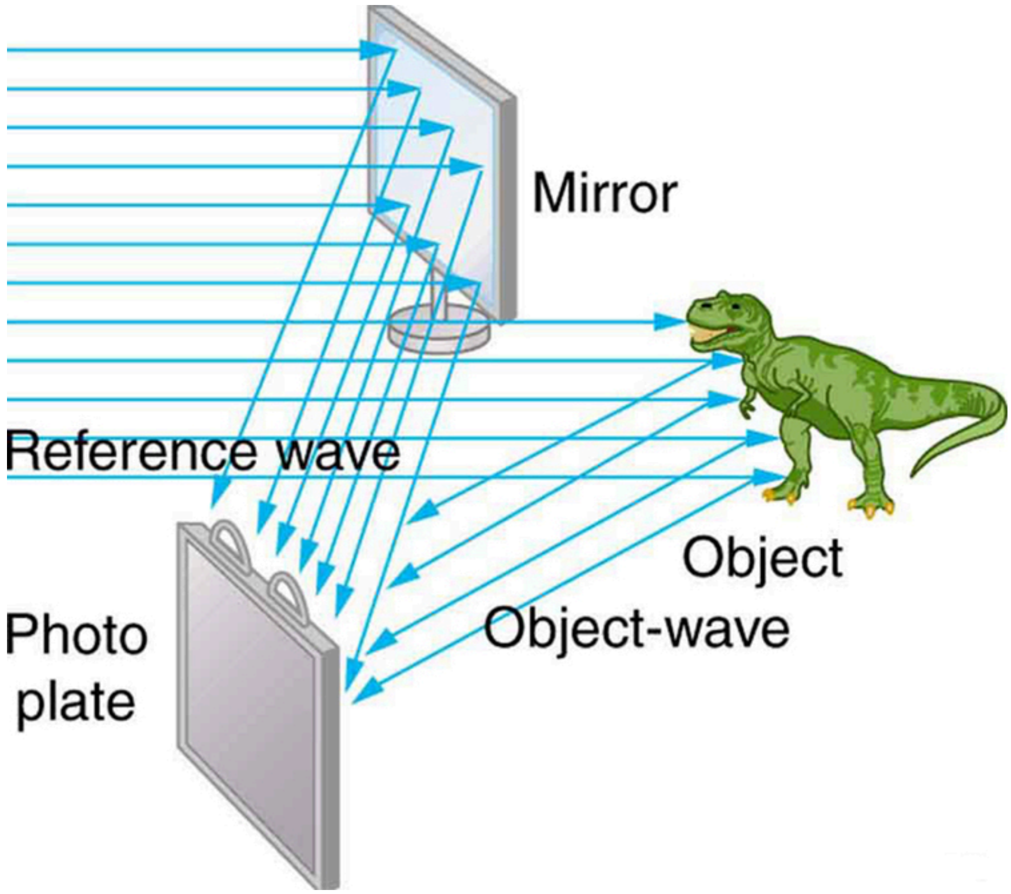
A CD has digital information stored in the form of laser-created pits on its surface. These in turn can be read by detecting the laser light scattered from the pit. Large information capacity is possible because of the precision of the laser. Shorter-wavelength lasers enable greater storage capacity.

Holograms, such as those in [Figure 14], are true three-dimensional images recorded on film by lasers. Holograms are used for amusement, decoration on novelty items and magazine covers, security on credit cards and driver’s licenses (a laser and other equipment is needed to reproduce them), and for serious three-dimensional information storage. You can see that a hologram is a true three-dimensional image, because objects change relative position in the image when viewed from different angles.



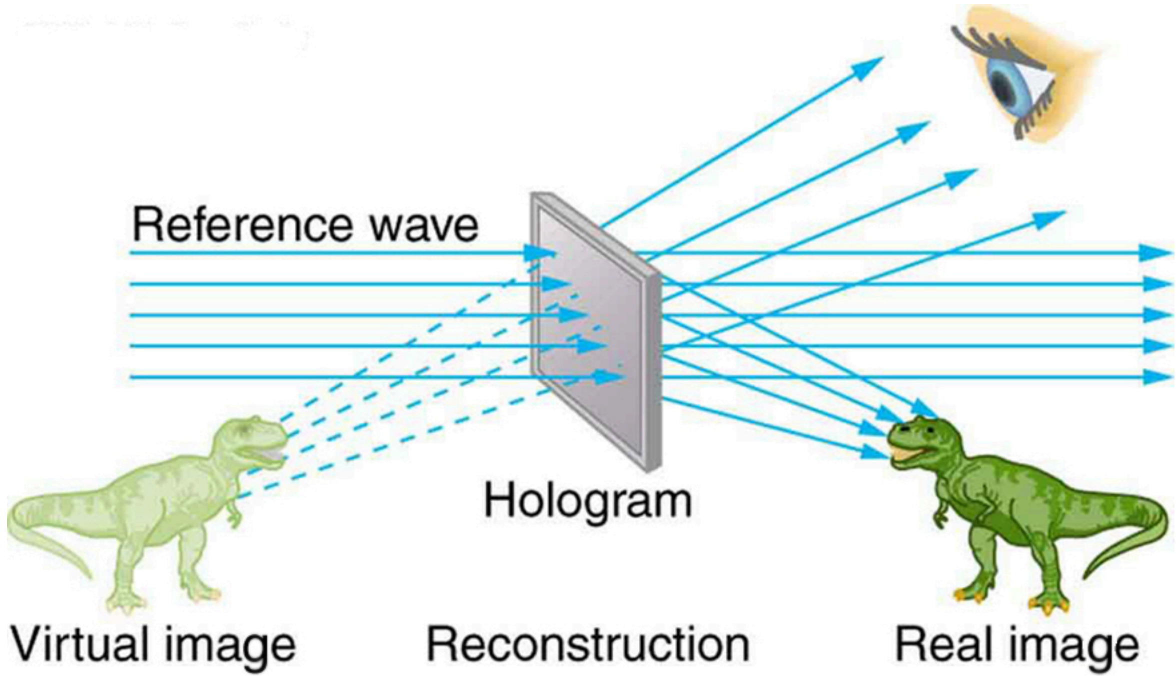
Credit cards commonly have holograms for logos, making them difficult to reproduce (credit: Dominic Alves, Flickr)

The name **hologram** means “entire picture” (from the Greek *holo*, as in holistic), because the image is three-dimensional. **Holography** is the process of producing holograms and, although they are recorded on photographic film, the process is quite different from normal photography. Holography uses light interference or wave optics, whereas normal photography uses geometric optics. [Figure 15] shows one method of producing a hologram. Coherent light from a laser is split by a mirror, with part of the light illuminating the object. The remainder, called the reference beam, shines directly on a piece of film. Light scattered from the object interferes with the reference beam, producing constructive and destructive interference. As a result, the exposed film looks foggy, but close examination reveals a complicated interference pattern stored on it. Where the interference was constructive, the film (a negative actually) is darkened. Holography is sometimes called lensless photography, because it uses the wave characteristics of light as contrasted to normal photography, which uses geometric optics and so requires lenses.



Production of a hologram. Single-wavelength coherent light from a laser produces a well-defined interference pattern on a piece of film. The laser beam is split by a partially silvered mirror, with part of the light illuminating the object and the remainder shining directly on the film.

Light falling on a hologram can form a three-dimensional image. The process is complicated in detail, but the basics can be understood as shown in [Figure 16], in which a laser of the same type that exposed the film is now used to illuminate it. The myriad tiny exposed regions of the film are dark and block the light, while less exposed regions allow light to pass. The film thus acts much like a collection of diffraction gratings with various spacings. Light passing through the hologram is diffracted in various directions, producing both real and virtual images of the object used to expose the film. The interference pattern is the same as that produced by the object. Moving your eye to various places in the interference pattern gives you different perspectives, just as looking directly at the object would. The image thus looks like the object and is three-dimensional like the object.



> A transmission hologram is one that produces real and virtual images when a laser of the same type as that which exposed the hologram is passed through it. Diffraction from various parts of the film produces the same interference pattern as the object that was used to expose it.

The hologram illustrated in [Figure 16] is a transmission hologram. Holograms that are viewed with reflected light, such as the white light holograms on credit cards, are reflection holograms and are more common. White light holograms often appear a little blurry with rainbow edges, because the diffraction patterns of various colors of light are at slightly different locations due to their different wavelengths. Further uses of holography include all types of 3-D information storage, such as of statues in museums and engineering studies of structures and 3-D images of human organs. Invented in the late 1940s by Dennis Gabor (1900–1979), who won the 1971 Nobel Prize in Physics for his work, holography became far more practical with the development of the laser. Since lasers produce coherent single-wavelength light, their interference patterns are more pronounced. The precision is so great that it is even possible to record numerous holograms on a single piece of film by just changing the angle of the film for each successive image. This is how the holograms that move as you walk by them are produced—a kind of lensless movie.

In a similar way, in the medical field, holograms have allowed complete 3-D holographic displays of objects from a stack of images. Storing these images for future use is relatively easy. With the use of an endoscope, high-resolution 3-D holographic images of internal organs and tissues can be made.

Section Summary

- An important atomic process is fluorescence, defined to be any process in which an atom or molecule is excited by absorbing a photon of a given energy and de-excited by emitting a photon of a lower energy.
- Some states live much longer than others and are termed metastable.
- Phosphorescence is the de-excitation of a metastable state.
- Lasers produce coherent single-wavelength EM radiation by stimulated emission, in which a metastable state is stimulated to decay.
- Lasing requires a population inversion, in which a majority of the atoms or molecules are in their metastable state.

Conceptual Questions

How do the allowed orbits for electrons in atoms differ from the allowed orbits for planets around the sun? Explain how the correspondence principle applies here.

Atomic and molecular spectra are discrete. What does discrete mean, and how are discrete spectra related to the quantization of energy and electron orbits in atoms and molecules?

Hydrogen gas can only absorb EM radiation that has an energy corresponding to a transition in the atom, just as it can only emit these discrete energies. When a spectrum is taken of the solar corona, in which a broad range of EM wavelengths are passed through very hot hydrogen gas, the absorption spectrum shows all the features of the emission spectrum. But when such EM radiation passes through room-temperature hydrogen gas, only the Lyman series is absorbed. Explain the difference.

Lasers are used to burn and read CDs. Explain why a laser that emits blue light would be capable of burning and reading more information than one that emits infrared.

The coating on the inside of fluorescent light tubes absorbs ultraviolet light and subsequently emits visible light. An inventor claims that he is able to do the reverse process. Is the inventor’s claim possible?

What is the difference between fluorescence and phosphorescence?

How can you tell that a hologram is a true three-dimensional image and that those in 3-D movies are not?

Problem Exercises

[Figure 10] shows the energy-level diagram for neon. (a) Verify that the energy of the photon emitted when neon goes from its metastable state to the one immediately below is equal to 1.96 eV. (b) Show that the wavelength of this radiation is 633 nm. (c) What wavelength is emitted when the neon makes a direct transition to its ground state?

Show Solution

Strategy

From [Figure 10], we can identify the energy levels of neon. The metastable state is at 20.66 eV, and the state immediately below it is at 18.70 eV. For part (a), we calculate the energy difference between these levels. For part (b), we use the relationship $\lambda = \frac{hc}{E}$ with the convenient form $hc = 1240 \text{ eV}\cdot\text{nm}$. For part (c), we calculate the energy difference from the metastable state to the ground state (0 eV) and find the corresponding wavelength.

Solution

(a) The energy of the photon emitted when neon transitions from the metastable state to the level immediately below is:

$$\Delta E = 20.66 \text{ eV} - 18.70 \text{ eV} = 1.96 \text{ eV}$$

This verifies that the photon energy is **1.96 eV**.

(b) The wavelength corresponding to this energy is:

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV}\cdot\text{nm}}{1.96 \text{ eV}} = 633 \text{ nm}$$

This confirms that the wavelength is **633 nm**, which is the characteristic red light of helium-neon lasers.

(c) For a direct transition from the metastable state to the ground state:

$$\Delta E = 20.66\text{ eV} - 0\text{ eV} = 20.66\text{ eV}$$

The wavelength is:

$$\lambda = \frac{hc}{E} = \frac{1240\text{ eV}\cdot\text{nm}}{20.66\text{ eV}} = 60.0\text{ nm}$$

The wavelength emitted is **60.0 nm**, which is in the extreme ultraviolet (EUV) region.

Discussion

The 633 nm red light from the neon transition is what makes helium-neon lasers produce their characteristic red beam, which is widely used in laser pointers, barcode scanners, and laboratory demonstrations. The metastable state at 20.66 eV is crucial for laser operation—its relatively long lifetime allows a population inversion to build up. The direct transition to ground state (60.0 nm) is in the UV range and is much more energetic than the visible red emission. This UV wavelength matches the energy needed to initially excite helium atoms in the He-Ne laser (as shown in Figure 10), creating a closed energy cycle. The fact that neon preferentially emits at 633 nm rather than 60 nm is due to the quantum mechanical selection rules and the availability of the intermediate energy level at 18.70 eV.

A helium-neon laser is pumped by electric discharge. What wavelength electromagnetic radiation would be needed to pump it? See [Figure 10] for energy-level information.

Show Solution

Strategy

From [Figure 10], we can see that helium atoms need to be excited from the ground state (0 eV) to the first metastable state at 20.61 eV. We'll use the relationship between photon energy and wavelength: $E = \frac{hc}{\lambda}$ or equivalently $\lambda = \frac{hc}{E}$, where we can use the convenient form $hc = 1240\text{ eV}\cdot\text{nm}$.

Solution

The energy required to pump helium from its ground state to the metastable state is:

$$\Delta E = 20.61\text{ eV} - 0\text{ eV} = 20.61\text{ eV}$$

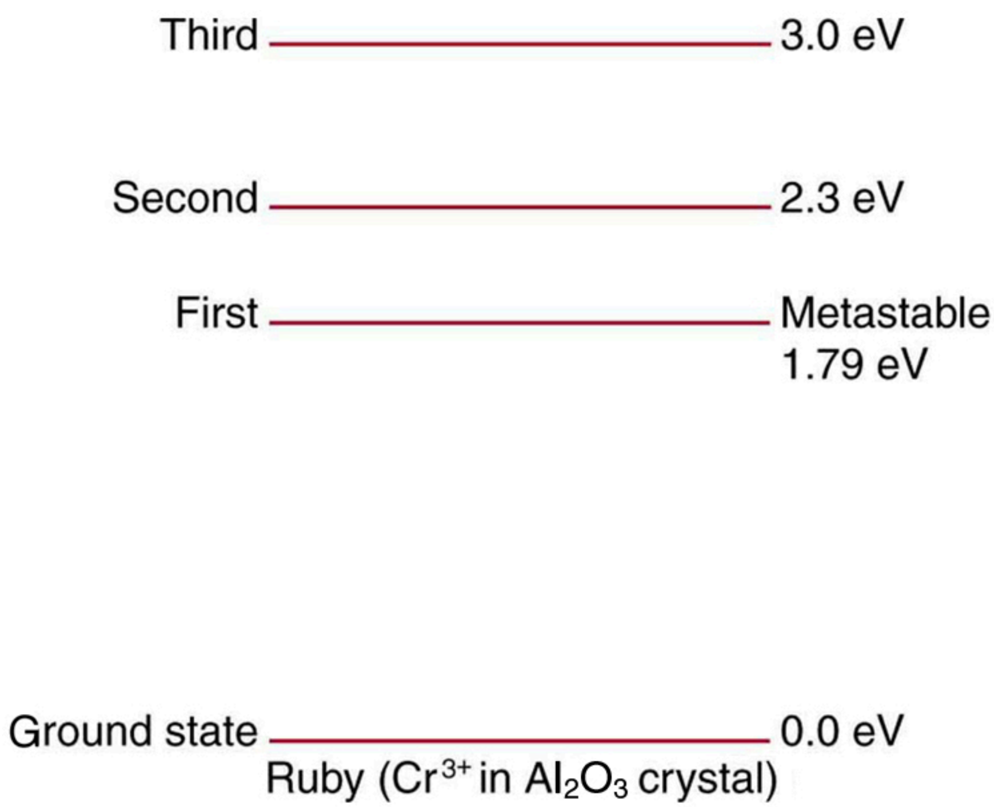
The wavelength of electromagnetic radiation with this energy is:

$$\lambda = \frac{hc}{E} = \frac{1240\text{ eV}\cdot\text{nm}}{20.61\text{ eV}} = 60.2\text{ nm}$$

Discussion

This wavelength of 60.2 nm is in the extreme ultraviolet (EUV) region of the electromagnetic spectrum, far below the visible range (400–700 nm). In practice, helium-neon lasers use electrical discharge rather than optical pumping because it's more practical to excite helium atoms through collisions with electrons in the discharge rather than trying to produce intense EUV radiation. Once the helium atoms are excited to their metastable state, they transfer energy to neon atoms through collisions, which is why helium-neon lasers work so efficiently. The laser output at 633 nm (red light) comes from the neon atoms, not the helium.

Ruby lasers have chromium atoms doped in an aluminum oxide crystal. The energy level diagram for chromium in a ruby is shown in [Figure 17]. What wavelength is emitted by a ruby laser?



Chromium atoms in an aluminum oxide crystal have these energy levels, one of which is metastable. This is the basis of a ruby laser. Visible light can pump the atom into an excited state above the metastable state to achieve a population inversion.

Show Solution

Strategy

From [Figure 17], we can see that the ruby laser operates by transitions from the metastable state at 1.79 eV down to the ground state at 0.0 eV. The wavelength of the emitted photon can be calculated using $\lambda = \frac{hc}{E}$, where $E = 1.79\text{ eV}$ is the energy difference and $hc = 1240\text{ eV}\cdot\text{nm}$.

Solution

The energy of the laser transition is:

$$E = 1.79\text{ eV} - 0.0\text{ eV} = 1.79\text{ eV}$$

The wavelength of the emitted radiation is:

$$\lambda = \frac{hc}{E} = \frac{1240\text{ eV}\cdot\text{nm}}{1.79\text{ eV}} = 693\text{ nm}$$

The ruby laser emits light with a wavelength of **693 nm**.

Discussion

This wavelength of 693 nm is in the deep red portion of the visible spectrum, which is why ruby lasers produce a characteristic red beam. Ruby was the material used in the very first laser, built by Theodore Maiman in 1960. The chromium atoms (which give ruby its red color) are the active lasing medium. Visible light (green and blue wavelengths around 400-600 nm) from a flash lamp is absorbed to pump chromium atoms from the ground state to higher excited states at 2.3 eV and 3.0 eV. These atoms quickly relax non-radiatively to the metastable state at 1.79 eV, where they remain for a relatively long time (on the order of milliseconds). This allows a population inversion to build up. When stimulated emission begins, all the stored energy is released coherently as intense red light at 693 nm. Ruby lasers typically operate in pulsed mode rather than continuously because it's difficult to maintain a population inversion continuously. Despite being the first type of laser ever built, ruby lasers are still used today in applications such as holography, tattoo removal, and as pump sources for other lasers.

(a) What energy photons can pump chromium atoms in a ruby laser from the ground state to its second and third excited states? (b) What are the wavelengths of these photons? Verify that they are in the visible part of the spectrum.

Show Solution

Strategy

From [Figure 17], we can identify the energy levels of chromium in aluminum oxide: ground state at 0.0 eV, first metastable state at 1.79 eV, second excited state at 2.3 eV, and third excited state at 3.0 eV. For part (a), we need the energy differences from the ground state to the second and third excited states. For part (b), we'll use $\lambda = \frac{hc}{E}$ with $hc = 1240\text{ eV}\cdot\text{nm}$ to find the wavelengths and verify they fall within the visible spectrum (approximately 400–700 nm).

Solution

(a) The energy required to excite chromium from the ground state to the second excited state is:

$$E_2 = 2.3\text{ eV} - 0.0\text{ eV} = 2.3\text{ eV}$$

The energy required to excite chromium from the ground state to the third excited state is:

$$E_3 = 3.0\text{ eV} - 0.0\text{ eV} = 3.0\text{ eV}$$

(b) The wavelength corresponding to the 2.3 eV photon is:

$$\lambda_2 = \frac{hc}{E_2} = \frac{1240\text{ eV}\cdot\text{nm}}{2.3\text{ eV}} = 539\text{ nm}$$

This is in the green portion of the visible spectrum.

The wavelength corresponding to the 3.0 eV photon is:

$$\lambda_3 = \frac{hc}{E_3} = \frac{1240\text{ eV}\cdot\text{nm}}{3.0\text{ eV}} = 413\text{ nm}$$

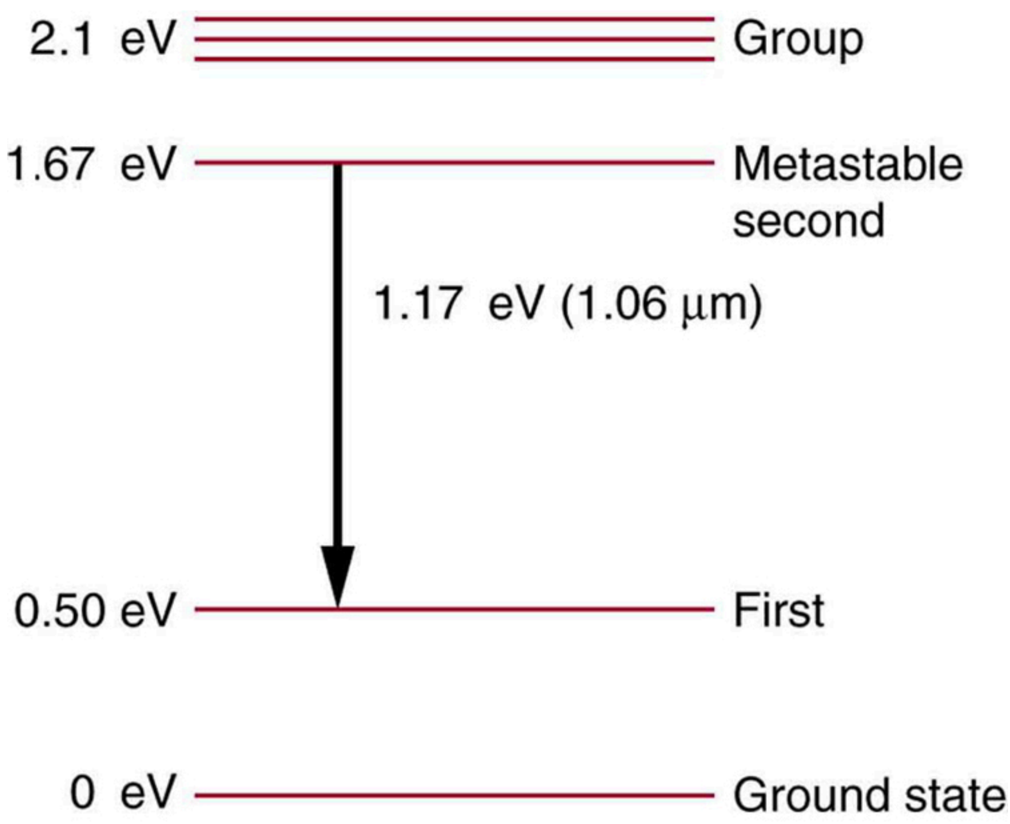
This is in the violet portion of the visible spectrum.

Both wavelengths fall within the visible range of approximately 400–700 nm, confirming that visible light can be used to optically pump a ruby laser.

Discussion

This is why ruby lasers can be pumped with visible light, typically using intense white light from a flash lamp. The chromium atoms absorb the green and violet/blue components of the white light, becoming excited to the higher energy states. They then quickly relax to the metastable state at 1.79 eV through non-radiative transitions. Once a population inversion is achieved in the metastable state, stimulated emission produces the characteristic red laser light at 693 nm (from the 1.79 eV transition back to ground state). The ability to pump with visible light makes ruby lasers relatively practical compared to systems requiring UV pumping. The green (539 nm) absorption is particularly strong, which is why ruby crystals appear red—they absorb the green light and reflect/transmit the red.

Some of the most powerful lasers are based on the energy levels of neodymium in solids, such as glass, as shown in [Figure 18]. (a) What average wavelength light can pump the neodymium into the levels above its metastable state? (b) Verify that the 1.17 eV transition produces $1.06\mu\text{m}$ radiation.



Neodymium atoms in glass have these energy levels, one of which is metastable. The group of levels above the metastable state is convenient for achieving a population inversion, since photons of many different energies can be absorbed by atoms in the ground state.

Show Solution

Strategy

From [Figure 18], we see that neodymium has energy levels at 0 eV (ground state), 0.50 eV, 1.67 eV (metastable state), and a group of levels around 2.1 eV. For part (a), we need the energy to pump atoms from ground state (0 eV) to the group above the metastable state (2.1 eV), then calculate the wavelength using $\lambda = \frac{hc}{E}$. For part (b), we verify that the 1.17 eV transition (from 1.67 eV metastable state to 0.50 eV first excited state) produces 1.06 μm radiation using $hc = 1240\ \text{eV}\cdot\text{nm}$.

Solution

(a) The average energy required to pump neodymium from ground state to the levels above the metastable state is:

$$E = 2.1\ \text{eV} - 0\ \text{eV} = 2.1\ \text{eV}$$

The wavelength of pump light is:

$$\lambda = \frac{hc}{E} = \frac{1240\ \text{eV}\cdot\text{nm}}{2.1\ \text{eV}} = 590\ \text{nm}$$

The average pump wavelength is **590 nm**, which is in the yellow-orange region of the spectrum.

(b) The laser transition is from the metastable state at 1.67 eV down to the first excited state at 0.50 eV:

$$E = 1.67\ \text{eV} - 0.50\ \text{eV} = 1.17\ \text{eV}$$

The wavelength is:

$$\lambda = \frac{hc}{E} = \frac{1240\ \text{eV}\cdot\text{nm}}{1.17\ \text{eV}} = 1060\ \text{nm} = 1.06\ \mu\text{m}$$

This verifies that the 1.17 eV transition produces **1.06 μm** radiation.

Discussion

Neodymium lasers (Nd:YAG or Nd:glass) are among the most powerful and versatile lasers available. The 1.06 μm wavelength is in the near-infrared, just beyond the visible red end of the spectrum. This makes the beam invisible to the human eye, which is both an advantage (no visible glare) and a safety concern (operators cannot see the beam). Neodymium lasers are widely used in materials processing (cutting, welding, marking), medical procedures (surgery, dentistry, dermatology), military applications (range finding, target designation), and scientific research. The pump wavelength of 590 nm (yellow-orange) is convenient because it can be provided by flashlamps or by laser diodes. The “group of levels” around 2.1 eV is particularly advantageous because atoms can be pumped to any of several closely-spaced levels, making the laser less sensitive to the exact pump wavelength—this is called broadband pumping. After being pumped to these upper levels, atoms quickly relax non-radiatively to the metastable state at 1.67 eV, where they accumulate to create a population inversion. The lasing transition brings atoms down to 0.50 eV, from which they quickly relax to the ground state, ready to be pumped again. This four-level laser scheme is very efficient and allows continuous operation at high power levels. Neodymium lasers can be Q-switched or mode-locked to produce extremely short, intense pulses, making them valuable for applications requiring precise energy delivery in very short time intervals.

Glossary

metastable
: a state whose lifetime is an order of magnitude longer than the most short-lived states

atomic excitation
: a state in which an atom or ion acquires the necessary energy to promote one or more of its electrons to electronic states higher in energy than their ground state

atomic de-excitation
: process by which an atom transfers from an excited electronic state back to the ground state electronic configuration; often occurs by emission of a photon

laser
: acronym for light amplification by stimulated emission of radiation

phosphorescence
: the de-excitation of a metastable state

population inversion
: the condition in which the majority of atoms in a sample are in a metastable state

stimulated emission
: emission by atom or molecule in which an excited state is stimulated to decay, most readily caused by a photon of the same energy that is necessary to excite the state

hologram
: means *entire picture* (from the Greek word *holo*, as in holistic), because the image produced is three dimensional

holography
: the process of producing holograms

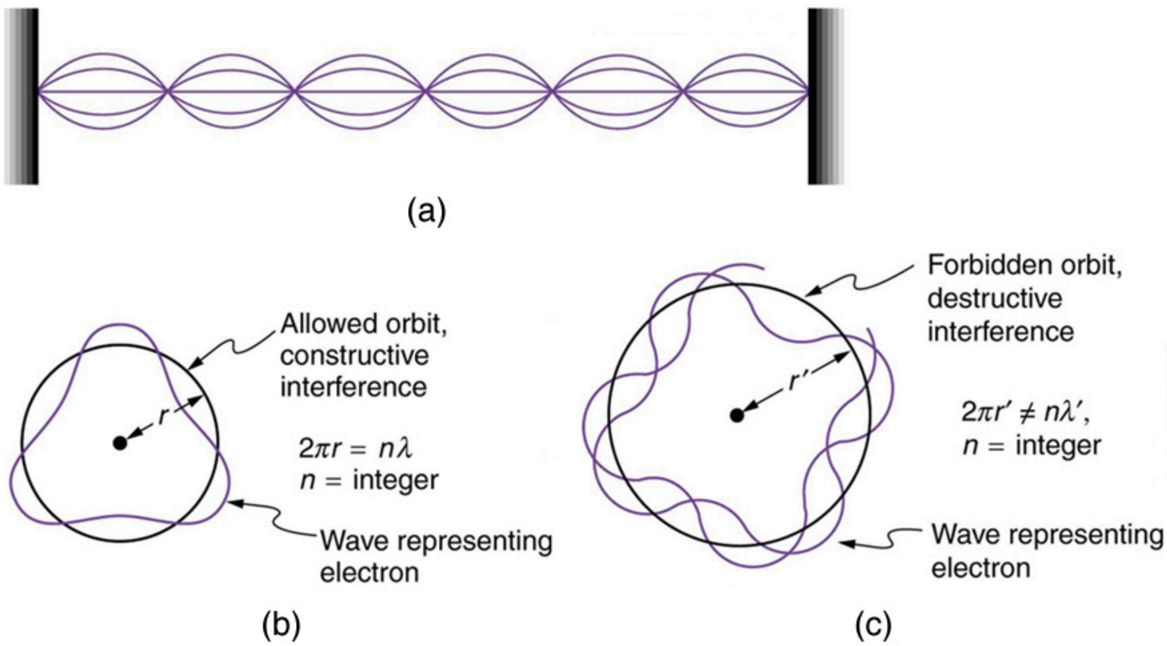
fluorescence
: any process in which an atom or molecule, excited by a photon of a given energy, de-excites by emission of a lower-energy photon

The Wave Nature of Matter Causes Quantization

- Explain Bohr's model of atom.
- Define and describe quantization of angular momentum.
- Calculate the angular momentum for an orbit of atom.
- Define and describe the wave-like properties of matter.

After visiting some of the applications of different aspects of atomic physics, we now return to the basic theory that was built upon Bohr's atom. Einstein once said it was important to keep asking the questions we eventually teach children not to ask. Why is angular momentum quantized? You already know the answer. Electrons have wave-like properties, as de Broglie later proposed. They can exist only where they interfere constructively, and only certain orbits meet proper conditions, as we shall see in the next module.

Following Bohr's initial work on the hydrogen atom, a decade was to pass before de Broglie proposed that matter has wave properties. The wave-like properties of matter were subsequently confirmed by observations of electron interference when scattered from crystals. Electrons can exist only in locations where they interfere constructively. How does this affect electrons in atomic orbits? When an electron is bound to an atom, its wavelength must fit into a small space, something like a standing wave on a string. (See [Figure 1].) Allowed orbits are those orbits in which an electron constructively interferes with itself. Not all orbits produce constructive interference. Thus only certain orbits are allowed—the orbits are quantized.



(a) Waves on a string have a wavelength related to the length of the string, allowing them to interfere constructively. (b) If we imagine the string bent into a closed circle, we get a rough idea of how electrons in circular orbits can interfere constructively. (c) If the wavelength does not fit into the circumference, the electron interferes destructively; it cannot exist in such an orbit.

For a circular orbit, constructive interference occurs when the electron's wavelength fits neatly into the circumference, so that wave crests always align with crests and wave troughs align with troughs, as shown in [Figure 1] (b). More precisely, when an integral multiple of the electron's wavelength equals the circumference of the orbit, constructive interference is obtained. In equation form, the *condition for constructive interference and an allowed electron orbit* is

$$n\lambda_n = 2\pi r_n (n=1,2,3...),$$

where λ_n is the electron's wavelength and r_n is the radius of that circular orbit. The de Broglie wavelength is $\lambda = h/p = h/mv$, and so here $\lambda = h/m_e v$. Substituting this into the previous condition for constructive interference produces an interesting result:

$$nhm_e v = 2\pi r_n.$$

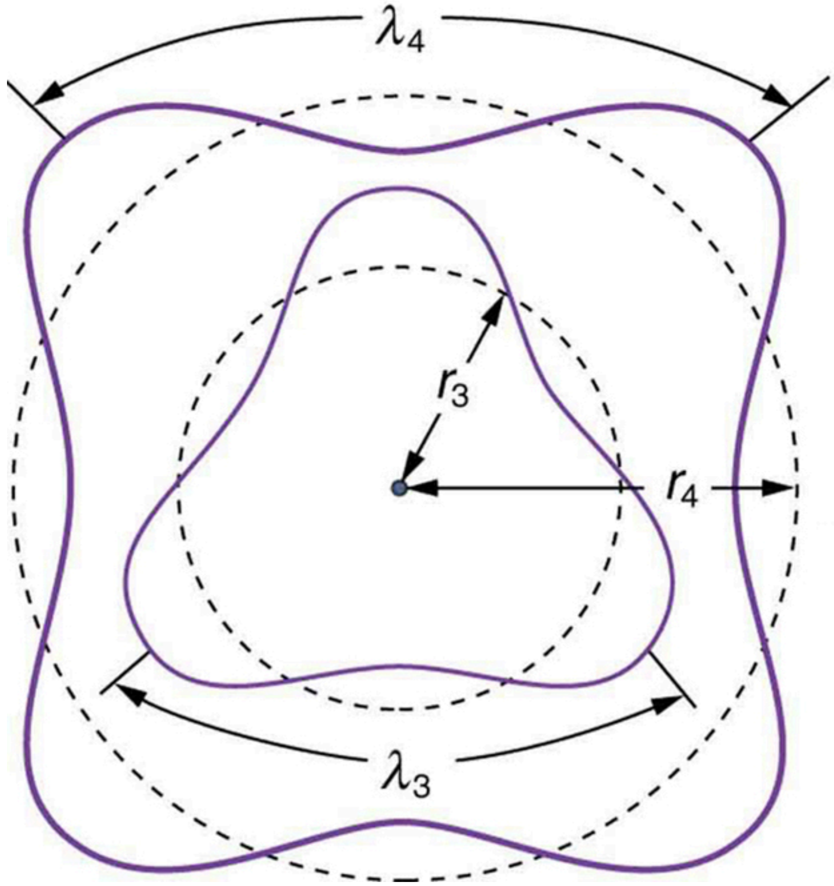
Rearranging terms, and noting that $L = mvr$ for a circular orbit, we obtain the quantization of angular momentum as the condition for allowed orbits:

$$L = m_e v r_n = nh2\pi (n=1,2,3...).$$

This is what Bohr was forced to hypothesize as the rule for allowed orbits, as stated earlier. We now realize that it is the condition for constructive interference of an electron in a circular orbit. [Figure 2] illustrates this for $n = 3$ and $n = 4$.

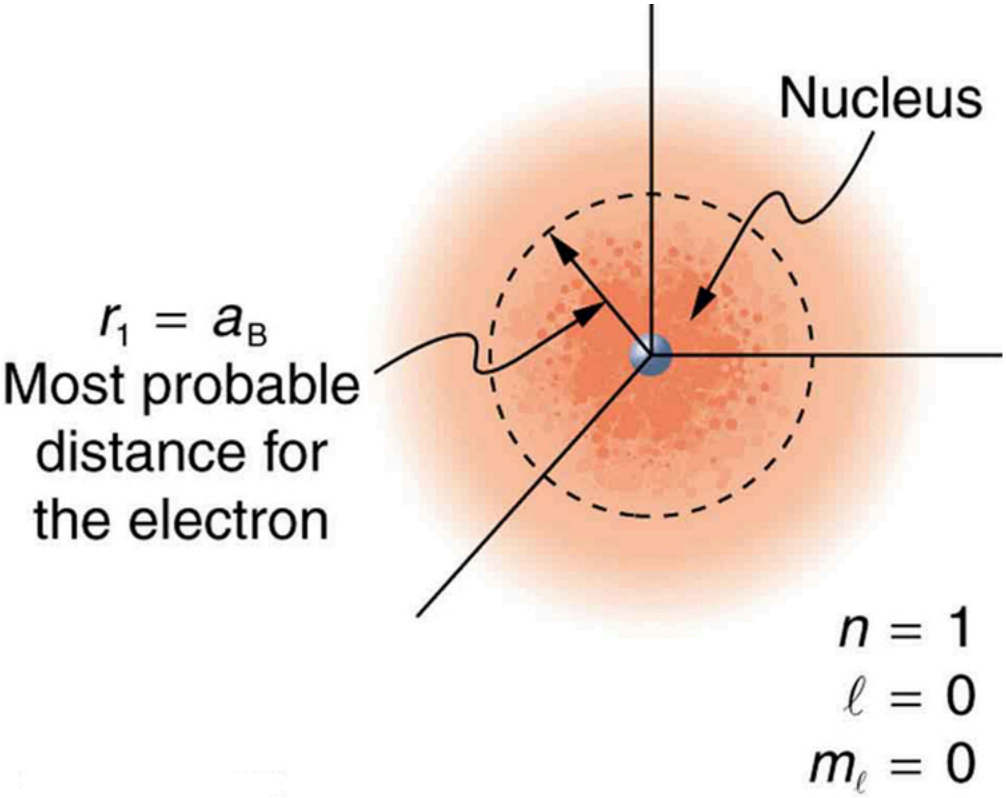
Waves and Quantization

The wave nature of matter is responsible for the quantization of energy levels in bound systems. Only those states where matter interferes constructively exist, or are “allowed.” Since there is a lowest orbit where this is possible in an atom, the electron cannot spiral into the nucleus. It cannot exist closer to or inside the nucleus. The wave nature of matter is what prevents matter from collapsing and gives atoms their sizes.



The third and fourth allowed circular orbits have three and four wavelengths, respectively, in their circumferences.

Because of the wave character of matter, the idea of well-defined orbits gives way to a model in which there is a cloud of probability, consistent with Heisenberg’s uncertainty principle. [Figure 3] shows how this applies to the ground state of hydrogen. If you try to follow the electron in some well-defined orbit using a probe that has a small enough wavelength to get some details, you will instead knock the electron out of its orbit. Each measurement of the electron’s position will find it to be in a definite location somewhere near the nucleus. Repeated measurements reveal a cloud of probability like that in the figure, with each speck the location determined by a single measurement. There is not a well-defined, circular-orbit type of distribution. Nature again proves to be different on a small scale than on a macroscopic scale.



The ground state of a hydrogen atom has a probability cloud describing the position of its electron. The probability of finding the electron is proportional to the darkness of the cloud. The electron can be closer or farther than the Bohr radius, but it is very unlikely to be a great distance from the nucleus.

There are many examples in which the wave nature of matter causes quantization in bound systems such as the atom. Whenever a particle is confined or bound to a small space, its allowed wavelengths are those which fit into that space. For example, the particle in a box model describes a particle free to move in a small space surrounded by impenetrable barriers. This is true in blackbody radiators (atoms and molecules) as well as in atomic and molecular spectra. Various atoms and molecules will have different sets of electron orbits, depending on the size and complexity of the system. When a system is large, such as a grain of sand, the tiny particle waves in it can fit in so many ways that it becomes impossible to see that the allowed states are discrete. Thus the correspondence principle is satisfied. As systems become large, they gradually look less grainy, and quantization becomes less evident. Unbound systems (small or not), such as an electron freed from an atom, do not have quantized energies, since their wavelengths are not constrained to fit in a certain volume.

PhET Explorations: Quantum Wave Interference

When do photons, electrons, and atoms behave like particles and when do they behave like waves? Watch waves spread out and interfere as they pass through a double slit, then get detected on a screen as tiny dots. Use quantum detectors to explore how measurements change the waves and the patterns they produce on the screen.

Section Summary

- Quantization of orbital energy is caused by the wave nature of matter. Allowed orbits in atoms occur for constructive interference of electrons in the orbit, requiring an integral number of wavelengths to fit in an orbit's circumference; that is,
  $n\lambda_n = 2\pi r_n$ where $(n = 1, 2, 3 \ldots)$,

  where λ_n is the electron's de Broglie wavelength.

- Owing to the wave nature of electrons and the Heisenberg uncertainty principle, there are no well-defined orbits; rather, there are clouds of probability.
- Bohr correctly proposed that the energy and radii of the orbits of electrons in atoms are quantized, with energy for transitions between orbits given by
  $\Delta E = hf = E_{\mathrm{i}} - E_{\mathrm{f}}$,

  where ΔE is the change in energy between the initial and final orbits and hf is the energy of an absorbed or emitted photon.

- It is useful to plot orbit energies on a vertical graph called an energy-level diagram.
- The allowed orbits are circular, Bohr proposed, and must have quantized orbital angular momentum given by
  $L = m_e v r_n = n\frac{h}{2\pi}$ where $(n = 1, 2, 3 \ldots)$,

  where L is the angular momentum, r_n is the radius of the n^{th} orbit , and h is Planck's constant.

Conceptual Questions

How is the de Broglie wavelength of electrons related to the quantization of their orbits in atoms and molecules?

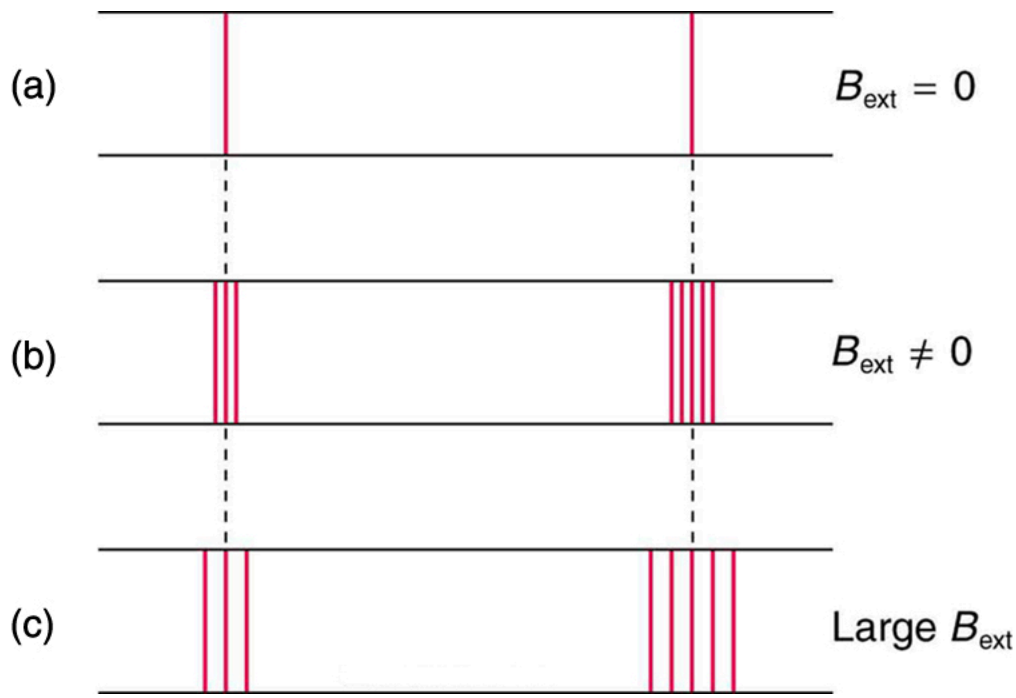
Patterns in Spectra Reveal More Quantization

- State and discuss the Zeeman effect.
- Define orbital magnetic field.
- Define orbital angular momentum.
- Define space quantization.

High-resolution measurements of atomic and molecular spectra show that the spectral lines are even more complex than they first appear. In this section, we will see that this complexity has yielded important new information about electrons and their orbits in atoms.

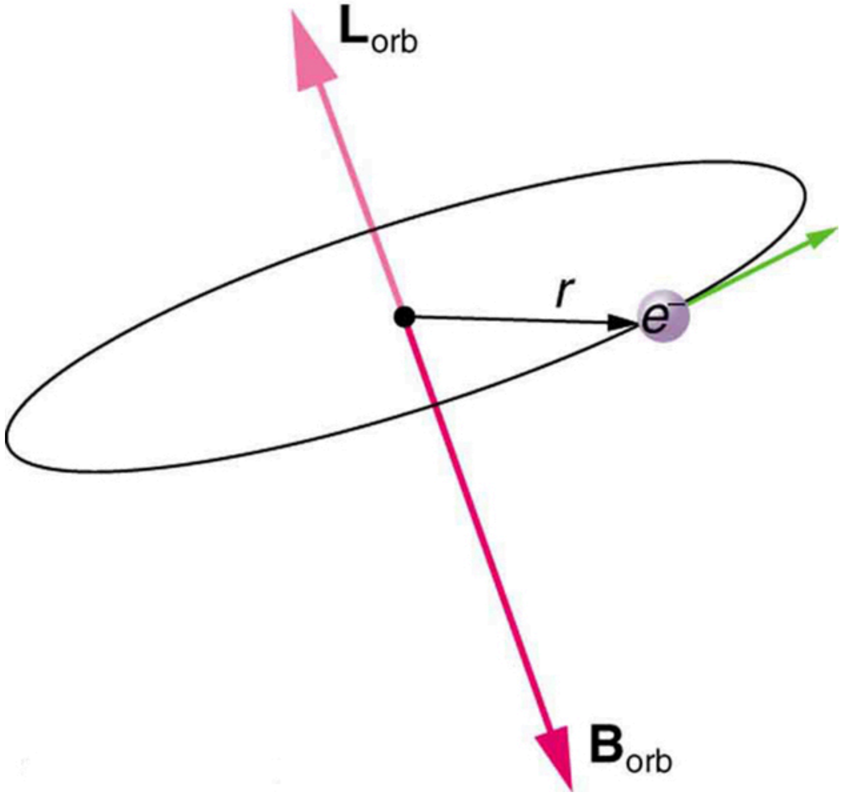
In order to explore the substructure of atoms (and knowing that magnetic fields affect moving charges), the Dutch physicist Hendrik Lorentz (1853–1930) suggested that his student Pieter Zeeman (1865–1943) study how spectra might be affected by magnetic fields. What they found became known as the **Zeeman effect**, which involved spectral lines being split into two or more separate emission lines by an external magnetic field, as shown in [Figure 1] . For their discoveries, Zeeman and Lorentz shared the 1902 Nobel Prize in Physics.

Zeeman splitting is complex. Some lines split into three lines, some into five, and so on. But one general feature is that the amount the split lines are separated is proportional to the applied field strength, indicating an interaction with a moving charge. The splitting means that the quantized energy of an orbit is affected by an external magnetic field, causing the orbit to have several discrete energies instead of one. Even without an external magnetic field, very precise measurements showed that spectral lines are doublets (split into two), apparently by magnetic fields within the atom itself.

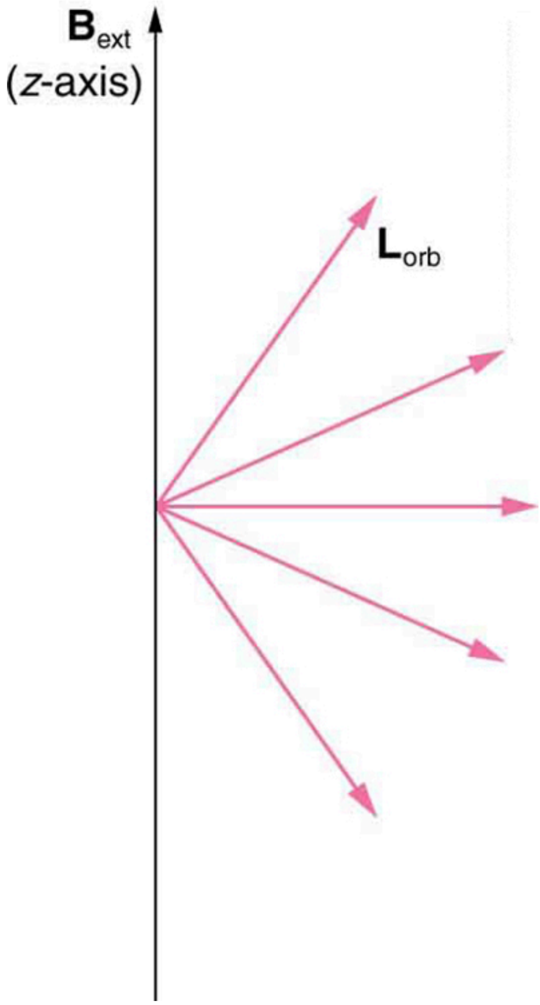


The Zeeman effect is the splitting of spectral lines when a magnetic field is applied. The number of lines formed varies, but the spread is proportional to the strength of the applied field. (a) Two spectral lines with no external magnetic field. (b) The lines split when the field is applied. (c) The splitting is greater when a stronger field is applied.

Bohr's theory of circular orbits is useful for visualizing how an electron's orbit is affected by a magnetic field. The circular orbit forms a current loop, which creates a magnetic field of its own, $\vec{\mathbf{B}}_{\text{orb}}$ as seen in [Figure 2]. Note that the **orbital magnetic field** $\vec{\mathbf{B}}_{\text{orb}}$ and the **orbital angular momentum** $\vec{\mathbf{L}}_{\text{orb}}$ are along the same line. The external magnetic field and the orbital magnetic field interact; a torque is exerted to align them. A torque rotating a system through some angle does work so that there is energy associated with this interaction. Thus, orbits at different angles to the external magnetic field have different energies. What is remarkable is that the energies are quantized—the magnetic field splits the spectral lines into several discrete lines that have different energies. This means that only certain angles are allowed between the orbital angular momentum and the external field, as seen in [Figure 3].



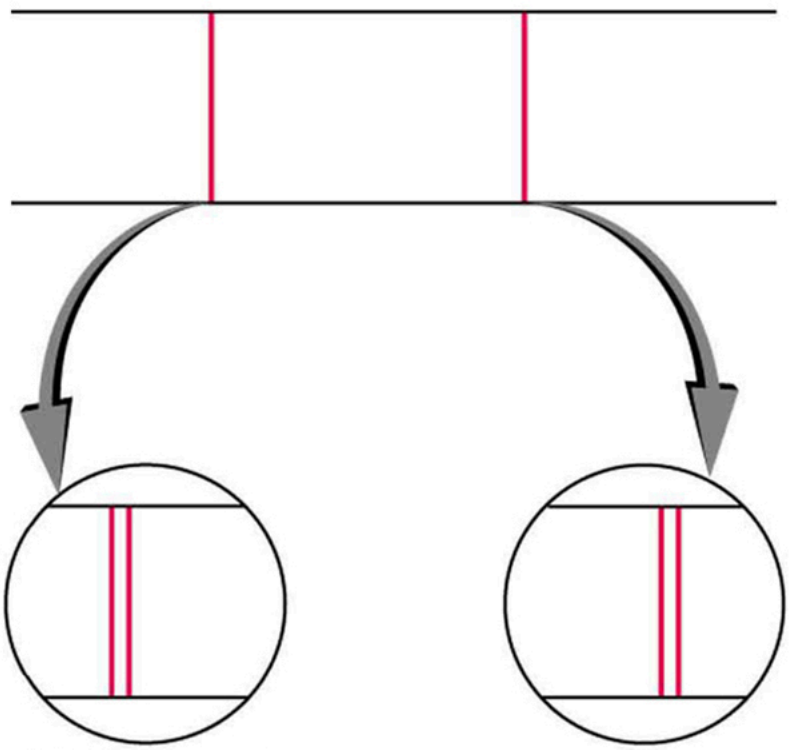
The approximate picture of an electron in a circular orbit illustrates how the current loop produces its own magnetic field, called B_{orb}. It also shows how B_{orb} is along the same line as the orbital angular momentum L_{orb}.



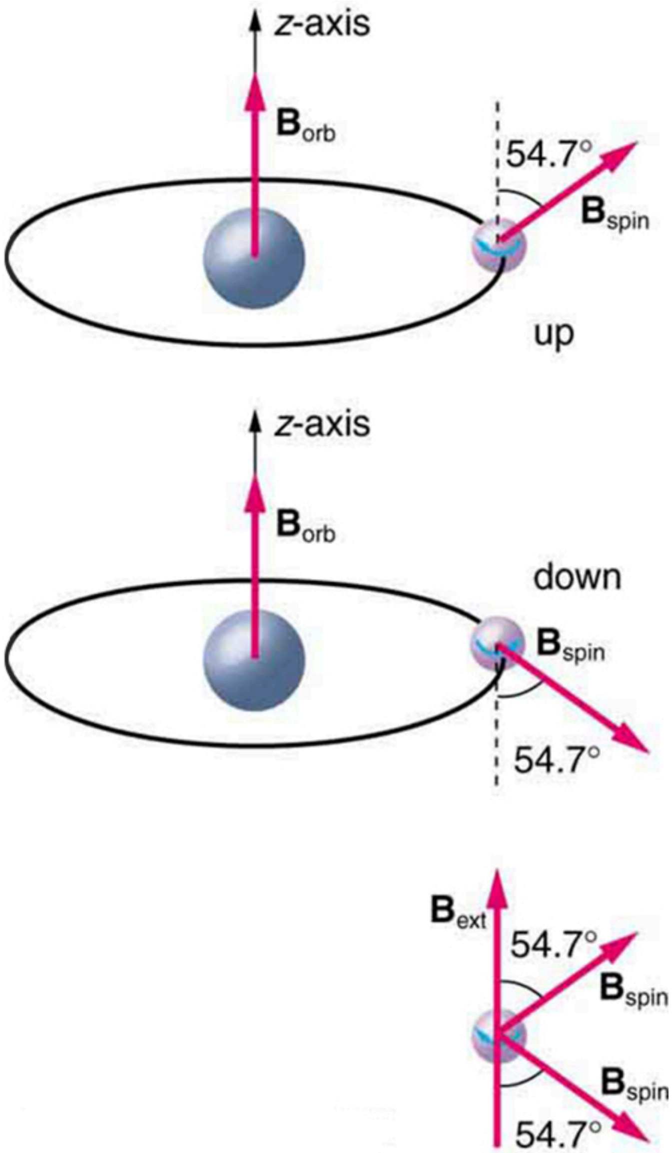
Only certain angles are allowed between the orbital angular momentum and an external magnetic field. This is implied by the fact that the Zeeman effect splits spectral lines into several discrete lines. Each line is associated with an angle between the external magnetic field and magnetic fields due to electrons and their orbits.

We already know that the magnitude of angular momentum is quantized for electron orbits in atoms. The new insight is that the *direction of the orbital angular momentum is also quantized*. The fact that the orbital angular momentum can have only certain directions is called **space quantization**. Like many aspects of quantum mechanics, this quantization of direction is totally unexpected. On the macroscopic scale, orbital angular momentum, such as that of the moon around the earth, can have any magnitude and be in any direction.

Detailed treatment of space quantization began to explain some complexities of atomic spectra, but certain patterns seemed to be caused by something else. As mentioned, spectral lines are actually closely spaced doublets, a characteristic called **fine structure**, as shown in [Figure 4]. The doublet changes when a magnetic field is applied, implying that whatever causes the doublet interacts with a magnetic field. In 1925, Sem Goudsmit and George Uhlenbeck, two Dutch physicists, successfully argued that electrons have properties analogous to a macroscopic charge spinning on its axis. Electrons, in fact, have an internal or intrinsic angular momentum called **intrinsic spin** $\vec{\mathbf{S}}$. Since electrons are charged, their intrinsic spin creates an **intrinsic magnetic field** $\vec{\mathbf{B}}_{\text{int}}$, which interacts with their orbital magnetic field $\vec{\mathbf{B}}_{\text{orb}}$. Furthermore, *electron* **intrinsic spin** is quantized in magnitude and direction*, analogous to the situation for orbital angular momentum. The spin of the electron can have only one magnitude, and its direction can be at only one of two angles relative to a magnetic field, as seen in [Figure 5]. We refer to this as spin up or spin down for the electron. Each spin direction has a different energy; hence, spectroscopic lines are split into two. Spectral doublets are now understood as being due to electron spin.



Fine structure. Upon close examination, spectral lines are doublets, even in the absence of an external magnetic field. The electron has an intrinsic magnetic field that interacts with its orbital magnetic field.



The intrinsic magnetic field B_{int} of an electron is attributed to its spin, S , roughly pictured to be due to its charge spinning on its axis. This is only a crude model, since electrons seem to have no size. The spin and intrinsic magnetic field of the electron can make only one of two angles with another magnetic field, such as that created by the electron's orbital motion. Space is quantized for spin as well as for orbital angular momentum.

These two new insights—that the direction of angular momentum, whether orbital or spin, is quantized, and that electrons have intrinsic spin—help to explain many of the complexities of atomic and molecular spectra. In magnetic resonance imaging, it is the way that the intrinsic magnetic field of hydrogen and biological atoms interact with an external field that underlies the diagnostic fundamentals.

Section Summary

- The Zeeman effect—the splitting of lines when a magnetic field is applied—is caused by other quantized entities in atoms.
- Both the magnitude and direction of orbital angular momentum are quantized.
- The same is true for the magnitude and direction of the intrinsic spin of electrons.

Conceptual Questions

What is the Zeeman effect, and what type of quantization was discovered because of this effect?

Glossary

Zeeman effect
: the effect of external magnetic fields on spectral lines

intrinsic spin
: the internal or intrinsic angular momentum of electrons

orbital angular momentum

an angular momentum that corresponds to the quantum analog of classical angular momentum

fine structure
: the splitting of spectral lines of the hydrogen spectrum when the spectral lines are examined at very high resolution

space quantization
: the fact that the orbital angular momentum can have only certain directions

intrinsic magnetic field
: the magnetic field generated due to the intrinsic spin of electrons

orbital magnetic field
: the magnetic field generated due to the orbital motion of electrons

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Quantum Numbers and Rules

- Define quantum number.
- Calculate angle of angular momentum vector with an axis.
- Define spin quantum number.

Physical characteristics that are quantized—such as energy, charge, and angular momentum—are of such importance that names and symbols are given to them. The values of quantized entities are expressed in terms of **quantum numbers**, and the rules governing them are of the utmost importance in determining what nature is and does. This section covers some of the more important quantum numbers and rules—all of which apply in chemistry, material science, and far beyond the realm of atomic physics, where they were first discovered. Once again, we see how physics makes discoveries which enable other fields to grow.

The *energy states of bound systems are quantized*, because the particle wavelength can fit into the bounds of the system in only certain ways. This was elaborated for the hydrogen atom, for which the allowed energies are expressed as $E_n \propto 1/n^2$, where $n = 1, 2, 3, ...$. We define n to be the principal quantum number that labels the basic states of a system. The lowest-energy state has $n = 1$, the first excited state has $n = 2$, and so on. Thus the allowed values for the principal quantum number are

$$n = 1, 2, 3, ...$$

This is more than just a numbering scheme, since the energy of the system, such as the hydrogen atom, can be expressed as some function of n , as can other characteristics (such as the orbital radii of the hydrogen atom).

The fact that the *magnitude of angular momentum is quantized* was first recognized by Bohr in relation to the hydrogen atom; it is now known to be true in general. With the development of quantum mechanics, it was found that the magnitude of angular momentum L can have only the values

$$L = \sqrt{l(l+1)}\frac{h}{2\pi} \quad (l = 0, 1, 2, ..., n-1),$$

where l is defined to be the **angular momentum quantum number**. The rule for l in atoms is given in the parentheses. Given n , the value of l can be any integer from zero up to $n - 1$. For example, if $n = 4$, then l can be 0, 1, 2, or 3.

Note that for $n = 1$, l can only be zero. This means that the ground-state angular momentum for hydrogen is actually zero, not $h/2\pi$ as Bohr proposed. The picture of circular orbits is not valid, because there would be angular momentum for any circular orbit. A more valid picture is the cloud of probability shown for the ground state of hydrogen in [Figure 3]. The electron actually spends time in and near the nucleus. The reason the electron does not remain in the nucleus is related to Heisenberg's uncertainty principle—the electron's energy would have to be much too large to be confined to the small space of the nucleus. Now the first excited state of hydrogen has $n = 2$, so that l can be either 0 or 1, according to the rule in $L = \sqrt{l(l+1)}\frac{h}{2\pi}$. Similarly, for $n = 3$, l can be 0, 1, or 2. It is often most convenient to state the value of l , a simple integer, rather than calculating the value of L from $L = \sqrt{l(l+1)}\frac{h}{2\pi}$. For example, for $l = 2$, we see that

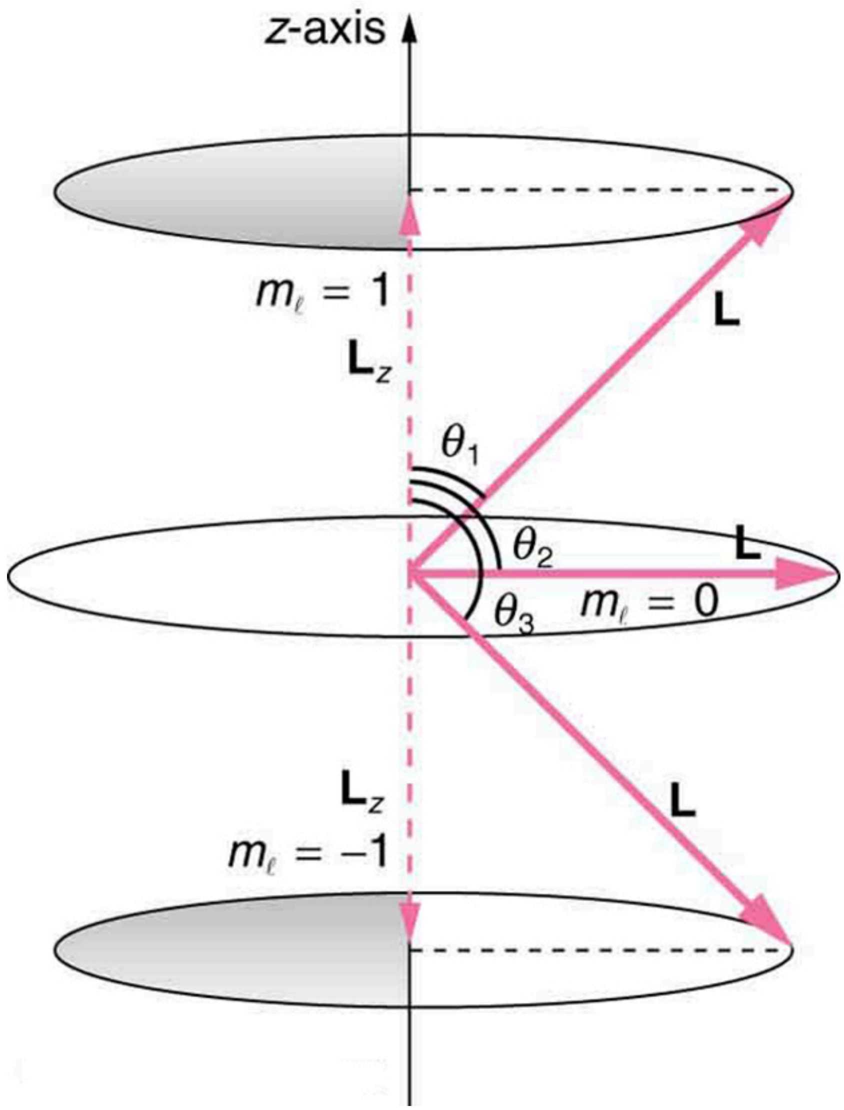
$$L = \sqrt{2(2+1)}\frac{h}{2\pi} = \sqrt{6}\frac{h}{2\pi} = 0.390h = 2.58 \times 10^{-34} \text{J} \cdot s.$$

It is much simpler to state $l = 2$.

As recognized in the Zeeman effect, the *direction of angular momentum is quantized*. We now know this is true in all circumstances. It is found that the component of angular momentum along one direction in space, usually called the z -axis, can have only certain values of L_z . The direction in space must be related to something physical, such as the direction of the magnetic field at that location. This is an aspect of relativity. Direction has no meaning if there is nothing that varies with direction, as does magnetic force. The allowed values of L_z are

$$L_z = m_l\frac{h}{2\pi} \quad (m_l = -l, -l+1, ..., -1, 0, 1, ... l-1, l),$$

where L_z is the z**-component of the angular momentum** and m_l is the angular momentum projection quantum number. The rule in parentheses for the values of m_l is that it can range from $-l$ to l in steps of one. For example, if $l = 2$, then m_l can have the five values –2, –1, 0, 1, and 2. Each m_l corresponds to a different energy in the presence of a magnetic field, so that they are related to the splitting of spectral lines into discrete parts, as discussed in the preceding section. If the z -component of angular momentum can have only certain values, then the angular momentum can have only certain directions, as illustrated in [Figure 1].



The component of a given angular momentum along the z-axis (defined by the direction of a magnetic field) can have only certain values; these are shown here for l=1, for which $m_\text{l}=-1, 0, \text{ and } +1$. The direction of L is quantized in the sense that it can have only certain angles relative to the z-axis.

What Are the Allowed Directions?

Calculate the angles that the angular momentum vector $\vec{\mathbf{L}}$ can make with the z -axis for $l=1$, as illustrated in [Figure 1].

Strategy

[Figure 1] represents the vectors $\vec{\mathbf{L}}$ and $\vec{\mathbf{L}}_z$ as usual, with arrows proportional to their magnitudes and pointing in the correct directions. $\vec{\mathbf{L}}$ and $\vec{\mathbf{L}}_z$ form a right triangle, with $\vec{\mathbf{L}}$ being the hypotenuse and $\vec{\mathbf{L}}_z$ the adjacent side. This means that the ratio of $\vec{\mathbf{L}}_z$ to $\vec{\mathbf{L}}$ is the cosine of the angle of interest. We can find $\vec{\mathbf{L}}$ and $\vec{\mathbf{L}}_z$ using $L=\sqrt{l(l+1)}\frac{h}{2\pi}$ and $L_z=m\frac{h}{2\pi}$.

Solution

We are given $l=1$, so that m_l can be +1, 0, or −1. Thus L has the value given by $L=\sqrt{l(l+1)}\frac{h}{2\pi}$.

$$L=\sqrt{l(l+1)}\frac{h}{2\pi}=\sqrt{2}\frac{h}{2\pi}$$

L_z can have three values, given by $L_z=m_l\frac{h}{2\pi}$.

$$L_z=m_l\frac{h}{2\pi}=\begin{cases}\frac{h}{2\pi}, & m_l=+1\\ 0, & m_l=0\\ -\frac{h}{2\pi}, & m_l=-1\end{cases}$$

As can be seen in [Figure 1], $\cos\theta = L_z L$, and so for $m_l = +1$, we have

$$\cos\theta_1 = L_Z L = h 2\pi \sqrt{2} h 2\pi = 1\sqrt{2} = 0.707.$$

Thus,

$$\theta_1 = \cos^{-1} 0.707 = 45.0º.$$

Similarly, for $m_l = 0$, we find $\cos\theta_2 = 0$; thus,

$$\theta_2 = \cos^{-1} 0 = 90.0º.$$

And for $m_l = -1$,

$$\cos\theta_3 = L_Z L = -h 2\pi \sqrt{2} h 2\pi = -1\sqrt{2} = -0.707,$$

so that

$$\theta_3 = \cos^{-1}(-0.707) = 135.0º.$$

Discussion

The angles are consistent with the figure. Only the angle relative to the z -axis is quantized. L can point in any direction as long as it makes the proper angle with the z -axis. Thus the angular momentum vectors lie on cones as illustrated. This behavior is not observed on the large scale. To see how the correspondence principle holds here, consider that the smallest angle (θ_1 in the example) is for the maximum value of $m_l = 0$, namely $m_l = l$. For that smallest angle,

$$\cos\theta = L_z L = l \sqrt{l(l+1)},$$

which approaches 1 as l becomes very large. If $\cos\theta = 1$, then $\theta = 0$ º. Furthermore, for large l, there are many values of m_l, so that all angles become possible as l gets very large.

Intrinsic Spin Angular Momentum Is Quantized in Magnitude and Direction

There are two more quantum numbers of immediate concern. Both were first discovered for electrons in conjunction with fine structure in atomic spectra. It is now well established that electrons and other fundamental particles have _ intrinsic spin_, roughly analogous to a planet spinning on its axis. This spin is a fundamental characteristic of particles, and only one magnitude of intrinsic spin is allowed for a given type of particle. Intrinsic angular momentum is quantized independently of orbital angular momentum. Additionally, the direction of the spin is also quantized. It has been found that the **magnitude of the intrinsic (internal) spin angular momentum**, S, of an electron is given by

$$S = \sqrt{s(s+1)} h 2\pi (s = 1/2 \text{for electrons}),$$

where s is defined to be the **spin quantum number**. This is very similar to the quantization of L given in $L = \sqrt{l(l+1)} h 2\pi$, except that the only value allowed for s for electrons is 1/2.

The *direction of intrinsic spin is quantized*, just as is the direction of orbital angular momentum. The direction of spin angular momentum along one direction in space, again called the z -axis, can have only the values

$$S_z = m_s h 2\pi (m_s = -12, +12)$$

for electrons. S_z is the ** z -component of spin angular momentum** and m_s is the **spin projection quantum number**. For electrons, s can only be 1/2, and m_s can be either +1/2 or -1/2. Spin projection $m_s = +1/2$ is referred to as *spin up*, whereas $m_s = -1/2$ is called *spin down*. These are illustrated in [Figure 5].

Intrinsic Spin

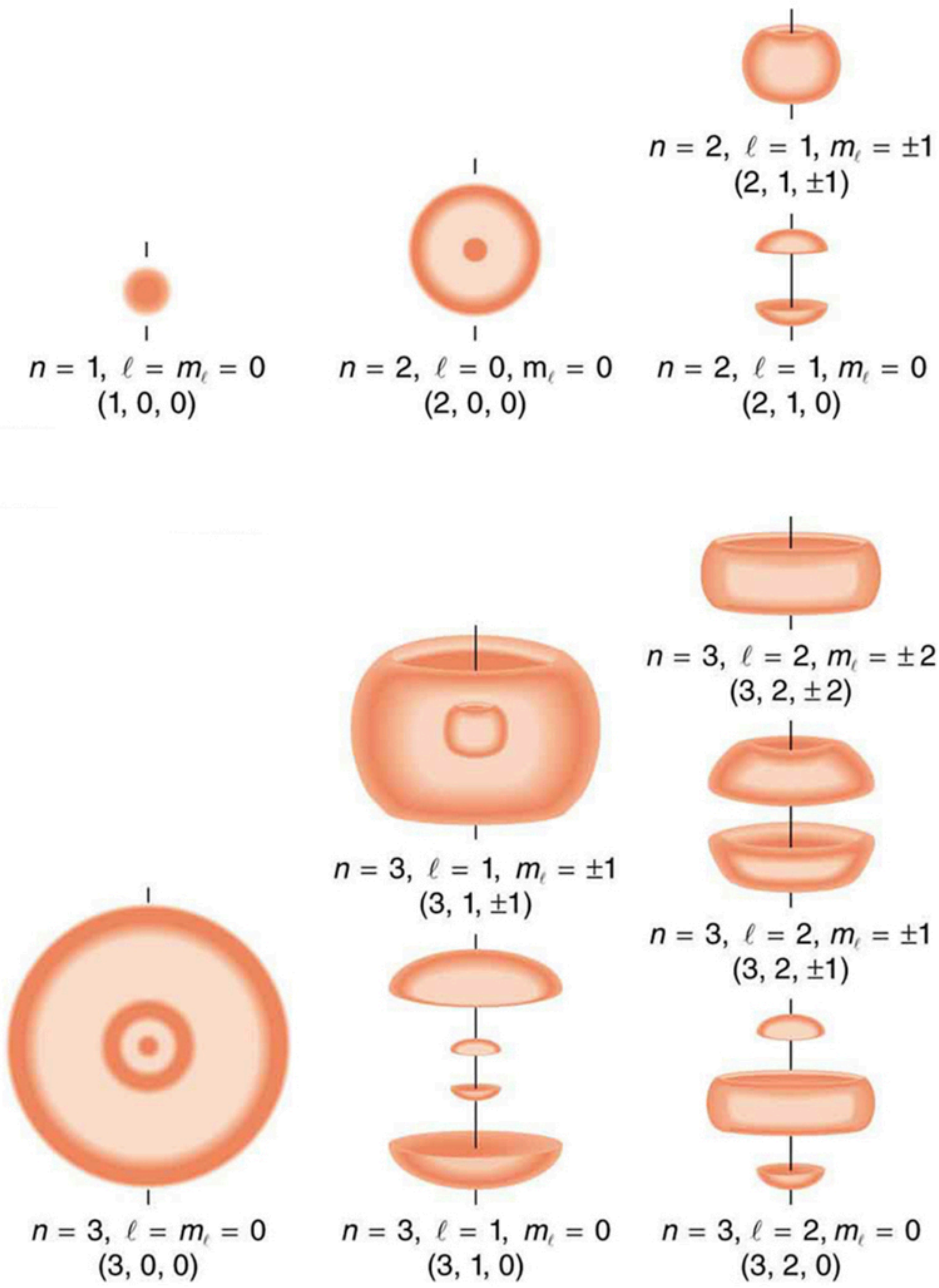
In later chapters, we will see that intrinsic spin is a characteristic of all subatomic particles. For some particles s is half-integral, whereas for others s is integral—there are crucial differences between half-integral spin particles and integral spin particles. Protons and neutrons, like electrons, have $s = 1/2$, whereas photons have $s = 1$, and other particles called pions have $s = 0$, and so on.

To summarize, the state of a system, such as the precise nature of an electron in an atom, is determined by its particular quantum numbers. These are expressed in the form (n, l, m_l, m_s) —see [Table 1] *For electrons in atoms*, the principal quantum number can have the values $n = 1, 2, 3, ...$. Once n is known, the values of the angular momentum quantum number are limited to $l = 1, 2, 3, ..., n - 1$. For a given value of l, the angular momentum projection quantum number can have only the values $m_l = -l, -l + 1, ..., -1, 0, 1, ..., l - 1, l$. Electron spin is independent of n,l, and m_l, always having $s = 1/2$. The spin projection quantum number can have two values, $m\_s = 1/2$ or $-1/2$.

Atomic Quantum Numbers

Name	Symbol	Allowed values
Principal quantum number	n	1,2,3,...
Angular momentum	l	0,1,2,...$n-1$
Angular momentum projection	m_l	$-l,-l+1,...,-1,0,1,...,l-1,l$ (or 0,±1,±2,...,$\pm l$)
Spin[1]	s	1/2(electrons)
Spin projection	m_s	−1/2,+1/2

[Figure 2] shows several hydrogen states corresponding to different sets of quantum numbers. Note that these clouds of probability are the locations of electrons as determined by making repeated measurements—each measurement finds the electron in a definite location, with a greater chance of finding the electron in some places rather than others. With repeated measurements, the pattern of probability shown in the figure emerges. The clouds of probability do not look like nor do they correspond to classical orbits. The uncertainty principle actually prevents us and nature from knowing how the electron gets from one place to another, and so an orbit really does not exist as such. Nature on a small scale is again much different from that on the large scale.



Probability clouds for the electron in the ground state and several excited states of hydrogen. The nature of these states is determined by their sets of quantum numbers, here given as n,l,m_l. The ground state is (0, 0, 0); one of the possibilities for the second excited state is (3, 2, 1). The probability of finding the electron is indicated by the shade of color; the darker the coloring the greater the chance of finding the electron.

We will see that the quantum numbers discussed in this section are valid for a broad range of particles and other systems, such as nuclei. Some quantum numbers, such as intrinsic spin, are related to fundamental classifications of subatomic particles, and they obey laws that will give us further insight into the substructure of matter and its interactions.

PhET Explorations: Stern-Gerlach Experiment

The classic Stern-Gerlach Experiment shows that atoms have a property called spin. Spin is a kind of intrinsic angular momentum, which has no classical counterpart. When the z-component of the spin is measured, one always gets one of two values: spin up or spin down.

At least Flash Player 8 required to run this simulation.

No Flash Player was detected

Attempt to view the simulation anyways

Section Summary

- Quantum numbers are used to express the allowed values of quantized entities. The principal quantum number n labels the basic states of a system and is given by

  $$n = 1, 2, 3, \ldots.$$

- The magnitude of angular momentum is given by

  $$L = \sqrt{l(l+1)}\frac{h}{2\pi} \quad (l = 0, 1, 2, \ldots, n-1),$$

  where l is the angular momentum quantum number. The direction of angular momentum is quantized, in that its component along an axis defined by a magnetic field, called the z -axis is given by

  $$L_z = m_l\frac{h}{2\pi} \quad (m_l = -l, -l+1, \ldots, -1, 0, 1, \ldots l-1, l),$$

  where L_z is the z -component of the angular momentum and m_l is the angular momentum projection quantum number. Similarly, the electron's intrinsic spin angular momentum S is given by

  $$S = \sqrt{s(s+1)}\frac{h}{2\pi} \quad (s = 1/2 \text{ for electrons}).$$

  s is defined to be the spin quantum number. Finally, the direction of the electron's spin along the z -axis is given by

  $$S_z = m_s\frac{h}{2\pi} \quad \left(m_s = -\frac{1}{2}, +\frac{1}{2}\right),$$

  where S_z is the z -component of spin angular momentum and m_s is the spin projection quantum number. Spin projection $m_s = +1/2$ is referred to as spin up, whereas $m_s = -1/2$ is called spin down. [Table 1] summarizes the atomic quantum numbers and their allowed values.

Conceptual Questions

Define the quantum numbers n, l, m_l, s , and m_s.

For a given value of n , what are the allowed values of l ?

For a given value of l , what are the allowed values of m_l ? What are the allowed values of m_l for a given value of n ? Give an example in each case.

List all the possible values of s and m_s for an electron. Are there particles for which these values are different? The same?

Problem Exercises

If an atom has an electron in the $n = 5$ state with $m_l = 3$, what are the possible values of l ?

Show Solution

Strategy

We need to apply the rules for quantum numbers. The angular momentum quantum number l must satisfy two constraints: (1) For a given principal quantum number n, the allowed values of l are $l = 0, 1, 2, ..., n - 1$; and (2) The angular momentum projection quantum number m_l is constrained by $$ m_l \le l .We need to find which values of l satisfy both conditions when n = 5 and m_l = 3 $$.

Solution

Given that $n = 5$, the first constraint tells us that:

l=0,1,2,3, or 4

Given that $m_l = 3$, the second constraint requires:

$|m_l| \le l$
$3 \le l$

Combining both constraints, we need $l \ge 3$ and $l \le 4$ (since $l < n = 5$).

Therefore, the possible values are: **l = 3 or l = 4**

If $l = 3$ (f-orbital), then m_l can be –3, –2, –1, 0, +1, +2, or +3, so $m_l = 3$ is allowed.

If $l = 4$ (g-orbital), then m_l can be –4, –3, –2, –1, 0, +1, +2, +3, or +4, so $m_l = 3$ is also allowed.

Discussion

This problem demonstrates the hierarchical relationship between quantum numbers in atoms. The principal quantum number $n = 5$ places an upper limit on l ($l \le 4$), while the value of $m_l = 3$ places a lower limit on l ($l \ge 3$). The fact that $m_l = 3$ rules out $l = 0, 1, 2$ is physically reasonable: m_l represents the component of angular momentum along a specific axis, and this component cannot exceed the total angular momentum magnitude (which depends on l). In spectroscopic notation, $l = 3$ corresponds to an f-orbital (named historically: s, p, d, f for $l = 0, 1, 2, 3$), and $l = 4$ corresponds to a g-orbital. These higher angular momentum states first appear in atoms with many electrons. For example, f-orbitals begin to fill in the lanthanide elements (starting with cerium, element 58), while g-orbitals are not occupied in the ground states of known elements but can be accessed in excited states. Understanding which quantum number combinations are allowed is crucial for predicting electron configurations, chemical bonding, and spectroscopic properties of atoms.

An atom has an electron with $m_l = 2$. What is the smallest value of n for this electron?

Show Solution

Strategy

The angular momentum projection quantum number m_l is constrained by the angular momentum quantum number l through the relationship $$ m_l \le l .Furthermore, for a given principal quantum number n, the angular momentum quantum number can only take values l = 0, 1, 2, …, n-1 .Therefore, we need to find the minimum value of n such that l can be large enough to accommodate m_l = 2 $$.

Solution

Since $m_l = 2$, we need $l \ge 2$ (because $$ m_l \le l $$).

The minimum value of l that satisfies this is $l = 2$.

For $l = 2$ to be an allowed value, we need $l \le n - 1$, which gives us $2 \le n - 1$, or $n \ge 3$.

Therefore, the smallest value of n is **n = 3**.

Discussion

This problem illustrates the hierarchical relationship among the quantum numbers in atoms. The principal quantum number n determines the energy level and sets an upper limit on the angular momentum quantum number l . In turn, l determines the magnitude of the orbital angular momentum and sets limits on the possible values of m_l , which specifies the orientation of the angular momentum vector. For $n = 3$, the allowed values of l are 0, 1, and 2. When $l = 2$ (called a d-orbital), m_l can take any of the five values: –2, –1, 0, +1, or +2. Each of these corresponds to a different orientation of the electron's orbital angular momentum relative to an external magnetic field. The fact that we need at least $n = 3$ for $m_l = 2$ explains why d-orbitals first appear in the third shell of atoms (the M shell), which is crucial for understanding the electron configurations of transition metals and their chemical properties.

What are the possible values of m_l for an electron in the $n = 4$ state?

Show Solution

Strategy

For a given principal quantum number n , the angular momentum quantum number l can range from 0 to $n - 1$. For each value of l , the angular momentum projection quantum number m_l can range from $-l$ to $+l$ in integer steps. We need to determine all possible values of m_l for any allowed l when $n = 4$.

Solution

For $n = 4$, the allowed values of l are:

$$l = 0, 1, 2, 3$$

For each value of l , the allowed values of m_l are:

For l = 0 (4s orbital):

$$m_l = 0$$

For l = 1 (4p orbitals):

$$m_l = -1, 0, +1$$

For l = 2 (4d orbitals):

$$m_l = -2, -1, 0, +1, +2$$

For l = 3 (4f orbitals):

$$m_l = -3, -2, -1, 0, +1, +2, +3$$

Combining all possibilities, the complete set of possible values of m_l for $n = 4$ is:

$$m_l = -3, -2, -1, 0, +1, +2, +3$$

Or more compactly: $\boldsymbol{m_l = 0, \pm 1, \pm 2, \pm 3}$

Discussion

This problem reveals an important feature of atomic structure: for a given principal quantum number n , there are multiple orbitals with different angular momentum properties. The $n = 4$ shell contains 4s (1 orbital), 4p (3 orbitals), 4d (5 orbitals), and 4f (7 orbitals), for a total of 16 orbitals. Since each orbital can hold two electrons (with opposite spins), the $n = 4$ shell can accommodate up to 32 electrons. The seven distinct values of m_l (–3 to +3) represent different orientations of the orbital angular momentum vector relative to an external magnetic field (the z -axis). In the absence of an external field, orbitals with the same n and l but different m_l values have the same energy (they are "degenerate"). However, when an external magnetic field is applied, this degeneracy is broken—the Zeeman effect causes each m_l value to correspond to a slightly different energy. This is why spectral lines split in magnetic fields. The maximum value of $$ m_l *equalsthemaximumvalueof* l *forthatshell,whichis* n - 1 $$. Understanding these relationships is essential for explaining atomic spectra, magnetic properties of atoms, and the structure of the periodic table.

What, if any, constraints does a value of $m_l = 1$ place on the other quantum numbers for an electron in an atom?

Show Solution

Strategy

We need to examine the relationships between the quantum numbers to determine what constraints $m_l = 1$ places on the other quantum numbers (n , l , and m_s). The key relationships are: (1) $$ m_l \le l ,(2) l \le n - 1 ,and(3) m_s $$ is independent of the orbital quantum numbers.

Solution

Given that $m_l = 1$:

Constraint on l: Since $$ m_l \le l ,wemusthave l \ge 1 .Therefore,theelectroncannotbeinans−orbital(l = 0)butcouldbeinap−orbital(l = 1),d−orbital(l = 2),f−orbital(l = 3 $$), or any higher angular momentum state.

Constraint on n: Since $l \geq 1$ and $l \leq n - 1$, we need $n - 1 \geq 1$, which gives $n \geq 2$. Therefore, the electron must be in the second shell or higher ($n = 2, 3, 4, \ldots$).

Constraint on m_s: The spin projection quantum number m_s can be either +1/2 or –1/2, independent of the value of m_l . There is no constraint on m_s .

In summary:

- $l \geq 1$ (p, d, f, or higher orbitals)
- $n \geq 2$ (second shell or higher)
- $m_s = \pm 1/2$ (no constraint)

Discussion

This problem demonstrates the interconnected nature of quantum numbers in atoms. The value of $m_l = 1$ tells us about the orientation of the electron’s orbital angular momentum but doesn’t uniquely specify the state of the electron. For example, the electron could be in a 2p orbital ($n = 2, l = 1$), a 3p orbital ($n = 3, l = 1$), a 3d orbital ($n = 3, l = 2$), or many other possibilities. Each of these states would have different energies and different probability distributions for the electron’s position. The fact that $m_l = 1$ excludes only s-orbitals ($l = 0$) but allows all others shows that m_l specifies orientation rather than the magnitude of angular momentum. The independence of m_s reflects the fact that electron spin is an intrinsic property not related to orbital motion. Understanding these constraints is essential for writing electron configurations and predicting the chemical and spectroscopic properties of atoms.

(a) Calculate the magnitude of the angular momentum for an $l = 1$ electron. (b) Compare your answer to the value Bohr proposed for the $n = 1$ state.

Show Solution

Strategy

(a) The magnitude of angular momentum is given by the quantum mechanical formula $L = \sqrt{l(l+1)}\frac{h}{2\pi}$, where l is the angular momentum quantum number and $h = 6.626 \times 10^{-34}$ J·s is Planck’s constant. We’ll substitute $l = 1$ and calculate.

(b) Bohr’s original model proposed that the angular momentum of an electron in orbit is quantized as $L = n\frac{h}{2\pi}$, where n is the principal quantum number. For the $n = 1$ state (ground state), Bohr predicted $L = \frac{h}{2\pi}$. We’ll calculate this value and compare it to the quantum mechanical result.

Solution

(a) For $l = 1$:

$$L=\sqrt{l(l+1)}\frac{h}{2\pi}=\sqrt{1(1+1)}\frac{h}{2\pi}=\sqrt{2}\frac{h}{2\pi}$$
$$L=\sqrt{2}\times\frac{6.626\times10^{-34}\text{ J·s}}{2\pi}=\sqrt{2}\times1.055\times10^{-34}\text{ J·s}$$
$$L=1.49\times10^{-34}\text{ J·s}$$

(b) Bohr’s value for the $n = 1$ state:

$$L_{\text{Bohr}}=n\frac{h}{2\pi}=1\times\frac{6.626\times10^{-34}\text{ J·s}}{2\pi}=1.055\times10^{-34}\text{ J·s}$$
$$L_{\text{Bohr}}=1.06\times10^{-34}\text{ J·s}$$

Comparing the two values:

$$\frac{L_{l=1}}{L_{\text{Bohr},n=1}}=\frac{1.49\times10^{-34}}{1.06\times10^{-34}}=1.41=\sqrt{2}$$

The quantum mechanical value for $l=1$ is $\sqrt{2}$ times larger than Bohr's prediction for $n=1$.

Discussion

This comparison reveals a fundamental difference between Bohr's semi-classical model and the full quantum mechanical treatment of the atom. Bohr correctly predicted that angular momentum is quantized, but his specific formula $L=n\frac{h}{2\pi}$ turned out to be incorrect. Modern quantum mechanics shows that (1) the magnitude of angular momentum depends on l, not n, and follows $L=\sqrt{l(l+1)}\frac{h}{2\pi}$; and (2) for a given n, l can range from 0 to $n-1$, meaning different orbitals in the same shell have different angular momenta.

Most surprisingly, quantum mechanics reveals that the ground state of hydrogen ($n=1, l=0$) has **zero** angular momentum, not $\frac{h}{2\pi}$ as Bohr proposed! This is completely incompatible with Bohr's picture of electrons orbiting in circular paths. Instead, the $l=0$ state (s-orbital) is spherically symmetric with no orbital angular momentum at all—the electron's probability cloud is centered on the nucleus with no preferred direction.

The $l=1$ state we calculated represents a p-orbital, which has angular momentum and a more complex spatial structure. The factor $\sqrt{l(l+1)}$ rather than just l appears because of the way angular momentum operators work in quantum mechanics—this same factor appears for all quantized angular momenta. The comparison shows that while Bohr's model was revolutionary and correct in predicting quantization, the details of quantum mechanics are more subtle and accurate. Nevertheless, Bohr's achievement in introducing quantization to atomic physics was a crucial step toward the modern theory.

(a) What is the magnitude of the angular momentum for an $l=1$ electron? (b) Calculate the magnitude of the electron's spin angular momentum. (c) What is the ratio of these angular momenta?

Show Solution

Strategy

We'll use the formulas for quantized angular momentum. For orbital angular momentum: $L=\sqrt{l(l+1)}\frac{h}{2\pi}$. For spin angular momentum: $S=\sqrt{s(s+1)}\frac{h}{2\pi}$, where $s=1/2$ for electrons. We'll calculate both values and then find their ratio.

Solution

(a) For $l=1$, the magnitude of the orbital angular momentum is:

$$L=\sqrt{l(l+1)}\frac{h}{2\pi}=\sqrt{1(1+1)}\frac{h}{2\pi}=\sqrt{2}\frac{h}{2\pi}$$

Substituting $h=6.626\times10^{-34}$ J·s :

$$L=\sqrt{2}\frac{6.626\times10^{-34}\text{ J·s}}{2\pi}=\sqrt{2}(1.055\times10^{-34}\text{ J·s})=1.49\times10^{-34}\text{ J·s}$$

(b) For electron spin with $s=1/2$:

$$S=\sqrt{s(s+1)}\frac{h}{2\pi}=\sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)}\frac{h}{2\pi}=\sqrt{\frac{1}{2}\cdot\frac{3}{2}}\frac{h}{2\pi}=\sqrt{\frac{3}{4}}\frac{h}{2\pi}$$
$$S=\frac{\sqrt{3}}{2}\frac{h}{2\pi}=\frac{\sqrt{3}}{2}(1.055\times10^{-34}\text{ J·s})=9.13\times10^{-35}\text{ J·s}$$

(c) The ratio of orbital to spin angular momentum is:

$$\frac{L}{S}=\frac{\sqrt{2}\frac{h}{2\pi}}{\frac{\sqrt{3}}{2}\frac{h}{2\pi}}=\frac{\sqrt{2}}{\frac{\sqrt{3}}{2}}=\frac{2\sqrt{2}}{\sqrt{3}}=\sqrt{\frac{4\cdot2}{3}}=\sqrt{\frac{8}{3}}=1.63$$

Discussion

This calculation reveals several important aspects of atomic structure. First, both orbital and spin angular momenta are on the order of 10^{-34} J·s , which is the same order as Planck's constant divided by 2π (often written as $\hbar$). This demonstrates that atomic-scale angular momentum is quantized in units of $\hbar$. Second, the ratio of about 1.63 shows that for an $l=1$ electron (a p-orbital), the orbital angular momentum is somewhat larger than the spin angular momentum, but they are of the same order of magnitude. This comparability is important because both contribute to the total angular momentum of the atom and both interact with external magnetic fields. The spin-orbit coupling—the interaction between the electron's orbital and spin magnetic moments—leads to fine structure in atomic spectra. For different values of l, this ratio changes: for $l=3$ (next problem), the ratio is 4, showing that as l increases, orbital angular momentum dominates over spin. Understanding these relationships is crucial for explaining magnetic properties of atoms, spectroscopic splitting patterns, and the behavior of atoms in external magnetic fields.

Repeat [the previous exercise] for $l=3$.

Show Solution

Strategy

Following the same approach as Exercise 7, we'll calculate: (a) the magnitude of orbital angular momentum for $l=3$ using $L=\sqrt{l(l+1)}\frac{h}{2\pi}$; (b) the magnitude of electron spin angular momentum using $S=\sqrt{s(s+1)}\frac{h}{2\pi}$ where $s=1/2$; (c) the ratio of these two quantities.

Solution

(a) For $l=3$, the magnitude of orbital angular momentum is:

$$L=\sqrt{l(l+1)}\frac{h}{2\pi}=\sqrt{3(3+1)}\frac{h}{2\pi}=\sqrt{12}\frac{h}{2\pi}$$
$$L=\sqrt{12}\times\frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{2\pi}=\sqrt{12}\times(1.055\times10^{-34}\ \text{J}\cdot\text{s})$$
$$L=3.464\times(1.055\times10^{-34}\ \text{J}\cdot\text{s})=3.66\times10^{-34}\ \text{J}\cdot\text{s}$$

(b) The electron spin angular momentum (same for all electrons regardless of l):

$$S=\sqrt{s(s+1)}\frac{h}{2\pi}=\sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)}\frac{h}{2\pi}=\sqrt{\frac{3}{4}}\frac{h}{2\pi}$$
$$S=\frac{\sqrt{3}}{2}\times(1.055\times10^{-34}\ \text{J}\cdot\text{s})=9.13\times10^{-35}\ \text{J}\cdot\text{s}$$

(c) The ratio of orbital to spin angular momentum:

$$\frac{L}{S}=\frac{\sqrt{12}\frac{h}{2\pi}}{\sqrt{3/4}\frac{h}{2\pi}}=\frac{\sqrt{12}}{\sqrt{3/4}}=\frac{\sqrt{12}\times2}{\sqrt{3}}=\frac{2\sqrt{12}}{\sqrt{3}}=\frac{2\times2\sqrt{3}}{\sqrt{3}}=4$$

The ratio is **4**.

Discussion

Comparing this result to Exercise 7 (where $l=1$ gave a ratio of 1.63), we see that as l increases, the orbital angular momentum becomes much larger relative to the spin angular momentum. For $l=3$ (an f-orbital), the orbital angular momentum is exactly 4 times the spin angular momentum, whereas for $l=1$ (a p-orbital) it was only 1.63 times larger. This makes sense because orbital angular momentum scales as $\sqrt{l(l+1)}$, which grows roughly as $\sqrt{l^2}=l$ for large l , while spin angular momentum is fixed at $\sqrt{3/4}\frac{h}{2\pi}$ for all electrons. The $l=3$ state corresponds to f-orbitals, which first appear in the fourth shell (n = 4) and are important in the chemistry of lanthanide and actinide elements. The large orbital angular momentum of f-electrons contributes significantly to the magnetic properties of these elements. Both orbital and spin angular momenta couple together to give the total angular momentum of the atom, and this coupling is responsible for fine structure in atomic spectra and for the magnetic moments of atoms. The fact that orbital angular momentum can vary widely (depending on l) while spin is always the same ($s=1/2$ for electrons) means that the relative importance of spin-orbit coupling varies greatly for different orbitals—it's more important for s and p electrons (low l) than for d and f electrons (high l) in terms of the fractional contribution.

(a) How many angles can L make with the z-axis for an $l=2$ electron? (b) Calculate the value of the smallest angle.

Show Solution

Strategy

For a given value of l , the angular momentum projection quantum number m_l can take values from $-l$ to $+l$ in integer steps, giving $2l+1$ possible values. Each value of m_l corresponds to a different angle that the angular momentum vector $\mathbf{L}$ makes with the z -axis. The angle can be found from $\cos\theta=\frac{L_z}{L}=\frac{m_l}{\sqrt{l(l+1)}}$.

Solution

(a) For $l=2$, the possible values of m_l are:

$$m_l=-2,-1,0,+1,+2$$

This gives $2l+1=2(2)+1=5$ possible orientations. Therefore, $\mathbf{L}$ can make **5 different angles** with the z -axis.

(b) The smallest angle occurs when m_l has its maximum value, $m_l=+2$ (or minimum value $m_l=-2$, which gives the same angle by symmetry). The magnitude of $\mathbf{L}$ is:

$$L=\sqrt{l(l+1)}\frac{h}{2\pi}=\sqrt{2(2+1)}\frac{h}{2\pi}=\sqrt{6}\frac{h}{2\pi}$$

The z -component is:

$$L_z=m_l\frac{h}{2\pi}=2\frac{h}{2\pi}$$

The angle is given by:

$$\cos\theta=\frac{L_z}{L}=\frac{2\frac{h}{2\pi}}{\sqrt{6}\frac{h}{2\pi}}=\frac{2}{\sqrt{6}}=\sqrt{\frac{4}{6}}=\sqrt{\frac{2}{3}}=0.8165$$
$$\theta=\cos^{-1}(0.8165)=35.3°$$

The smallest angle is **35.3°** (or **0.616 radians**).

Discussion

This problem illustrates the concept of space quantization—the remarkable fact that the angular momentum vector can only point in certain discrete directions relative to an external reference direction (the z -axis, typically defined by an external magnetic field). For $l = 2$ (a d-orbital), the five allowed angles correspond to the five d-orbitals in chemistry ($d_{z^2}, d_{xz}, d_{yz}, d_{xy}, d_{x^2-y^2}$). The smallest angle of 35.3° represents the most closely aligned orientation—even at this “closest” alignment, the angular momentum vector cannot point directly along the z -axis. This is fundamentally different from classical physics, where a rotating object could have its angular momentum pointing in any direction. The fact that even the maximum value of $m_l = l$ doesn’t allow perfect alignment ($\theta = 0°$) is due to the quantum mechanical relationship $L = \sqrt{l(l+1)}\frac{h}{2\pi}$ rather than $l\frac{h}{2\pi}$. This “ $\sqrt{l(l+1)}$ “ factor, rather than just “ l ,” ensures that $L > L_z$ for all states, preventing perfect alignment and reflecting the uncertainty principle’s constraint on simultaneously knowing all components of angular momentum.

What angles can the spin S of an electron make with the z -axis?

Show Solution

Strategy

For electron spin, the spin quantum number is $s = 1/2$, and the spin projection quantum number m_s can be either $+1/2$ (spin up) or $-1/2$ (spin down). The magnitude of the spin angular momentum is $S = \sqrt{s(s+1)}\frac{h}{2\pi}$, and its z -component is $S_z = m_s\frac{h}{2\pi}$. The angle between $\mathbf{S}$ and the z -axis can be found from $\cos\theta = \frac{S_z}{S}$.

Solution

The magnitude of electron spin is:

$$S=\sqrt{s(s+1)}\frac{h}{2\pi}=\sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)}\frac{h}{2\pi}=\sqrt{\frac{3}{4}}\frac{h}{2\pi}=\frac{\sqrt{3}}{2}\frac{h}{2\pi}$$

The z -component can have two values:

$$S_z=m_s\frac{h}{2\pi}=\pm\frac{1}{2}\frac{h}{2\pi}$$

For $m_s = +1/2$ (spin up):

$$\cos\theta_1=\frac{S_z}{S}=\frac{+\frac{1}{2}\frac{h}{2\pi}}{\frac{\sqrt{3}}{2}\frac{h}{2\pi}}=\frac{1/2}{\sqrt{3}/2}=\frac{1}{\sqrt{3}}=0.5774$$
$$\theta_1=\cos^{-1}(0.5774)=54.7°$$

For $m_s = -1/2$ (spin down):

$$\cos\theta_2=\frac{S_z}{S}=\frac{-\frac{1}{2}\frac{h}{2\pi}}{\frac{\sqrt{3}}{2}\frac{h}{2\pi}}=\frac{-1/2}{\sqrt{3}/2}=\frac{-1}{\sqrt{3}}=-0.5774$$
$$\theta_2=\cos^{-1}(-0.5774)=125.3°$$

The electron spin can make only two angles with the z -axis: **54.7°** and **125.3°**.

Discussion

This result demonstrates the quantization of spin direction—a purely quantum mechanical phenomenon with no classical analogue. Unlike orbital angular momentum, where the number of allowed angles depends on l (giving $2l + 1$ orientations), electron spin always has exactly two allowed orientations regardless of the atom or state. These two orientations are commonly called “spin up” and “spin down,” though these names are somewhat misleading since the spin is never aligned parallel or antiparallel to the z -axis.

Notice that even for the “spin up” state ($m_s = +1/2$), the angle is 54.7° rather than 0°—the spin cannot point directly along the z -axis. This is a consequence of the $\sqrt{s(s+1)}$ factor in the magnitude formula: since $S = \frac{\sqrt{3}}{2}\frac{h}{2\pi}$ is always larger than $$ S_z = \frac{1}{2} \frac{h}{2\pi} $$, perfect alignment is impossible. This reflects a fundamental aspect of quantum mechanics: you cannot simultaneously know all three components of angular momentum (Heisenberg uncertainty principle for angular momentum).

The two angles are symmetric about 90°: one is 90° – 35.3° = 54.7°, and the other is 90° + 35.3° = 125.3°. This symmetry reflects the fact that the two spin states have equal magnitude angular momenta, just pointing in generally opposite directions.

Electron spin is fundamental to understanding: (1) the Pauli exclusion principle—two electrons in the same orbital must have opposite spins; (2) magnetic properties of materials—unpaired electron spins create magnetic moments; (3) fine structure in atomic spectra—spin-orbit coupling splits spectral lines; (4) modern technologies like MRI, which depends on nuclear spin (similar to electron spin but for protons), and spintronics, which exploits electron spin for information storage and processing.

Footnotes

- 1 The spin quantum number *s* is usually not stated, since it is always 1/2 for electrons { data-list-type=”bulleted” data-bullet-style=”none”}

Glossary

quantum numbers
: the values of quantized entities, such as energy and angular momentum

angular momentum quantum number
: a quantum number associated with the angular momentum of electrons

spin quantum number
: the quantum number that parameterizes the intrinsic angular momentum (or spin angular momentum, or simply spin) of a given particle

spin projection quantum number
: quantum number that can be used to calculate the intrinsic electron angular momentum along the z -axis

z-component of spin angular momentum
: component of intrinsic electron spin along the z -axis

magnitude of the intrinsic (internal) spin angular momentum
: given by $S = \sqrt{s(s+1)}\frac{h}{2\pi}$

z-component of the angular momentum
: component of orbital angular momentum of electron along the z -axis

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The Pauli Exclusion Principle

- Define the composition of an atom along with its electrons, neutrons, and protons.
- Explain the Pauli exclusion principle and its application to the atom.
- Specify the shell and subshell symbols and their positions.
- Define the position of electrons in different shells of an atom.
- State the position of each element in the periodic table according to shell filling.

Multiple-Electron Atoms

All atoms except hydrogen are multiple-electron atoms. The physical and chemical properties of elements are directly related to the number of electrons a neutral atom has. The periodic table of the elements groups elements with similar properties into columns. This systematic organization is related to the number of electrons in a neutral atom, called the **atomic number**, Z . We shall see in this section that the exclusion principle is key to the underlying explanations, and that it applies far beyond the realm of atomic physics.

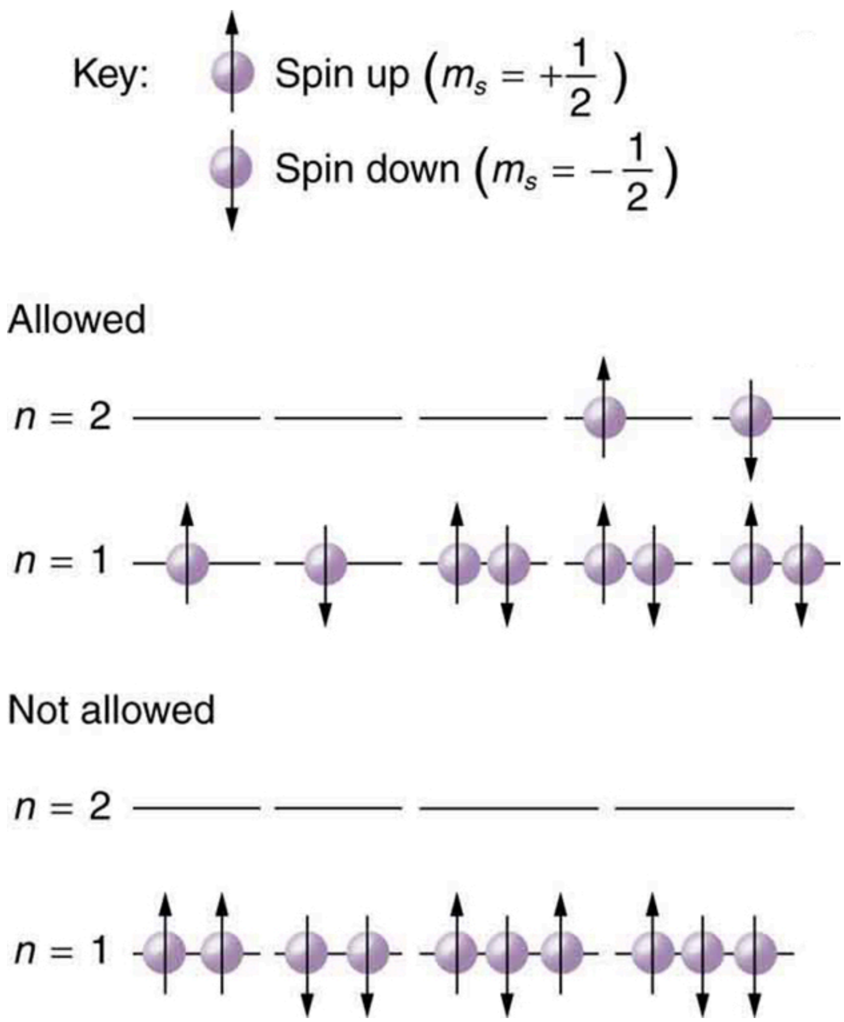
In 1925, the Austrian physicist Wolfgang Pauli (see [Figure 1]) proposed the following rule: No two electrons can have the same set of quantum numbers. That is, no two electrons can be in the same state. This statement is known as the **Pauli exclusion principle**, because it excludes electrons from being in the same state. The Pauli exclusion principle is extremely powerful and very broadly applicable. It applies to any identical particles with half-integral intrinsic spin—that is, having $s = 1/2, 3/2, \ldots$ Thus no two electrons can have the same set of quantum numbers.

Pauli Exclusion Principle

No two electrons can have the same set of quantum numbers. That is, no two electrons can be in the same state.

The Austrian physicist Wolfgang Pauli (1900–1958) played a major role in the development of quantum mechanics. He proposed the exclusion principle; hypothesized the existence of an important particle, called the neutrino, before it was directly observed; made fundamental contributions to several areas of theoretical physics; and influenced many students who went on to do important work of their own. (credit: Nobel Foundation, via Wikimedia Commons)

Let us examine how the exclusion principle applies to electrons in atoms. The quantum numbers involved were defined in Quantum Numbers and Rules as n, l, m_l, s , and m_s . Since s is always $1/2$ for electrons, it is redundant to list s , and so we omit it and specify the state of an electron by a set of four numbers (n, l, m_l, m_s) . For example, the quantum numbers $(2, 1, 0, -1/2)$ completely specify the state of an electron in an atom. Since no two electrons can have the same set of quantum numbers, there are limits to how many of them can be in the same energy state. Note that n determines the energy state in the absence of a magnetic field. So we first choose n , and then we see how many electrons can be in this energy state or energy level. Consider the $n = 1$ level, for example. The only value l can have is 0 (see [Table 1] for a list of possible values once n is known), and thus m_l can only be 0. The spin projection m_s can be either $+1/2$ or $-1/2$, and so there can be two electrons in the $n = 1$ state. One has quantum numbers $(1, 0, 0, +1/2)$, and the other has $(1, 0, 0, -1/2)$. [Figure 2] illustrates that there can be one or two electrons having $n = 1$, but not three.



The Pauli exclusion principle explains why some configurations of electrons are allowed while others are not. Since electrons cannot have the same set of quantum numbers, a maximum of two can be in the $n=1$ level, and a third electron must reside in the higher-energy $n=2$ level. If there are two electrons in the $n=1$ level, their spins must be in opposite directions. (More precisely, their spin projections must differ.)

Shells and Subshells

Because of the Pauli exclusion principle, only hydrogen and helium can have all of their electrons in the $n=1$ state. Lithium (see the periodic table) has three electrons, and so one must be in the $n=2$ level. This leads to the concept of shells and shell filling. As we progress up in the number of electrons, we go from hydrogen to helium, lithium, beryllium, boron, and so on, and we see that there are limits to the number of electrons for each value of n. Higher values of the shell n correspond to higher energies, and they can allow more electrons because of the various combinations of l, m_l, and m_s that are possible. Each value of the principal quantum number n thus corresponds to an atomic **shell** into which a limited number of electrons can go. Shells and the number of electrons in them determine the physical and chemical properties of atoms, since it is the outermost electrons that interact most with anything outside the atom.

The probability clouds of electrons with the lowest value of l are closest to the nucleus and, thus, more tightly bound. Thus when shells fill, they start with $l=0$, progress to $l=1$, and so on. Each value of l thus corresponds to a **subshell**.

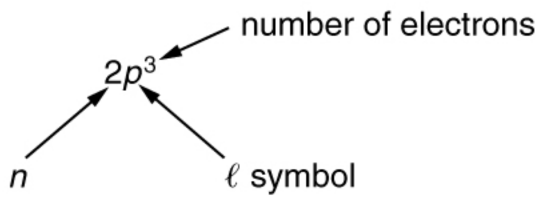
The table given below lists symbols traditionally used to denote shells and subshells.

Shell and Subshell Symbols

Shell	**Subshell**	
n	l	***Symbol***
1	0	s
2	1	p
3	2	d

Shell	Subshell	
4	3	f
5	4	g
	5	h
	6[1]	i

To denote shells and subshells, we write nl with a number for n and a letter for l . For example, an electron in the $n=1$ state must have $l=0$, and it is denoted as a $1s$ electron. Two electrons in the $n=1$ state is denoted as $1s^2$. Another example is an electron in the $n=2$ state with $l=1$, written as $2p$. The case of three electrons with these quantum numbers is written $2p^3$. This notation, called spectroscopic notation, is generalized as shown in [Figure 3] .



Counting the number of possible combinations of quantum numbers allowed by the exclusion principle, we can determine how many electrons it takes to fill each subshell and shell.

How Many Electrons Can Be in This Shell?

List all the possible sets of quantum numbers for the $n=2$ shell, and determine the number of electrons that can be in the shell and each of its subshells.

Strategy

Given $n=2$ for the shell, the rules for quantum numbers limit l to be 0 or 1. The shell therefore has two subshells, labeled $2s$ and $2p$. Since the lowest l subshell fills first, we start with the $2s$ subshell possibilities and then proceed with the $2p$ subshell.

Solution

It is convenient to list the possible quantum numbers in a table, as shown below.

n	ℓ	m_l	m_s	Subshell	Total in subshell	Total in shell
2	0	0	+1/2	2s	2	8
2	0	0	−1/2			
2	1	1	+1/2	2p	6	
2	1	1	−1/2			
2	1	0	+1/2			
2	1	0	−1/2			
2	1	−1	+1/2			
2	1	−1	−1/2			

Discussion

It is laborious to make a table like this every time we want to know how many electrons can be in a shell or subshell. There exist general rules that are easy to apply, as we shall now see.

The number of electrons that can be in a subshell depends entirely on the value of l . Once l is known, there are a fixed number of values of m_l , each of which can have two values for m_s First, since m_l goes from $-l$ to l in steps of 1, there are $2l+1$ possibilities. This number is multiplied by 2, since each electron can be spin up or spin down. Thus the *maximum number of electrons that can be in a subshell* is $2(2l+1)$.

For example, the $2s$ subshell in [Figure 4] has a maximum of 2 electrons in it, since $2(2l+1) = 2(0+1) = 2$ for this subshell. Similarly, the $2p$ subshell has a maximum of 6 electrons, since $2(2l+1) = 2(2+1) = 6$. For a shell, the maximum number is the sum of what can fit in the subshells. Some algebra shows that the *maximum number of electrons that can be in a shell* is $2n^2$.

For example, for the first shell $n = 1$, and so $2n^2 = 2$. We have already seen that only two electrons can be in the $n = 1$ shell. Similarly, for the second shell, $n = 2$, and so $2n^2 = 8$. As found in [Figure 4], the total number of electrons in the $n = 2$ shell is 8.

Subshells and Totals for $n=3$

How many subshells are in the $n = 3$ shell? Identify each subshell, calculate the maximum number of electrons that will fit into each, and verify that the total is $2n^2$.

Strategy

Subshells are determined by the value of l ; thus, we first determine which values of l are allowed, and then we apply the equation "maximum number of electrons that can be in a subshell $= 2(2l+1)$ " to find the number of electrons in each subshell.

Solution

Since $n = 3$, we know that l can be $0, 1$, or 2 ; thus, there are three possible subshells. In standard notation, they are labeled the $3s$, $3p$, and $3d$ subshells. We have already seen that 2 electrons can be in an s state, and 6 in a p ** state, but let us use the equation "maximum number of electrons that can be in a subshell = $2(2l+1)$ " to calculate the maximum number in each:

$$3s \text{ has } l=0; \text{thus}, 2(2l+1)=2(0+1)=2 \quad 3p \text{ has } l=1; \text{ thus, } 2(2l+1)=2(2+1)=6 \quad 3d \text{ has } l=2; \text{ thus, } 2(2l+1)=2(4+1)=10 \quad \text{Total}=18 \quad (\text{in the } n=3 \text{ shell})$$

The equation "maximum number of electrons that can be in a shell = $2n^2$ " gives the maximum number in the $n = 3$ shell to be

$$\text{Maximum number of electrons} = 2n^2 = 2(3)^2 = 2(9) = 18.$$

Discussion

The total number of electrons in the three possible subshells is thus the same as the formula $2n^2$. In standard (spectroscopic) notation, a filled $n = 3$ shell is denoted as $3s^2 3p^6 3d^{10}$. Shells do not fill in a simple manner. Before the $n = 3$ shell is completely filled, for example, we begin to find electrons in the $n = 4$ shell.

Shell Filling and the Periodic Table

[Table 2] shows electron configurations for the first 20 elements in the periodic table, starting with hydrogen and its single electron and ending with calcium. The Pauli exclusion principle determines the maximum number of electrons allowed in each shell and subshell. But the order in which the shells and subshells are filled is complicated because of the large numbers of interactions between electrons.

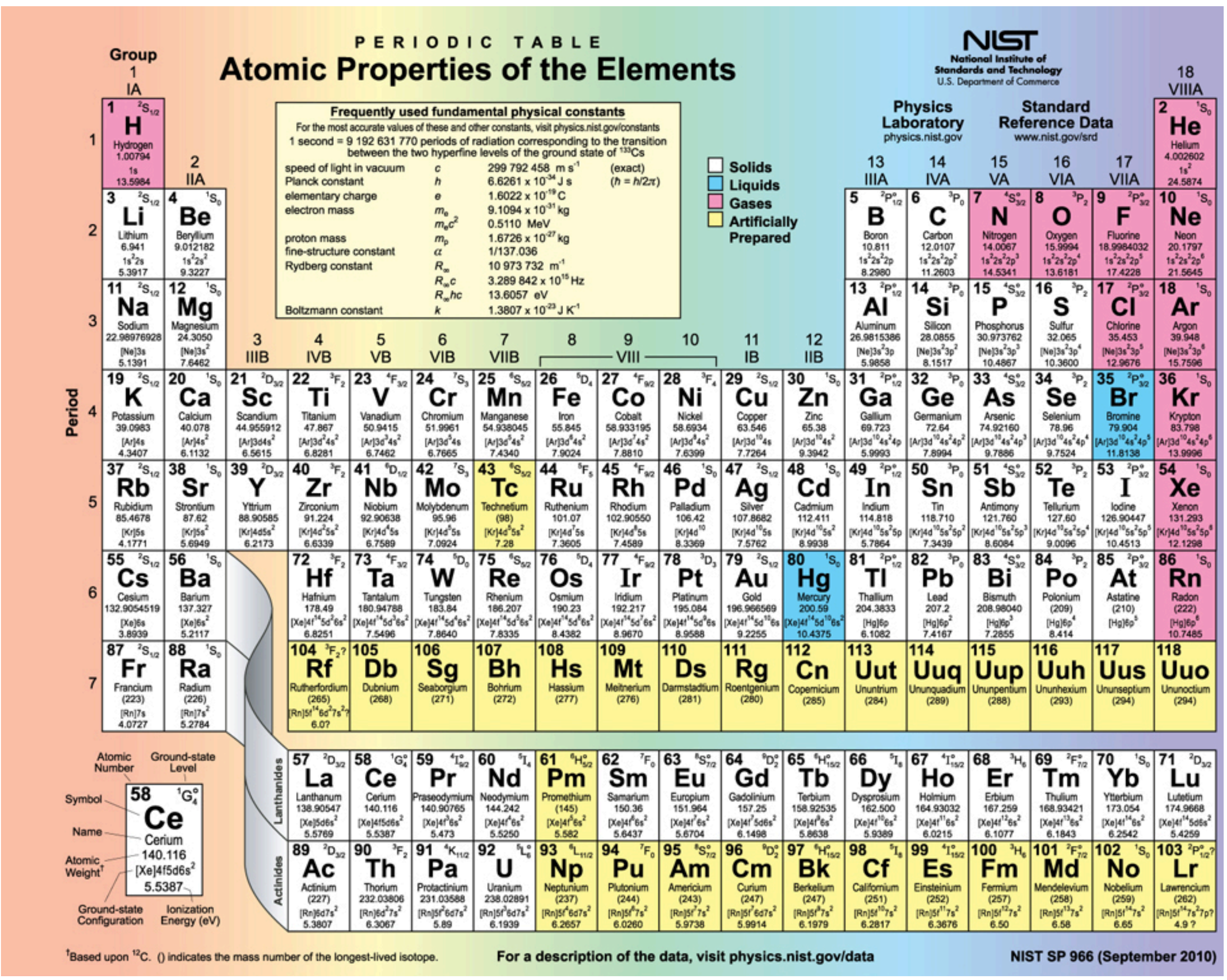
Electron Configurations of Elements Hydrogen Through Calcium

Element	Number of electrons (Z)	Ground state configuration
H	1	$1s^1$
He	2	$1s^2$
Li	3	$1s^2\ 2s^1$
Be	4	" $2s^2$
B	5	" $2s^2\ 2p^1$
C	6	" $2s^2\ 2p^2$
N	7	" $2s^2\ 2p^3$
O	8	" $2s^2\ 2p^4$
F	9	" $2s^2\ 2p^5$
Ne	10	" $2s^2\ 2p^6$
Na	11	" $2s^2\ 2p^6\ 3s^1$
Mg	12	" " " $3s^2$

Element	Number of electrons (Z)	Ground state configuration
Al	13	" " " $3s^2\,3p^1$
Si	14	" " " $3s^2\,3p^2$
P	15	" " " $3s^2\,3p^3$
S	16	" " " $3s^2\,3p^4$
Cl	17	" " " $3s^2\,3p^5$
Ar	18	" " " $3s^2\,3p^6$
K	19	" " " $3s^2\,3p^6\,4s^1$
Ca	20	" " " " " $4s^2$

Examining the above table, you can see that as the number of electrons in an atom increases from 1 in hydrogen to 2 in helium and so on, the lowest-energy shell gets filled first—that is, the $n=1$ shell fills first, and then the $n=2$ shell begins to fill. Within a shell, the subshells fill starting with the lowest l , or with the s subshell, then the p , and so on, usually until all subshells are filled. The first exception to this occurs for potassium, where the $4s$ subshell begins to fill before any electrons go into the $3d$ subshell. The next exception is not shown in [Table 2]; it occurs for rubidium, where the $5s$ subshell starts to fill before the $4d$ subshell. The reason for these exceptions is that $l=0$ electrons have probability clouds that penetrate closer to the nucleus and, thus, are more tightly bound (lower in energy).

[Figure 5] shows the periodic table of the elements, through element 118. Of special interest are elements in the main groups, namely, those in the columns numbered 1, 2, 13, 14, 15, 16, 17, and 18.



Periodic table of the elements (credit: National Institute of Standards and Technology, U.S. Department of Commerce)

The number of electrons in the outermost subshell determines the atom's chemical properties, since it is these electrons that are farthest from the nucleus and thus interact most with other atoms. If the outermost subshell can accept or give up an electron easily, then the atom will be highly reactive chemically. Each group in the periodic table is characterized by its outermost electron configuration. Perhaps the most familiar is Group 18 (Group VIII), the noble gases (helium, neon, argon, etc.). These gases are all characterized by a filled outer subshell that is particularly stable. This means that they have large ionization energies and do not readily give up an electron. Furthermore, if they were to accept an extra electron, it would be in a significantly higher level and thus loosely bound. Chemical reactions often involve sharing electrons. Noble gases can be forced into unstable chemical compounds only under high pressure and temperature.

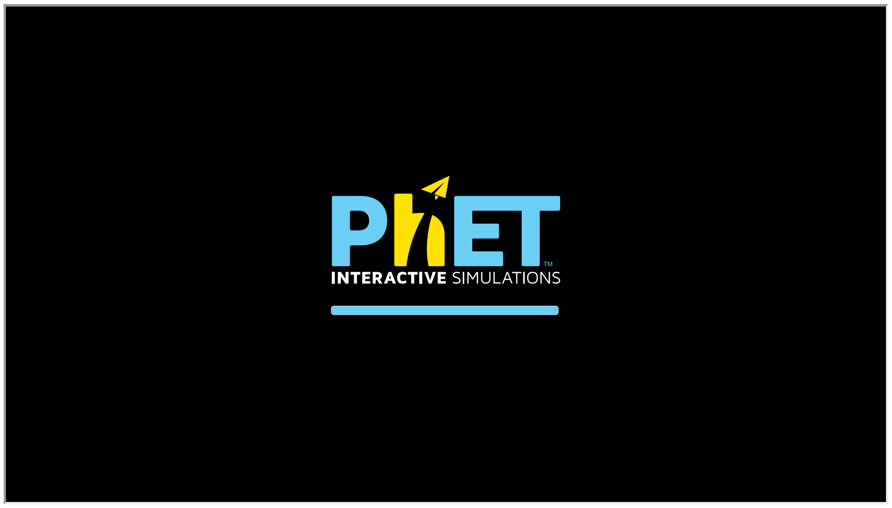
Group 17 (Group VII) contains the halogens, such as fluorine, chlorine, iodine and bromine, each of which has one less electron than a neighboring noble gas. Each halogen has 5 p electrons (a p^5 configuration), while the p subshell can hold 6 electrons. This means the halogens have one vacancy in their outermost subshell. They thus readily accept an extra electron (it becomes tightly bound, closing the shell as in noble gases) and are highly reactive chemically. The halogens are also likely to form singly negative ions, such as $C1^{-}$, fitting an extra electron into the vacancy in the outer subshell. In contrast, alkali metals, such as sodium and potassium, all have a single s electron in their outermost subshell (an s^1 configuration) and are members of Group 1 (Group I). These elements easily give up their extra electron and are thus highly reactive chemically. As you might expect, they also tend to form singly positive ions, such as Na^{+}

, by losing their loosely bound outermost electron. They are metals (conductors), because the loosely bound outer electron can move freely.

Of course, other groups are also of interest. Carbon, silicon, and germanium, for example, have similar chemistries and are in Group 4 (Group IV). Carbon, in particular, is extraordinary in its ability to form many types of bonds and to be part of long chains, such as inorganic molecules. The large group of what are called transitional elements is characterized by the filling of the d subshells and crossing of energy levels. Heavier groups, such as the lanthanide series, are more complex—their shells do not fill in simple order. But the groups recognized by chemists such as Mendeleev have an explanation in the substructure of atoms.

PhET Explorations: Stern-Gerlach Experiment

Build an atom out of protons, neutrons, and electrons, and see how the element, charge, and mass change. Then play a game to test your ideas!



Section Summary

- The state of a system is completely described by a complete set of quantum numbers. This set is written as (n, l, m_l, m_s).
- The Pauli exclusion principle says that no two electrons can have the same set of quantum numbers; that is, no two electrons can be in the same state.
- This exclusion limits the number of electrons in atomic shells and subshells. Each value of n corresponds to a shell, and each value of l corresponds to a subshell.
- The maximum number of electrons that can be in a subshell is $2(2l+1)$.
- The maximum number of electrons that can be in a shell is $2n^2$.

Conceptual Questions

Identify the shell, subshell, and number of electrons for the following: (a) $2p^3$. (b) $4d^9$. (c) $3s^1$. (d) $5g^{16}$.

Show Solution

Solution

(a) $2p^3$: Shell $n=2$, subshell p (which means $l=1$), 3 electrons.

(b) $4d^9$: Shell $n = 4$, subshell d (which means $l = 2$), 9 electrons.

(c) $3s^1$: Shell $n = 3$, subshell s (which means $l = 0$), 1 electron.

(d) $5g^{16}$: Shell $n = 5$, subshell g (which means $l = 4$), 16 electrons.

Discussion

In spectroscopic notation nl^x , the number gives the principal quantum number n (the shell), the letter gives the orbital angular momentum quantum number l (the subshell: $s, p, d, f, g, h...$ correspond to $l = 0, 1, 2, 3, 4, 5...$), and the superscript gives the number of electrons in that subshell.

Which of the following are not allowed? State which rule is violated for any that are not allowed. (a) $1p^3$ (b) $2p^8$ (c) $3g^{11}$ (d) $4f^2$

Show Solution

Solution

(a) $1p^3$ is **not allowed**. For $n = 1$, the only allowed value of l is $l = 0$ (the s subshell). The rule $l is violated, since p corresponds to $l = 1$, but l must be less than $n = 1$.

(b) $2p^8$ is **not allowed**. The p subshell has $l = 1$, so it can hold a maximum of $2(2l+1) = 2(2 \cdot 1+1) = 6$ electrons. Having 8 electrons exceeds this maximum.

(c) $3g^{11}$ is **not allowed**. For $n = 3$, the maximum value of l is $l = n - 1 = 2$ (the d subshell). The g subshell corresponds to $l = 4$, which violates the rule $l .

(d) $4f^2$ is **allowed**. For $n = 4$, l can range from 0 to 3. The f subshell corresponds to $l = 3$, which satisfies $l . The f subshell can hold up to $2(2l+1) = 2(2 \cdot 3+1) = 14$ electrons, so having 2 electrons is allowed.

Problem Exercises

(a) How many electrons can be in the $n = 4$ shell?

(b) What are its subshells, and how many electrons can be in each?

Show Solution

(a) 32. (b) $2 \mathrm{in} s, 6 \mathrm{in} p, 10 \mathrm{in} d,$ and 14 in f , for a total of 32.

(a) What is the minimum value of 1 for a subshell that has 11 electrons in it?

(b) If this subshell is in the $n = 5$ shell, what is the spectroscopic notation for this atom?

Show Solution

Strategy

The maximum number of electrons that can occupy a subshell is given by $2(2l+1)$. We need to find the minimum value of l such that this maximum is at least 11.

Solution

(a) We need $2(2l+1) \geq 11$. Solving for l :

$$2(2l+1) \geq 11 \quad 2l+1 \geq 5.5 \quad 2l \geq 4.5 \quad l \geq 2.25$$

Since l must be an integer, the minimum value is $l = 3$. We can verify: $2(2 \cdot 3+1) = 2(7) = 14$ electrons maximum, which can indeed hold 11 electrons. Note that $l = 2$ would only allow $2(2 \cdot 2+1) = 10$ electrons, which is insufficient.

(b) For $n = 5$ and $l = 3$, the subshell is f (since $l = 3$ corresponds to the f subshell). The spectroscopic notation is $5f^{11}$.

Discussion

This problem illustrates how the Pauli exclusion principle limits the number of electrons in each subshell. The f subshell ($l = 3$) is the smallest subshell that can accommodate 11 electrons. This is relevant for understanding the electron configurations of heavy elements in the lanthanide and actinide series, where f orbitals begin to fill.

(a) If one subshell of an atom has 9 electrons in it, what is the minimum value of l ? (b) What is the spectroscopic notation for this atom, if this subshell is part of the $n = 3$ shell?

Show Solution

(a) 2

(b) $3d^9$

(a) List all possible sets of quantum numbers (n, l, m_l, m_s) for the $n = 3$ shell, and determine the number of electrons that can be in the shell and each of its subshells.

(b) Show that the number of electrons in the shell equals $2n^2$ and that the number in each subshell is $2(2l + 1)$.

Show Solution

Strategy

For $n = 3$, the allowed values of l are $0, 1,$ and 2 (corresponding to $3s, 3p,$ and $3d$ subshells). For each value of l , m_l ranges from $-l$ to $+l$, and m_s can be $\pm 1/2$.

Solution

(a) The complete list of quantum numbers for the $n = 3$ shell:

3s subshell ($l = 0$):

- $(3, 0, 0, +1/2)$
- $(3, 0, 0, -1/2)$

Total: 2 electrons in $3s$

3p subshell ($l = 1$):

- $(3, 1, -1, +1/2)$
- $(3, 1, -1, -1/2)$
- $(3, 1, 0, +1/2)$
- $(3, 1, 0, -1/2)$
- $(3, 1, +1, +1/2)$
- $(3, 1, +1, -1/2)$

Total: 6 electrons in $3p$

3d subshell ($l = 2$):

- $(3, 2, -2, +1/2)$
- $(3, 2, -2, -1/2)$
- $(3, 2, -1, +1/2)$
- $(3, 2, -1, -1/2)$
- $(3, 2, 0, +1/2)$
- $(3, 2, 0, -1/2)$
- $(3, 2, +1, +1/2)$
- $(3, 2, +1, -1/2)$
- $(3, 2, +2, +1/2)$
- $(3, 2, +2, -1/2)$

Total: 10 electrons in $3d$

Total electrons in $n = 3$ shell: $2 + 6 + 10 = 18$

(b) For each subshell, we can verify $2(2l + 1)$:

- $3s$: $2(2 \cdot 0 + 1) = 2(1) = 2$ ✓
- $3p$: $2(2 \cdot 1 + 1) = 2(3) = 6$ ✓
- $3d$: $2(2 \cdot 2 + 1) = 2(5) = 10$ ✓

For the entire shell: $2n^2 = 2(3)^2 = 2(9) = 18$ ✓

Discussion

This systematic enumeration demonstrates how the Pauli exclusion principle limits electron capacity. Each unique set of quantum numbers can hold exactly one electron. The factor of 2 in $2(2l+1)$ comes from the two possible spin states ($m_s = \pm 1/2$), while $2l + 1$ represents the number of possible m_l values for a given l . The formula $2n^2$ emerges from summing over all possible l values for a given n .

Which of the following spectroscopic notations are not allowed? (a) $5s^1$ (b) $1d^1$ (c) $4s^3$ (d) $3p^7$ (e) $5g^{15}$. State which rule is violated for each that is not allowed.

Show Solution

(b) $n \geq l$ is violated, (c) cannot have 3 electrons in s subshell since $3 > (2l+1) = 2$ (d) cannot have 7 electrons in p subshell since $7 > (2l+1) = 2(2+1) = 6$

Which of the following spectroscopic notations are allowed (that is, which violate none of the rules regarding values of quantum numbers)? (a) $1s^1$ (b) $1d^3$ (c) $4s^2$ (d) $3p^7$ (e) $6h^{20}$

Show Solution

Strategy

For each notation, we must check two conditions: (1) $l , and (2) the number of electrons does not exceed $2(2l+1)$ for that subshell.

Solution

(a) $1s^1$ is **allowed**. For $n = 1$, we can have $l = 0$ (s subshell). The maximum number of electrons is $2(2 \cdot 0+1) = 2$, so having 1 electron is allowed.

(b) $1d^3$ is **not allowed**. The d subshell corresponds to $l = 2$. For $n = 1$, the maximum value of l is $n - 1 = 0$. Since $l = 2 > n = 1$ violates the rule $l , this configuration is forbidden.

(c) $4s^2$ is **allowed**. For $n = 4$, we can have $l = 0$ (s subshell). The maximum number of electrons is $2(2 \cdot 0+1) = 2$, so having 2 electrons exactly fills this subshell.

(d) $3p^7$ is **not allowed**. The p subshell has $l = 1$, which is allowed for $n = 3$. However, the maximum number of electrons in a p subshell is $2(2 \cdot 1+1) = 6$. Having 7 electrons exceeds this limit.

(e) $6h^{20}$ is **allowed**. The h subshell corresponds to $l = 5$. For $n = 6$, the maximum value of l is $n - 1 = 5$, so $l = 5$ is allowed. The maximum number of electrons is $2(2 \cdot 5+1) = 2(11) = 22$, so having 20 electrons is within the limit.

Discussion

The allowed configurations are **(a), (c), and (e)**. This problem reinforces the two fundamental constraints on electron configurations: the orbital angular momentum quantum number must satisfy $l \leq n - 1$, and the number of electrons in a subshell cannot exceed $2(2l+1)$. These constraints arise directly from the Pauli exclusion principle and the rules governing quantum numbers.

(a) Using the Pauli exclusion principle and the rules relating the allowed values of the quantum numbers (n, l, m_l, m_s) , prove that the maximum number of electrons in a subshell is $2n^2$.

(b) In a similar manner, prove that the maximum number of electrons in a shell is $2n^2$.

Show Solution

(a) The number of different values of m_l is $\pm l, \pm(l-1), \ldots, 0$ for each $l > 0$ and one for $l = 0 \Rightarrow (2l+1)$. Also an overall factor of 2 since each m_l can have m_s equal to either $+1/2$ or $-1/2 \Rightarrow 2(2l+1)$.

(b) for each value of l, you get $2(2l+1) = 0, 1, 2, \ldots, (n-1) \Rightarrow 2\{[(2)(0)+1]+[(2)(1)+1]+\ldots.+[(2)(n-1)+1]\} = 2[1+3+\ldots+(2n-3)+(2n-1)]$ $n$terms to see that the expression in the box is $= n^2$, imagine taking $(n-1)$ from the last term and adding it to first term $= 2[1+(n-1)+3+\ldots+(2n-3)+(2n-1)-(n-1)] = 2[n+3+\ldots.+(2n-3)+n]$. Now take $(n-3)$ from penultimate term and add to the second term $2[n+n+\ldots+n+n]$ $n$terms $= 2n^2$.

Integrated Concepts

Estimate the density of a nucleus by calculating the density of a proton, taking it to be a sphere 1.2 fm in diameter. Compare your result with the value estimated in this chapter.

Show Solution

Strategy

We can estimate nuclear density by treating a proton as a sphere with diameter $d = 1.2\text{ fm} = 1.2 \times 10^{-15}\text{ m}$. The density is $\rho = m/V$, where the mass of a proton is $m_p = 1.67 \times 10^{-27}\text{ kg}$ and the volume is $V = \frac{4}{3}\pi r^3$.

Solution

The radius of the proton is:

$$r = \frac{d}{2} = \frac{1.2 \times 10^{-15}\text{ m}}{2} = 6.0 \times 10^{-16}\text{ m}$$

The volume is:

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi (6.0 \times 10^{-16}\text{ m})^3 = 9.05 \times 10^{-46}\text{ m}^3$$

The density is:

$$\rho = \frac{m}{V} = \frac{1.67 \times 10^{-27}\text{ kg}}{9.05 \times 10^{-46}\text{ m}^3} = 1.8 \times 10^{17}\text{ kg/m}^3$$

Discussion

This density is extraordinarily large—about 10^{14} times greater than the density of water (10^3 kg/m^3) or solid matter ($\sim 10^4\text{ kg/m}^3$). This result is typical of nuclear densities and demonstrates that nuclei are incredibly compact objects. All atomic nuclei have approximately the same density because they are composed of closely packed nucleons (protons and neutrons). This uniform density is a key feature of the nuclear structure and validates the liquid drop model of the nucleus.

Integrated Concepts

The electric and magnetic forces on an electron in the CRT in [Figure 4] are supposed to be in opposite directions. Verify this by determining the direction of each force for the situation shown. Explain how you obtain the directions (that is, identify the rules used).

Show Solution

The electric force on the electron is up (toward the positively charged plate). The magnetic force is down (by the RHR).

(a) What is the distance between the slits of a diffraction grating that produces a first-order maximum for the first Balmer line at an angle of 20.0° ?

(b) At what angle will the fourth line of the Balmer series appear in first order?

(c) At what angle will the second-order maximum be for the first line?

Show Solution

Strategy

The diffraction grating equation is $d\sin\theta = m\lambda$, where d is the slit separation, θ is the angle, m is the order, and λ is the wavelength. The first Balmer line ($n_i = 3$ to $n_f = 2$) has $\lambda = 656\text{ nm}$. The fourth line ($n_i = 6$ to $n_f = 2$) has $\lambda = 410\text{ nm}$.

Solution

(a) For $m = 1$, $\theta = 20.0^\circ$, and $\lambda = 656\text{ nm}$:

$$d = \frac{m\lambda}{\sin\theta} = \frac{(1)(656 \times 10^{-9}\text{ m})}{\sin 20.0^\circ} = \frac{656 \times 10^{-9}\text{ m}}{0.342} = 1.92 \times 10^{-6}\text{ m} = 1.92\ \mu\text{m}$$

(b) For the fourth Balmer line with $\lambda = 410\text{ nm}$, $m = 1$, and $d = 1.92 \times 10^{-6}\text{ m}$:

$$\sin\theta = \frac{m\lambda}{d} = \frac{(1)(410 \times 10^{-9}\text{ m})}{1.92 \times 10^{-6}\text{ m}} = 0.214$$
$$\theta = \sin^{-1}(0.214) = 12.4^\circ$$

(c) For $m = 2$ (second order), $\lambda = 656\text{ nm}$, and $d = 1.92 \times 10^{-6}\text{ m}$:

$$\sin\theta = \frac{m\lambda}{d} = \frac{(2)(656 \times 10^{-9}\text{ m})}{1.92 \times 10^{-6}\text{ m}} = 0.683$$

$$\theta=\sin^{-1}(0.683)=43.1^\circ$$

Discussion

Diffraction gratings are powerful tools for analyzing spectral lines. The angular separation of different wavelengths increases with order m, but higher orders also require larger angles. Note that shorter wavelengths (like the fourth Balmer line at 410 nm) appear at smaller angles than longer wavelengths (like the first line at 656 nm), which is why dispersive elements can separate colors. This technique is fundamental to spectroscopy and was crucial in the early studies of atomic structure.

Integrated Concepts

A galaxy moving away from the earth has a speed of $0.0100c$. What wavelength do we observe for an $n_i=7$ to $n_f=2$ transition for hydrogen in that galaxy?

Show Solution

401 nm

Integrated Concepts

Calculate the velocity of a star moving relative to the earth if you observe a wavelength of 91.0 nm for ionized hydrogen capturing an electron directly into the lowest orbital (that is, a $n_i=\infty$ to $n_f=1$, or a Lyman series transition).

Show Solution

Strategy

First, we need to find the rest wavelength for the transition $n_i=\infty$ to $n_f=1$ using the Rydberg formula. Then we can use the Doppler shift formula for non-relativistic velocities: $\frac{\Delta\lambda}{\lambda}=\frac{v}{c}$.

Solution

The Rydberg formula gives:

$$\frac{1}{\lambda}=R\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$$

For $n_f=1$ and $n_i=\infty$:

$$\frac{1}{\lambda}=R\left(\frac{1}{1^2}-\frac{1}{\infty^2}\right)=R(1-0)=R$$

With $R=1.097\times10^7\ \text{m}^{-1}$:

$$\lambda_0=\frac{1}{R}=\frac{1}{1.097\times10^7\ \text{m}^{-1}}=9.116\times10^{-8}\ \text{m}=91.16\ \text{nm}$$

The observed wavelength is $\lambda=91.0\ \text{nm}$, which is less than λ_0, indicating a blueshift (star moving toward us).

The velocity is:

$$v=c\frac{\Delta\lambda}{\lambda_0}=c\frac{\lambda_0-\lambda}{\lambda_0}=(3.00\times10^8\ \text{m/s})\frac{91.16-91.0}{91.16}$$
$$v=(3.00\times10^8\ \text{m/s})(0.00175)=5.26\times10^5\ \text{m/s}=526\ \text{km/s}$$

The star is moving **toward** Earth at approximately **526 km/s**.

Discussion

This large velocity is typical of stellar motions in our galaxy. The blueshift (shorter observed wavelength) indicates motion toward us. This Lyman series limit transition ($n=\infty$ to $n=1$) corresponds to the ionization edge of hydrogen and is an important feature in stellar spectra. Such Doppler measurements are crucial for understanding galactic dynamics and stellar velocities.

Integrated Concepts

In a Millikan oil-drop experiment using a setup like that in [Figure 6], a 500-V potential difference is applied to plates separated by 2.50 cm. (a) What is the mass of an oil drop having two extra electrons that is suspended motionless by the field between the plates? (b) What is the diameter of the drop, assuming it is a sphere with the density of olive oil?

Show Solution

(a) $6.54\times10^{-16}\text{kg}$ (b) $5.54\times10^{-7}\text{m}$

Integrated Concepts

What double-slit separation would produce a first-order maximum at 3.00° for 25.0-keV X-rays? The small answer indicates that the wave character of X-rays is best determined by having them interact with very small objects such as atoms and molecules.

Show Solution

Strategy

First, we need to find the wavelength of 25.0-keV X-rays using $E = hf = hc/\lambda$. Then we can use the double-slit equation $d\sin\theta = m\lambda$ to find the slit separation d.

Solution

The wavelength of the X-rays is:

$$\lambda = \frac{hc}{E} = \frac{(6.63\times10^{-34}\text{ J·s})(3.00\times10^{8}\text{ m/s})}{25.0\times10^{3}\text{ eV}\times1.60\times10^{-19}\text{ J/eV}}$$
$$\lambda = \frac{1.989\times10^{-25}\text{ J·m}}{4.00\times10^{-15}\text{ J}} = 4.97\times10^{-11}\text{ m} = 0.0497\text{ nm}$$

For the first-order maximum ($m = 1$) at $\theta = 3.00^{\circ}$:

$$d = \frac{m\lambda}{\sin\theta} = \frac{(1)(4.97\times10^{-11}\text{ m})}{\sin 3.00^{\circ}} = \frac{4.97\times10^{-11}\text{ m}}{0.0523}$$
$$d = 9.50\times10^{-10}\text{ m} = 0.950\text{ nm}$$

Discussion

This slit separation of about 1 nm is on the order of atomic dimensions (typical atomic radii are 0.1–0.3 nm, and atomic spacings in crystals are 0.2–0.5 nm). This explains why X-ray diffraction is performed using crystal lattices rather than manufactured gratings—the regular spacing of atoms in crystals provides the necessary small-scale structure. X-ray crystallography has been instrumental in determining molecular structures, including the double helix structure of DNA. The wave nature of X-rays at these energy levels is essential for such applications.

Integrated Concepts

In a laboratory experiment designed to duplicate Thomson's determination of q_e/m_e , a beam of electrons having a velocity of $6.00\times10^{7}\text{m/s}$ enters a $5.00\times10^{-3}\text{T}$ magnetic field. The beam moves perpendicular to the field in a path having a 6.80-cm radius of curvature. Determine q_e/m_e from these observations, and compare the result with the known value.

Show Solution

$1.76\times10^{11}\text{C/kg}$, which agrees with the known value of $1.759\times10^{11}\text{C/kg}$ to within the precision of the measurement

Integrated Concepts

Find the value of l , the orbital angular momentum quantum number, for the moon around the earth. The extremely large value obtained implies that it is impossible to tell the difference between adjacent quantized orbits for macroscopic objects.

Show Solution

Strategy

The orbital angular momentum of the moon is $L = mvr$, where m is the moon's mass, v is its orbital velocity, and r is its orbital radius. According to quantum mechanics, $L = \sqrt{l(l+1)}\hbar$. For large l , this simplifies to $L \approx l\hbar$, so $l \approx L/\hbar$.

Solution

First, we need the moon's orbital angular momentum. Using:

- Moon's mass: $m = 7.35\times10^{22}$ kg
- Orbital radius: $r = 3.84\times10^{8}$ m
- Orbital period: $T = 27.3\text{ days} = 2.36\times10^{6}$ s

The orbital velocity is:

$$v = \frac{2\pi r}{T} = \frac{2\pi(3.84\times10^{8}\text{ m})}{2.36\times10^{6}\text{ s}} = 1.02\times10^{3}\text{ m/s}$$

The orbital angular momentum is:

$$L = mvr = (7.35\times10^{22}\text{ kg})(1.02\times10^{3}\text{ m/s})(3.84\times10^{8}\text{ m})$$
$$L = 2.88\times10^{34}\text{ kg·m}^2\text{/s}$$

The quantum number l is:

$$l \approx \frac{L}{\hbar} = \frac{2.88\times10^{34}\ \text{kg}\cdot\text{m}^2/\text{s}}{1.055\times10^{-34}\ \text{J}\cdot\text{s}} = 2.73\times10^{68}$$

Discussion

This extraordinarily large quantum number demonstrates why quantum effects are unobservable for macroscopic objects. The difference between adjacent quantum states would be $\Delta L = \hbar = 1.055 \times 10^{-34}\ \text{J}\cdot\text{s}$, which is completely negligible compared to the total angular momentum of $2.88 \times 10^{34}\ \text{kg}\cdot\text{m}^2/\text{s}$. The fractional change would be $\Delta L / L \sim 10^{-68}$, far too small to measure. This illustrates the correspondence principle: quantum mechanics gives the same results as classical mechanics for macroscopic systems.

Integrated Concepts

Particles called muons exist in cosmic rays and can be created in particle accelerators. Muons are very similar to electrons, having the same charge and spin, but they have a mass 207 times greater. When muons are captured by an atom, they orbit just like an electron but with a smaller radius, since the mass in $a_B = \frac{h^2}{4\pi^2 m_e k q_e^2} = 0.529 \times 10^{-10}\text{m}$ is 207 m_e.

(a) Calculate the radius of the $n = 1$ orbit for a muon in a uranium ion ($Z = 92$).

(b) Compare this with the 7.5-fm radius of a uranium nucleus. Note that since the muon orbits inside the electron, it falls into a hydrogen-like orbit. Since your answer is less than the radius of the nucleus, you can see that the photons emitted as the muon falls into its lowest orbit can give information about the nucleus.

Show Solution

(a) 2.78 fm

(b) 0.37 of the nuclear radius.

Integrated Concepts

Calculate the minimum amount of energy in joules needed to create a population inversion in a helium-neon laser containing 1.00×10^{-4} moles of neon.

Show Solution

Strategy

A helium-neon laser produces red light at $\lambda = 632.8\ \text{nm}$. To create a population inversion, at least half of the neon atoms must be excited to the upper laser level. The energy per photon is $E = hc/\lambda$. We need to find the total energy to excite half of the neon atoms.

Solution

The energy per photon is:

$$E = \frac{hc}{\lambda} = \frac{(6.63\times10^{-34}\ \text{J}\cdot\text{s})(3.00\times10^{8}\ \text{m/s})}{632.8\times10^{-9}\ \text{m}}$$
$$E = \frac{1.989\times10^{-25}\ \text{J}\cdot\text{m}}{6.328\times10^{-7}\ \text{m}} = 3.14\times10^{-19}\ \text{J}$$

The number of neon atoms is:

$$N = nN_A = (1.00\times10^{-4}\ \text{mol})(6.022\times10^{23}\ \text{atoms/mol}) = 6.022\times10^{19}\ \text{atoms}$$

For a population inversion, at least half the atoms must be in the excited state:

$$N_{\text{excited}} = \frac{N}{2} = \frac{6.022\times10^{19}}{2} = 3.01\times10^{19}\ \text{atoms}$$

The total energy required is:

$$E_{\text{total}} = N_{\text{excited}} \times E = (3.01\times10^{19})(3.14\times10^{-19}\ \text{J}) = 9.45\ \text{J}$$

Discussion

This represents the minimum energy needed to create a population inversion. In practice, more energy is required because not all the pump energy efficiently excites the neon atoms, and some atoms spontaneously emit before stimulated emission occurs. Helium-neon lasers use electrical discharge to excite helium atoms first, which then transfer energy to neon atoms through collisions, creating the population inversion necessary for laser action. The relatively modest energy requirement (less than 10 J) explains why He-Ne lasers are efficient and commonly used in laboratories and laser pointers.

Integrated Concepts

A carbon dioxide laser used in surgery emits infrared radiation with a wavelength of $10.6 \mu m$. In 1.00 ms, this laser raised the temperature of $1.00 cm^3$ of flesh to $100^{o}C$ and evaporated it.

(a) How many photons were required? You may assume flesh has the same heat of vaporization as water. (b) What was the minimum power output during the flash?

Show Solution

(a) 1.34×10^{23} (b) 2.52 MW

Integrated Concepts

Suppose an MRI scanner uses 100-MHz radio waves.

(a) Calculate the photon energy.

(b) How does this compare to typical molecular binding energies?

Show Solution

Strategy

The photon energy is given by $E = hf$, where $h = 6.63 \times 10^{-34}$ J·s and $f = 100$ MHz $= 1.00 \times 10^{8}$ Hz. We can compare this to typical molecular binding energies, which are on the order of a few eV.

Solution

(a) The photon energy is:

$$E=hf=(6.63\times10^{-34}\ \text{J}\cdot\text{s})(1.00\times10^{8}\ \text{Hz})=6.63\times10^{-26}\ \text{J}$$

Converting to eV:

$$E=6.63\times10^{-26}\ \text{J}1.60\times10^{-19}\ \text{J/eV}=4.14\times10^{-7}\ \text{eV}=0.414\ \mu\text{eV}$$

(b) Typical molecular binding energies range from about 1 eV to 10 eV. The ratio is:

$$E_{\text{binding}}E_{\text{photon}}\sim1\ \text{eV}4.14\times10^{-7}\ \text{eV}\sim2.4\times10^{6}$$

The MRI photon energy is about **2.4 million times smaller** than typical molecular binding energies.

Discussion

This enormous difference explains why MRI is safe and non-destructive. The radio wave photons used in MRI have far too little energy to break chemical bonds or ionize atoms. Instead, MRI works by detecting the magnetic resonance of hydrogen nuclei (protons) in the body's water and fat molecules. The low-energy photons can flip nuclear spins without damaging tissue, making MRI an ideal diagnostic tool. In contrast, X-rays and gamma rays have energies comparable to or greater than binding energies, which is why they can damage DNA and living tissue.

Integrated Concepts

(a) An excimer laser used for vision correction emits 193-nm UV. Calculate the photon energy in eV.

(b) These photons are used to evaporate corneal tissue, which is very similar to water in its properties. Calculate the amount of energy needed per molecule of water to make the phase change from liquid to gas. That is, divide the heat of vaporization in kJ/kg by the number of water molecules in a kilogram.

(c) Convert this to eV and compare to the photon energy. Discuss the implications.

Show Solution

(a) 6.42 eV

(b) 7.27×10^{-20} J/molecule (c) 0.454 eV, 14.1 times less than a single UV photon. Therefore, each photon will evaporate approximately 14 molecules of tissue. This gives the surgeon a rather precise method of removing corneal tissue from the surface of the eye.

Integrated Concepts

A neighboring galaxy rotates on its axis so that stars on one side move toward us as fast as 200 km/s, while those on the other side move away as fast as 200 km/s. This causes the EM radiation we receive to be Doppler shifted by velocities over the entire range of ±200 km/s. What range of wavelengths will we observe for the 656.0-nm line in the Balmer series of hydrogen emitted by stars in this galaxy. (This is called line broadening.)

Show Solution

Strategy

For non-relativistic velocities, the Doppler shift is $\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$. For stars moving toward us ($v = -200\text{ km/s}$), we observe a blueshift (shorter wavelength). For stars moving away ($v = +200\text{ km/s}$), we observe a redshift (longer wavelength).

Solution

The rest wavelength is $\lambda_0 = 656.0\text{ nm}$.

For stars moving **toward** us at $v = 200\text{ km/s} = 2.00 \times 10^5\text{ m/s}$:

$$\Delta\lambda = \lambda_0 \frac{v}{c} = (656.0\text{ nm}) \frac{2.00 \times 10^5\text{ m/s}}{3.00 \times 10^8\text{ m/s}} = (656.0\text{ nm})(6.67 \times 10^{-4}) = 0.437\text{ nm}$$
$$\lambda_{\text{blue}} = \lambda_0 - \Delta\lambda = 656.0 - 0.437 = 655.6\text{ nm}$$

For stars moving **away** from us at $v = 200\text{ km/s}$:

$$\lambda_{\text{red}} = \lambda_0 + \Delta\lambda = 656.0 + 0.437 = 656.4\text{ nm}$$

The observed wavelength range is **655.6 nm to 656.4 nm**.

Discussion

This line broadening effect is a powerful tool for astronomers to study galactic rotation. By measuring the width of spectral lines, we can determine rotation velocities and infer the mass distribution of galaxies. The broader the spectral line, the faster the galaxy is rotating. This technique has been crucial in discovering dark matter—galaxies rotate faster than expected based on visible matter alone, suggesting the presence of unseen mass. The symmetric broadening around the rest wavelength indicates that we're observing the galaxy edge-on, with equal numbers of stars moving toward and away from us.

Integrated Concepts

A pulsar is a rapidly spinning remnant of a supernova. It rotates on its axis, sweeping hydrogen along with it so that hydrogen on one side moves toward us as fast as 50.0 km/s, while that on the other side moves away as fast as 50.0 km/s. This means that the EM radiation we receive will be Doppler shifted over a range of $\pm 50.0\text{km/s}$. What range of wavelengths will we observe for the 91.20-nm line in the Lyman series of hydrogen? (Such line broadening is observed and actually provides part of the evidence for rapid rotation.)

Show Solution

91.18 nm to 91.22 nm

Integrated Concepts

Prove that the velocity of charged particles moving along a straight path through perpendicular electric and magnetic fields is $v = E/B$. Thus crossed electric and magnetic fields can be used as a velocity selector independent of the charge and mass of the particle involved.

Show Solution

Strategy

For a charged particle to move in a straight line through crossed electric and magnetic fields, the electric force and magnetic force must exactly balance. The electric force is $F_E = qE$ and the magnetic force is $F_B = qvB$ (for perpendicular velocity and field).

Solution

For a particle with charge q moving with velocity v :

The **electric force** is:

$$F_E = qE$$

The **magnetic force** is:

$$F_B = qvB$$

For the particle to travel in a straight line, these forces must be equal in magnitude and opposite in direction:

$$F_E = F_B$$
$$qE = qvB$$

Dividing both sides by q and B :

$$v = \frac{E}{B}$$

Notice that this result is **independent of both charge q and mass m** of the particle.

Discussion

This elegant result shows that only particles with velocity $v = E/B$ will pass through undeflected, regardless of their charge or mass. Particles moving faster will be deflected more by the magnetic force, while slower particles will be deflected more by the electric force. This principle is used in mass spectrometers and particle physics experiments to select particles with specific velocities. J.J. Thomson used this technique in his cathode ray experiments to measure the charge-to-mass ratio of electrons. The velocity selector is a beautiful example of how electric and magnetic forces can be balanced to achieve precise control over charged particle beams.

Unreasonable Results

(a) What voltage must be applied to an X-ray tube to obtain 0.0100-fm-wavelength X-rays for use in exploring the details of nuclei? (b) What is unreasonable about this result? (c) Which assumptions are unreasonable or inconsistent?

Show Solution

(a) $1.24 \times 10^{11} \mathrm{V}$ (b) The voltage is extremely large compared with any practical value.

(c) The assumption of such a short wavelength by this method is unreasonable.

Unreasonable Results

A student in a physics laboratory observes a hydrogen spectrum with a diffraction grating for the purpose of measuring the wavelengths of the emitted radiation. In the spectrum, she observes a yellow line and finds its wavelength to be 589 nm. (a) Assuming this is part of the Balmer series, determine n_{i} , the principal quantum number of the initial state. (b) What is unreasonable about this result? (c) Which assumptions are unreasonable or inconsistent?

Show Solution

Strategy

The Balmer series has $n_{\mathrm{f}} = 2$. We can use the Rydberg formula $1\lambda = R(1n_{2\mathrm{f}} - 1n_{2\mathrm{i}})$ to solve for n_{i} .

Solution

(a) Using the Rydberg formula with $\lambda = 589 \text{ nm} = 5.89 \times 10^{-7} \text{ m}$, $R = 1.097 \times 10^{7} \text{ m}^{-1}$, and $n_{\mathrm{f}} = 2$:

$$1\lambda = R(1n_{2\mathrm{f}} - 1n_{2\mathrm{i}})$$
$$15.89\times10^{-7} \text{ m} = 1.097\times10^{7} \text{ m}^{-1}(14 - 1n_{2\mathrm{i}})$$
$$1.698\times10^{6} \text{ m}^{-1} = 1.097\times10^{7} \text{ m}^{-1}(14 - 1n_{2\mathrm{i}})$$
$$0.1548 = 14 - 1n_{2\mathrm{i}}$$
$$1n_{2\mathrm{i}} = 0.250 - 0.1548 = 0.0952$$
$$n_{2\mathrm{i}} = 10.5$$
$$n_{\mathrm{i}} = 3.24$$

(b) The result $n_{\mathrm{i}} = 3.24$ is **unreasonable** because the principal quantum number n must be a positive integer (1, 2, 3, 4, …). A non-integer value violates the fundamental quantization condition.

(c) The assumption that the 589-nm yellow line is part of the hydrogen Balmer series is **incorrect**. In reality, 589 nm is the famous sodium D-line doublet (589.0 nm and 589.6 nm), not a hydrogen line. The student likely observed sodium contamination in the hydrogen lamp or confused the spectral lines. The Balmer series wavelengths are: 656 nm (red), 486 nm (cyan), 434 nm (blue), and 410 nm (violet)—none are yellow.

Discussion

This problem illustrates the importance of recognizing when results don't make physical sense. Non-integer quantum numbers should immediately signal an error in assumptions or calculations. The 589-nm sodium lines are extremely common in laboratory spectra because sodium is ubiquitous (from salt, fingerprints, etc.) and has a very strong emission at this wavelength, which is why sodium vapor lamps produce characteristic yellow light. This highlights the need for careful spectral identification and understanding of atomic physics principles.

Construct Your Own Problem

The solar corona is so hot that most atoms in it are ionized. Consider a hydrogen-like atom in the corona that has only a single electron. Construct a problem in which you calculate selected spectral energies and wavelengths of the Lyman, Balmer, or other series of this atom that could be used to identify its presence in a very hot gas. You will need to choose the atomic number of the atom, identify the element, and choose which spectral lines to consider.

Show Solution

Strategy

To construct this problem, we need to: (1) choose a highly ionized element likely present in the solar corona, (2) identify specific spectral transitions, and (3) calculate wavelengths and energies that could be observed.

Solution

Example Problem Construction:

The solar corona has temperatures of 1–2 million kelvin, hot enough to fully ionize lighter elements. Consider **helium II** (He^+), which is singly ionized helium with $Z=2$ and one electron, making it hydrogen-like.

Part (a): Calculate the wavelength of the first line in the Lyman series (transition from $n=2$ to $n=1$) for He^+.

For hydrogen-like ions, the wavelengths are modified by Z^2:

$$\frac{1}{\lambda}=Z^2R\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)=2^2(1.097\times10^7\ \text{m}^{-1})\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$$
$$\frac{1}{\lambda}=4(1.097\times10^7)\left(\frac{3}{4}\right)=3.29\times10^7\ \text{m}^{-1}$$
$$\lambda=30.4\ \text{nm (extreme UV)}$$

Part (b): Calculate the photon energy for this transition.

$$E=\frac{hc}{\lambda}=\frac{(6.63\times10^{-34})(3.00\times10^8)}{30.4\times10^{-9}}=6.54\times10^{-18}\ \text{J}=40.9\ \text{eV}$$

Discussion

This approach can be extended to other ions like O^{7+} ($Z=8$) or Fe^{25+} ($Z=26$), which are observed in coronal spectra. The high Z values shift spectral lines to much shorter wavelengths (UV and X-ray regions), which is why space-based telescopes are essential for coronal observations. Students could construct variations by: (1) choosing different elements, (2) examining different series (Balmer, Paschen), (3) calculating ionization energies, or (4) comparing multiple transitions to create a spectral fingerprint for element identification. This problem teaches how atomic spectroscopy is used to determine the composition and temperature of stellar atmospheres.

Construct Your Own Problem

Consider the Doppler-shifted hydrogen spectrum received from a rapidly receding galaxy. Construct a problem in which you calculate the energies of selected spectral lines in the Balmer series and examine whether they can be described with a formula like that in the equation $\frac{1}{\lambda}=R\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$, but with a different constant R.

Show Solution

Strategy

To construct this problem, we need to: (1) choose a recession velocity for the galaxy, (2) calculate Doppler-shifted wavelengths for several Balmer series lines, (3) determine if the shifted wavelengths follow the Rydberg formula with a modified constant, and (4) relate the modified constant to the recession velocity.

Solution

Example Problem Construction:

A distant galaxy is receding from Earth at $v=0.050c$ (1.50×10^7 m/s or 5% the speed of light).

Part (a): Calculate the observed wavelengths for the first three Balmer series lines.

For non-relativistic Doppler shift: $\lambda_{\text{observed}}=\lambda_0\left(1+\frac{v}{c}\right)$

First Balmer line ($n=3$ to $n=2$): $\lambda_0=656$ nm

$$\lambda_{\text{obs}}=656(1+0.050)=656\times1.050=689\ \text{nm}$$

Second Balmer line ($n=4$ to $n=2$): $\lambda_0=486$ nm

$$\lambda_{\text{obs}}=486\times1.050=510\ \text{nm}$$

Third Balmer line ($n=5$ to $n=2$): $\lambda_0=434$ nm

$$\lambda_{\text{obs}}=434\times1.050=456\ \text{nm}$$

Part (b): Determine if these wavelengths follow $\frac{1}{\lambda}=R'\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$ with a modified Rydberg constant R'.

Since $\lambda_{\text{obs}}=\lambda_0\left(1+\frac{v}{c}\right)$, we have:

$$\frac{1}{\lambda_{\text{obs}}}=\frac{1}{\lambda_0(1+v/c)}=\frac{1}{1+v/c}\cdot\frac{1}{\lambda_0}=\frac{R}{1+v/c}\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$$

Therefore, the modified Rydberg constant is:

$$R' = \frac{R}{1+v/c} = \frac{1.097\times10^7\ \text{m}^{-1}}{1.050} = 1.045\times10^7\ \text{m}^{-1}$$

Discussion

This demonstrates that Doppler-shifted spectra from receding galaxies follow the same Rydberg formula but with an effective Rydberg constant reduced by the factor $1/(1+v/c)$. This principle is crucial for measuring cosmological redshifts and determining distances to galaxies. Students could vary this problem by: (1) using different recession velocities, (2) examining other spectral series, (3) considering relativistic velocities requiring the formula $z = \sqrt{\frac{1+v/c}{1-v/c}} - 1$, or (4) investigating how astronomers use multiple spectral lines to confirm redshift measurements. This illustrates how fundamental atomic physics combined with the Doppler effect enables the study of galactic motion and cosmological expansion.

Footnotes

- 1 It is unusual to deal with subshells having l greater than 6, but when encountered, they continue to be labeled in alphabetical order. { data-list-type="bulleted" data-bullet-style="none"}

Glossary

atomic number
: the number of protons in the nucleus of an atom

Pauli exclusion principle
: a principle that states that no two electrons can have the same set of quantum numbers; that is, no two electrons can be in the same state

shell
: a probability cloud for electrons that has a single principal quantum number

subshell
: the probability cloud for electrons that has a single angular momentum quantum number l

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